Fpt University

TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REPORT

Report 4 – Software Design Description

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IV. Software Design Description

1. System Design

1.1 System Architecture

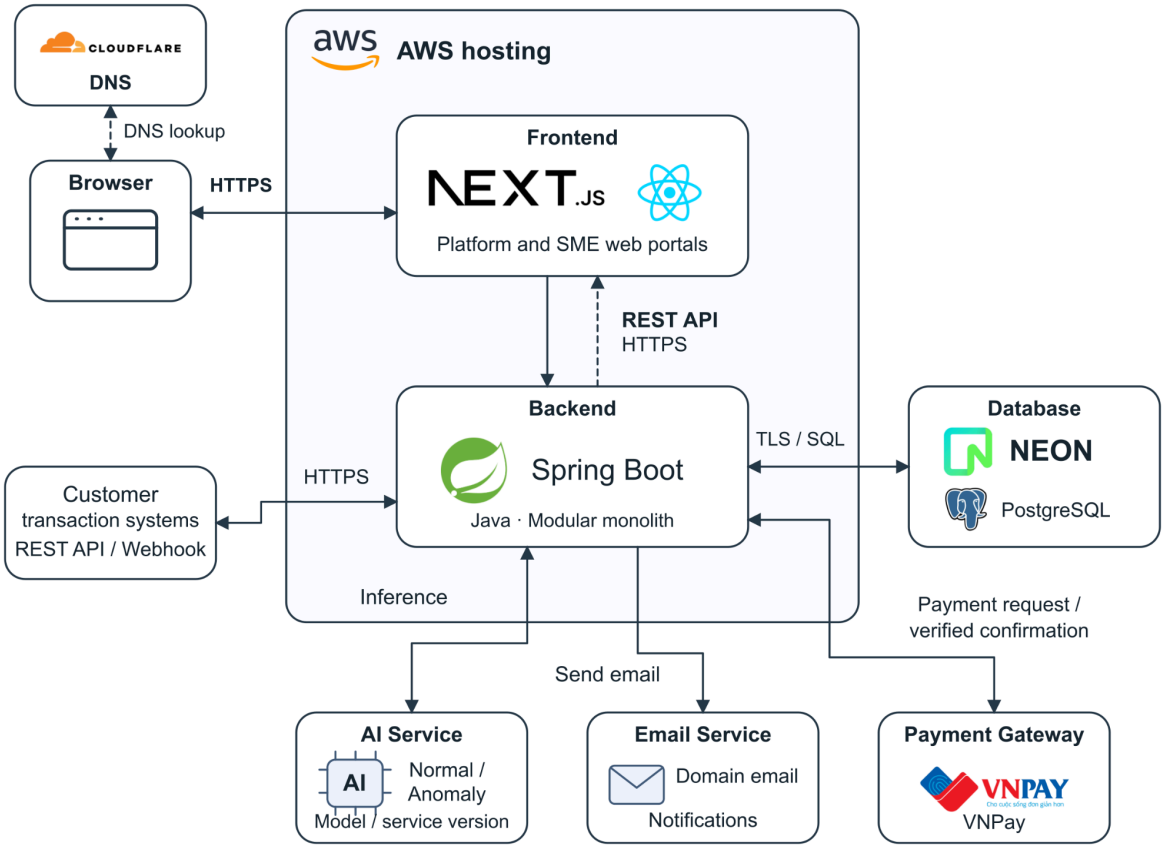


Figure 4.1. SME-FraudGuard System Architecture.

1.2 Package Diagram

The backend is organized as one Spring Boot monolith under com.fraudguard. Eleven business modules are accompanied by common infrastructure and external integration adapters. The packages below reflect the current codebase scaffold; business implementations are added in later development.

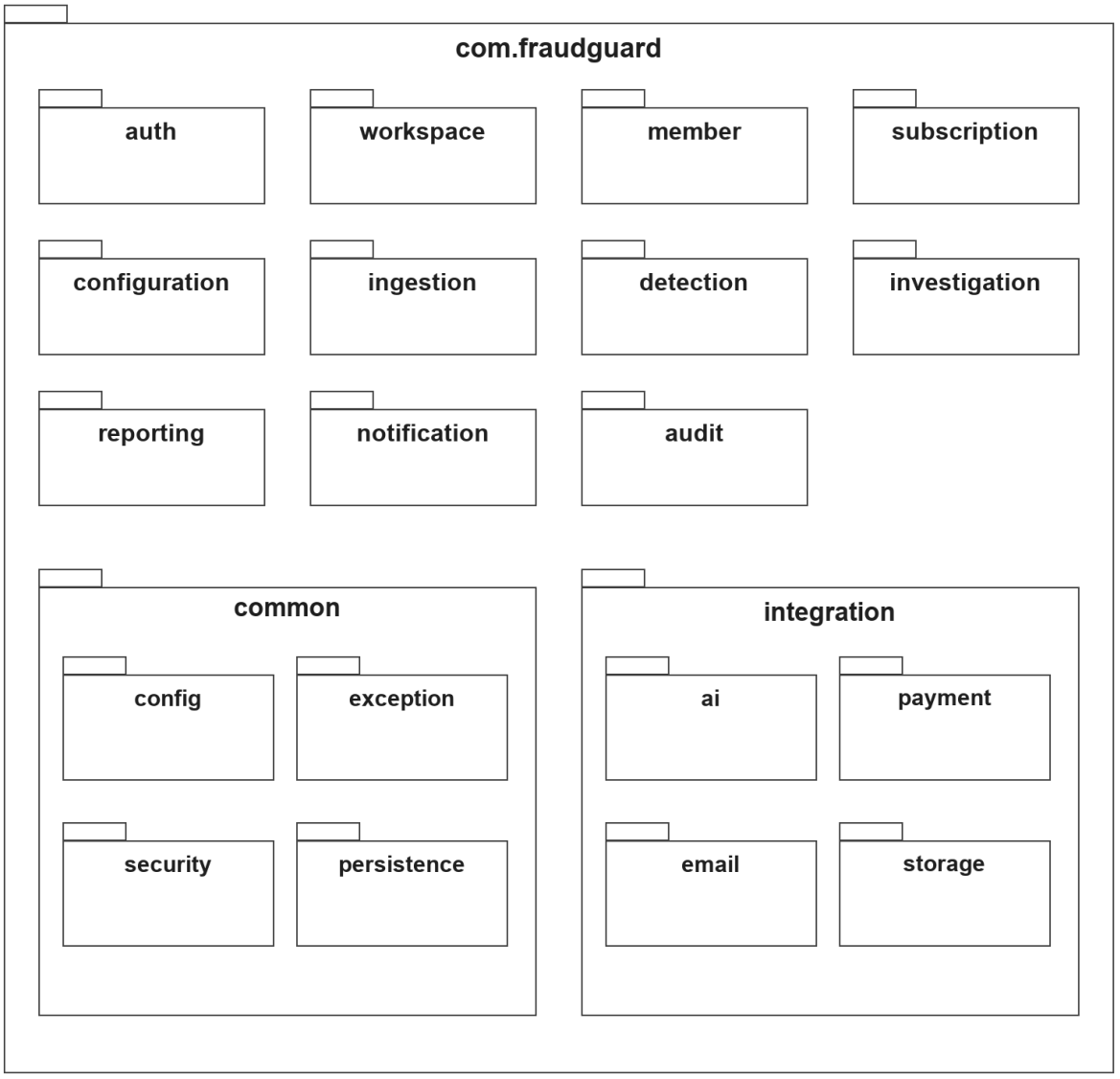


Figure 4.1.2.1. Backend Package Overview.

Each business module follows the layered structure below. Dashed arrows show permitted design dependencies, not implemented runtime calls. Other modules access service interfaces and DTOs; repositories, entities, mappers and service implementations remain private to their owning module. AI training remains outside this backend.

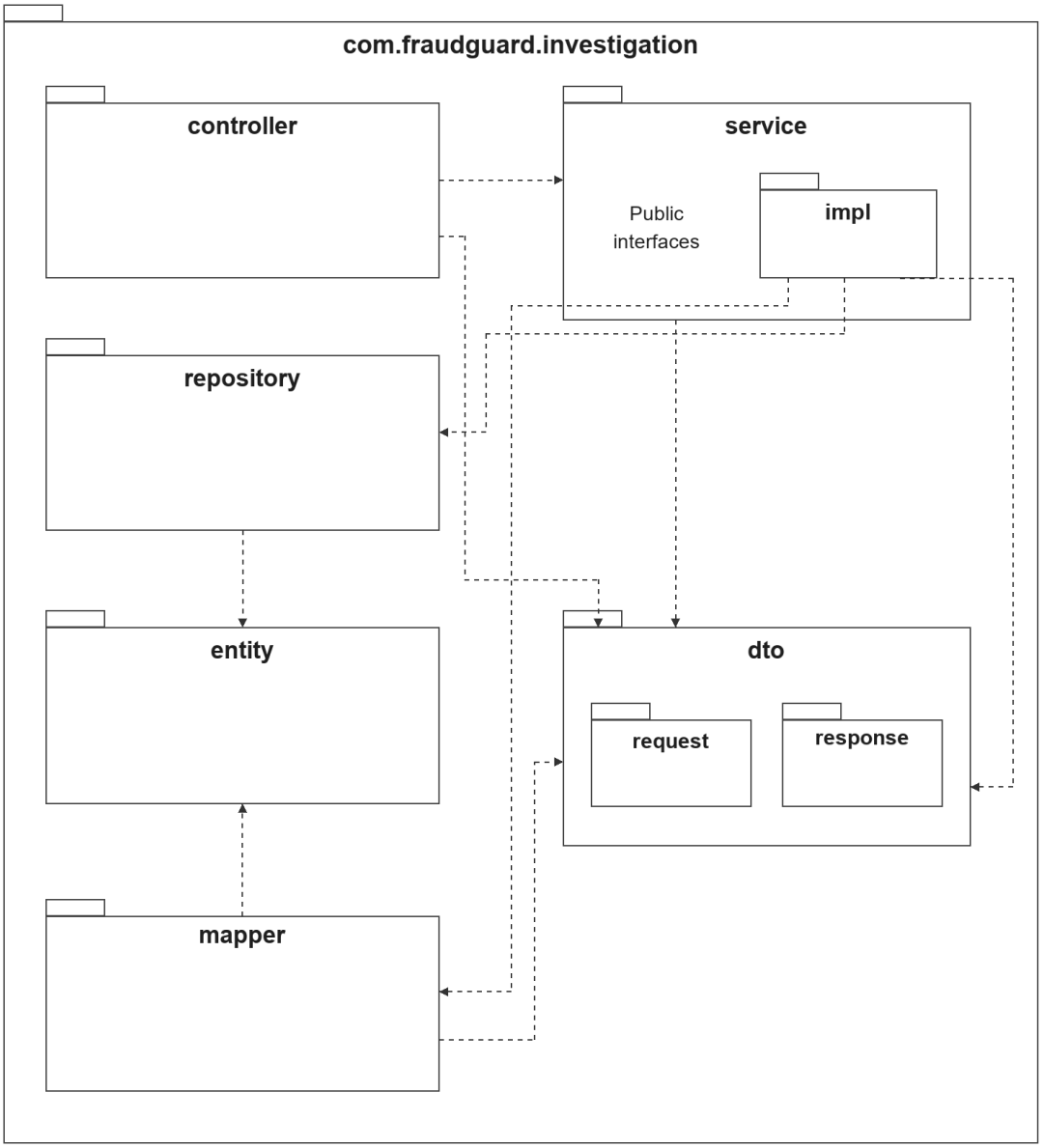


Figure 4.1.2.2. Layered Package Structure within a Business Module.

2. Database Design

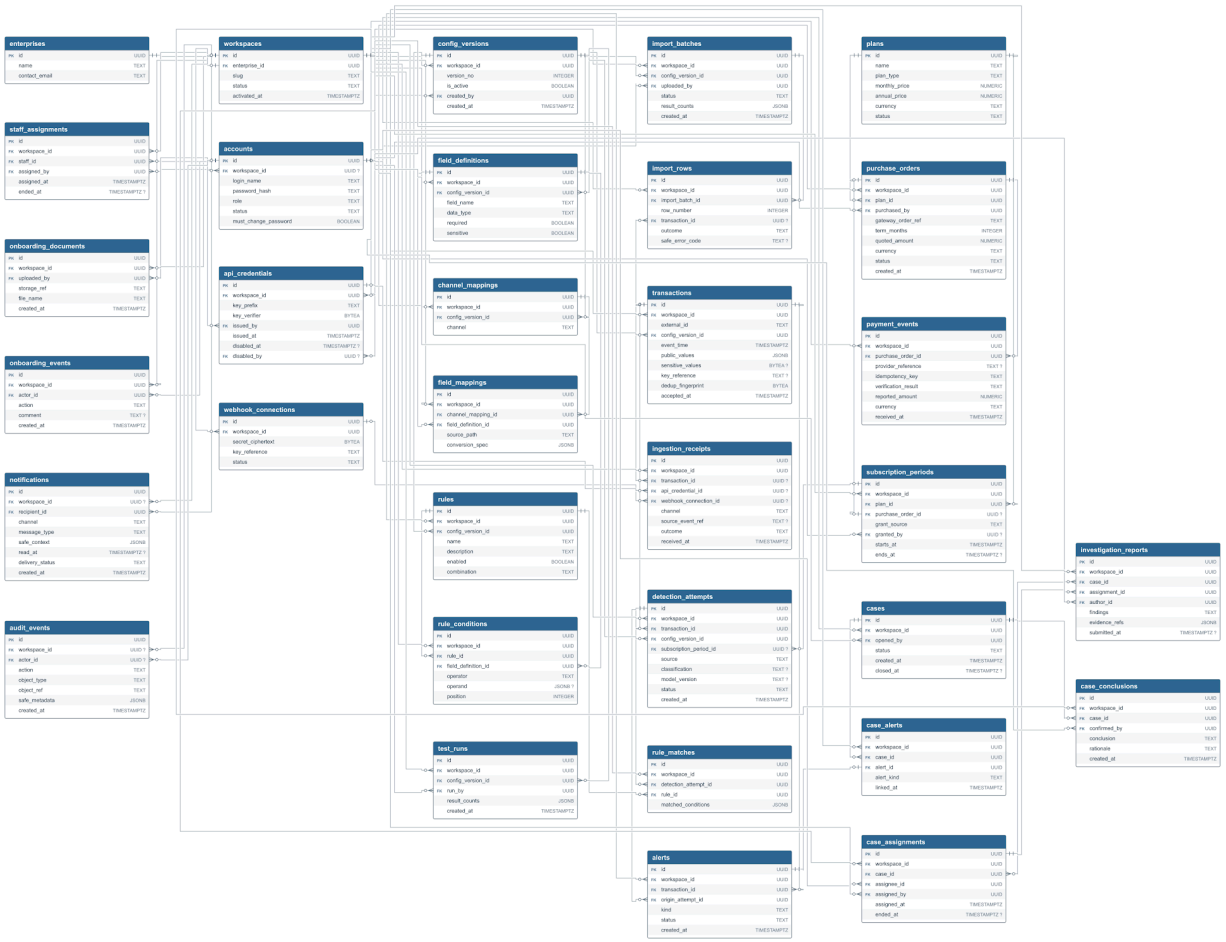


Figure 4.2. SME-FraudGuard Database Design.

3. Detailed Design

This section presents the target detailed design derived from Report 3 functional requirements, Sections 3.2-3.15. Class diagrams identify controllers/listeners, service interfaces and implementations, repositories, DTOs and domain types. Sequence diagrams use the same class names and operations, with explicit calls, returns, activation bars and repository/database interactions. These proposed design signatures are not claims of implemented Java classes or approved HTTP routes; cross-module calls use public APIs/ports within the modular monolith.

All operations enforce current identity, role, workspace and object/assignment scope on the backend. Integration authentication establishes workspace scope before mapping or acceptance. Sensitive values remain protected in ordinary results and diagnostics; required writes retain history atomically. The headings named State Diagram follow the supplied reference: their figures use processing activities, decision diamonds, guarded branches and start/end nodes. Exact external wire formats, upload limits, timeout values and retry schedules remain subject to their confirmed interface specifications.

3.1 Authentication and Account Access

SRS reference: 3.2.1-3.2.4. Sessions bind verified identity to platform or enterprise scope. Mandatory initial password change restricts access until saved; logout ends authorization without deleting business history.

3.1.1 Class Diagram

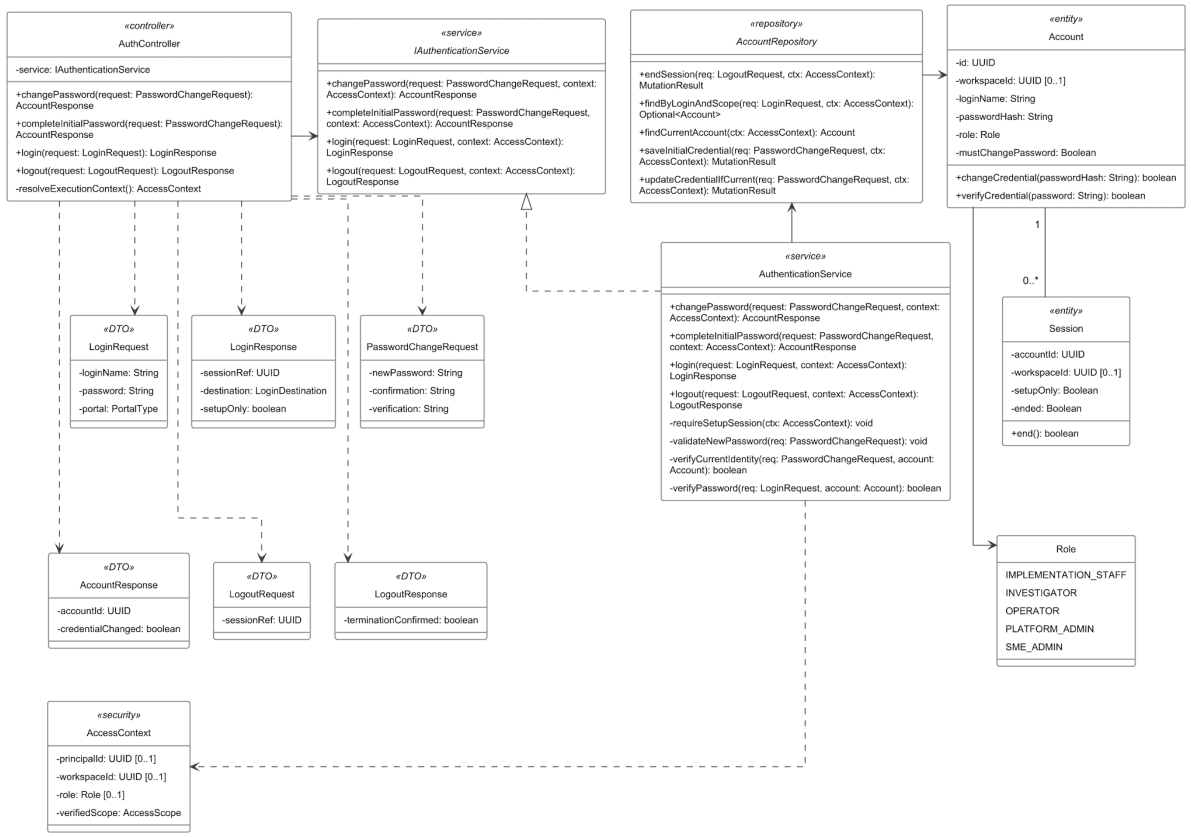


Figure 4.3.1.1.1. Authentication and Account Access Class Diagram.

3.1.2 Sequence Diagrams

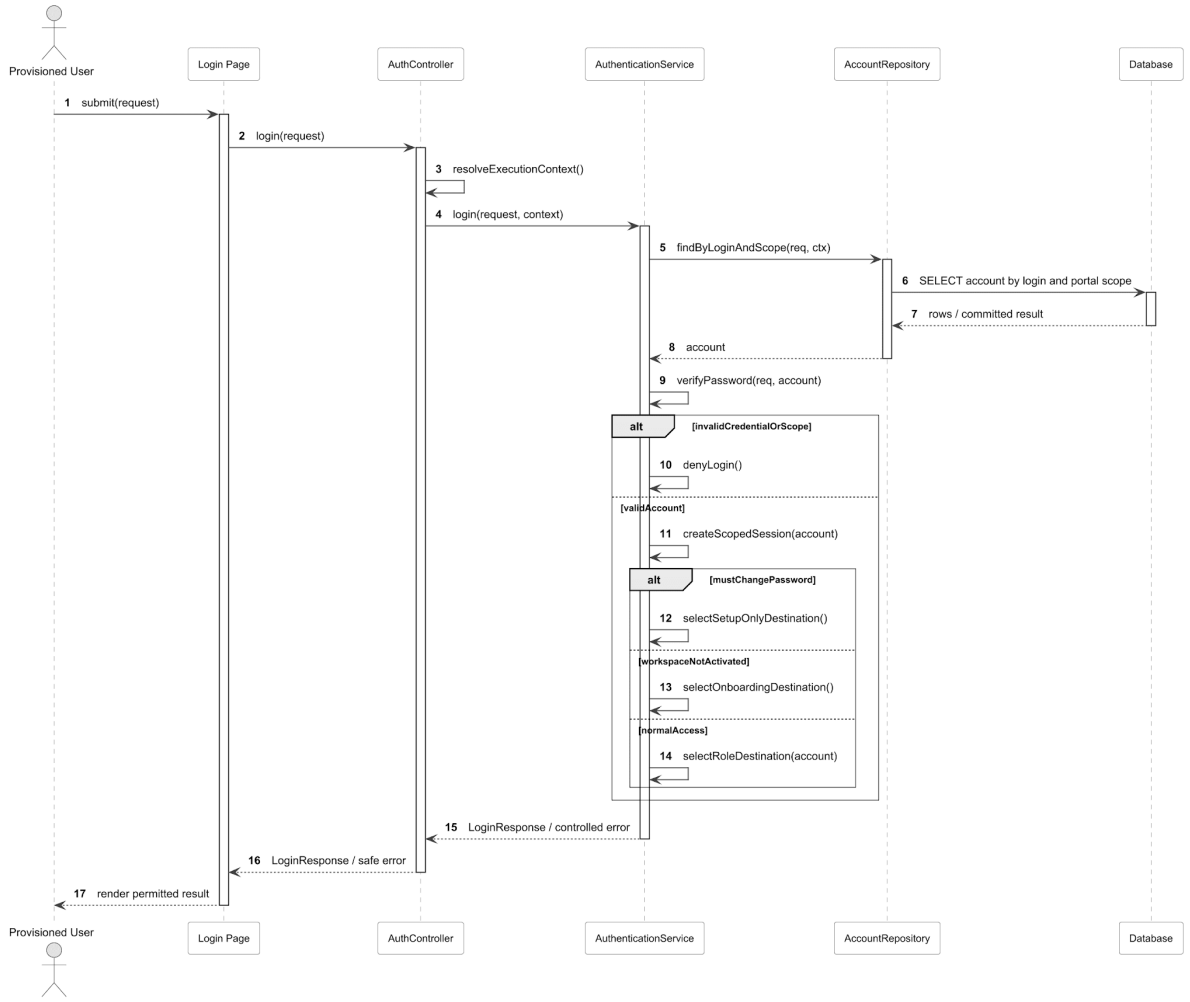


Figure 4.3.1.2.1. Log In and Select the Permitted Destination Sequence Diagram.

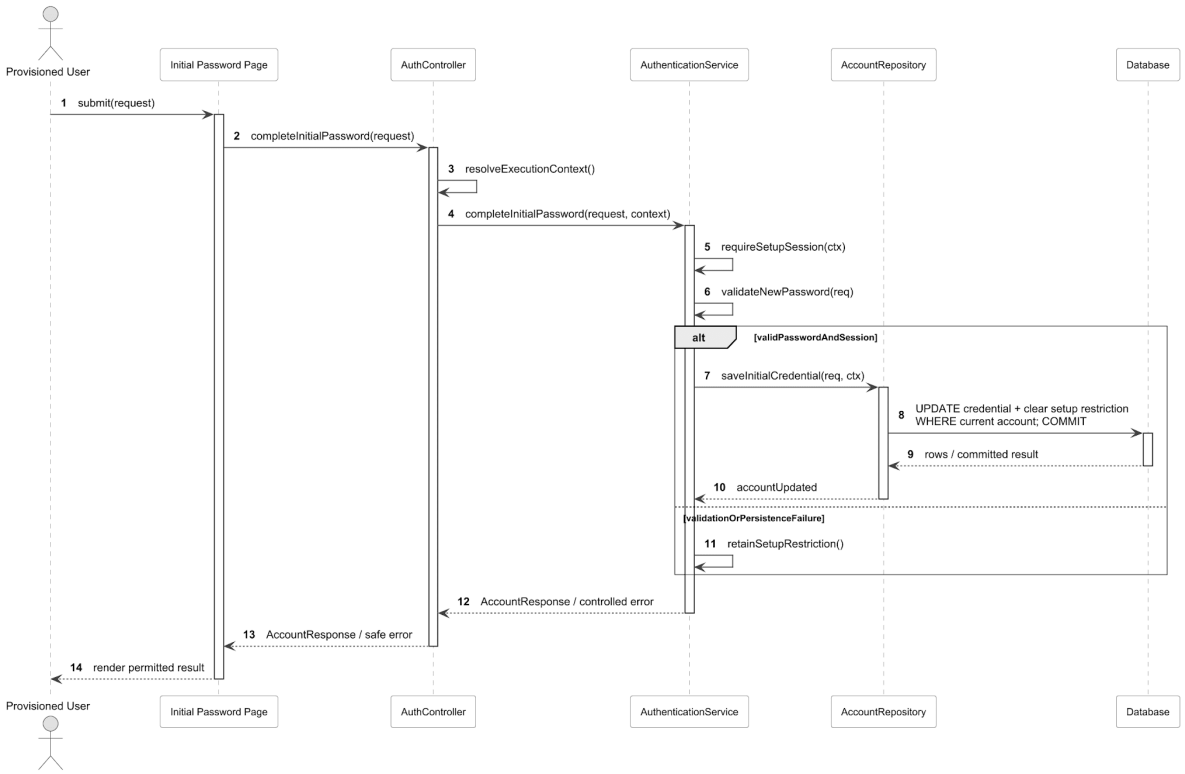


Figure 4.3.1.2.2. Complete Mandatory Initial Password Change Sequence Diagram.

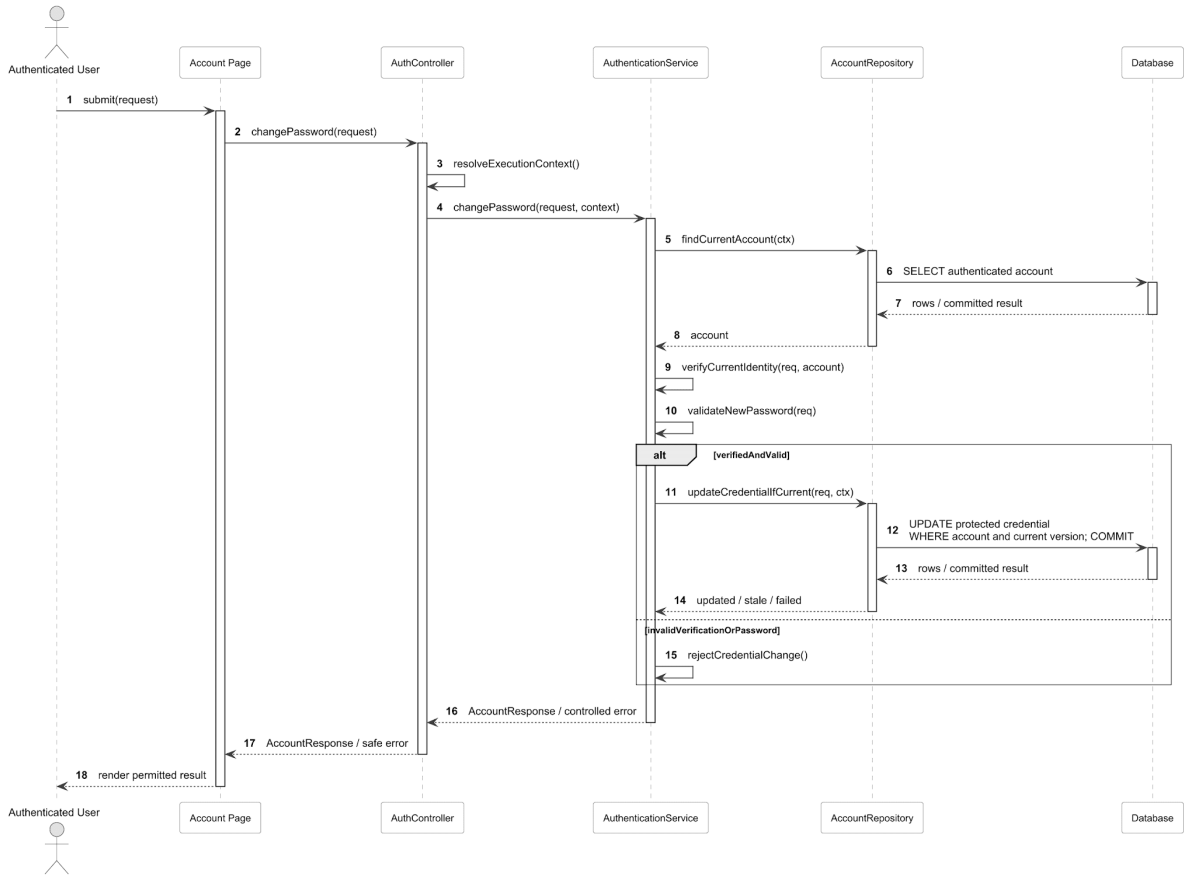


Figure 4.3.1.2.3. Change Current Password Sequence Diagram.

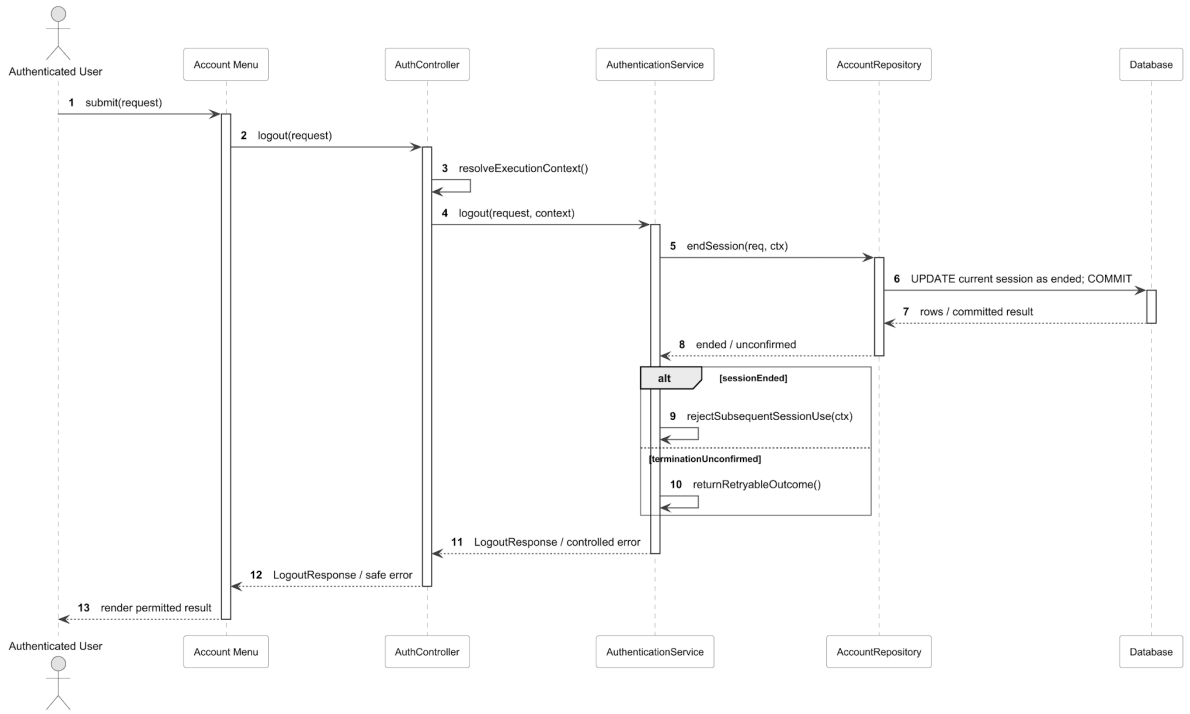


Figure 4.3.1.2.4. Log Out and Reject the Ended Session Sequence Diagram.

3.1.3 State Diagrams

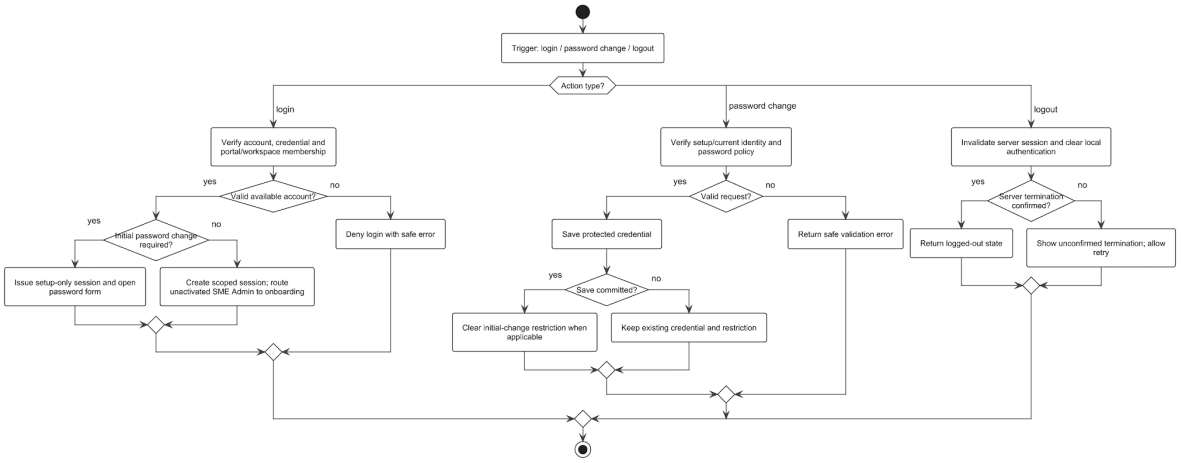


Figure 4.3.1.3.1. Authenticated Session Lifecycle State Diagram.

3.2 Customer and Workspace Management

SRS reference: 3.3.1-3.3.4. Platform Admin provisions one workspace per enterprise and its initial SME Admin. Staff accesses assigned setup metadata only; assignment changes never grant production-data access.

3.2.1 Class Diagram

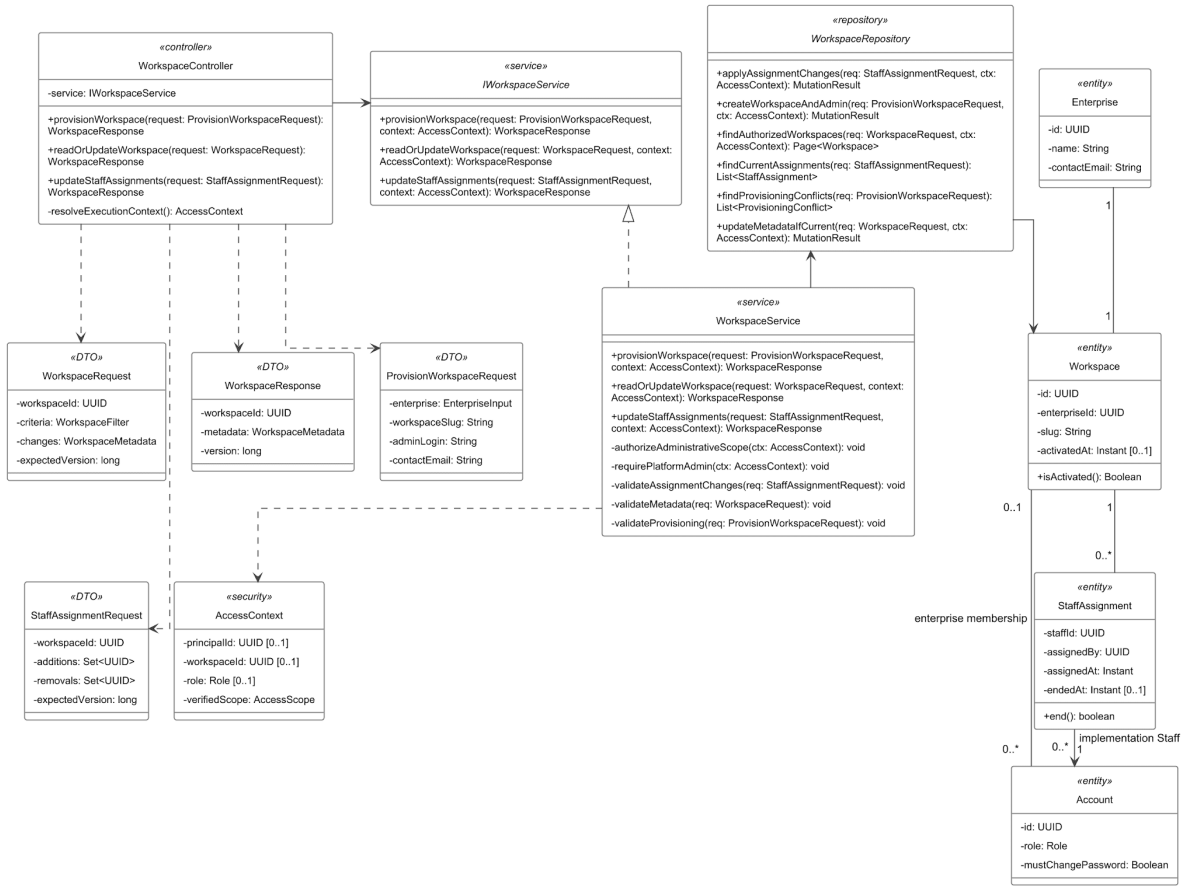


Figure 4.3.2.1.1. Customer and Workspace Management Class Diagram.

3.2.2 Sequence Diagrams

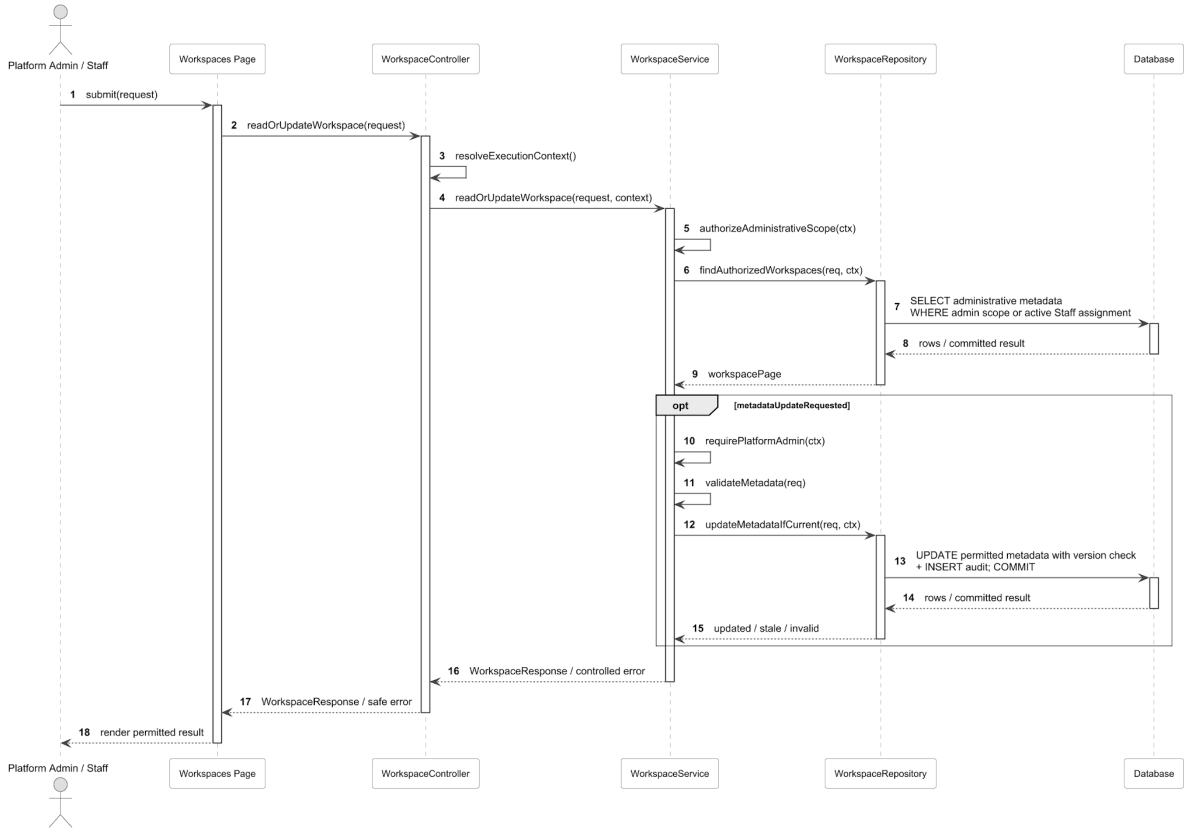


Figure 4.3.2.2.1. List, Inspect and Update Administrative Workspace Information Sequence Diagram.

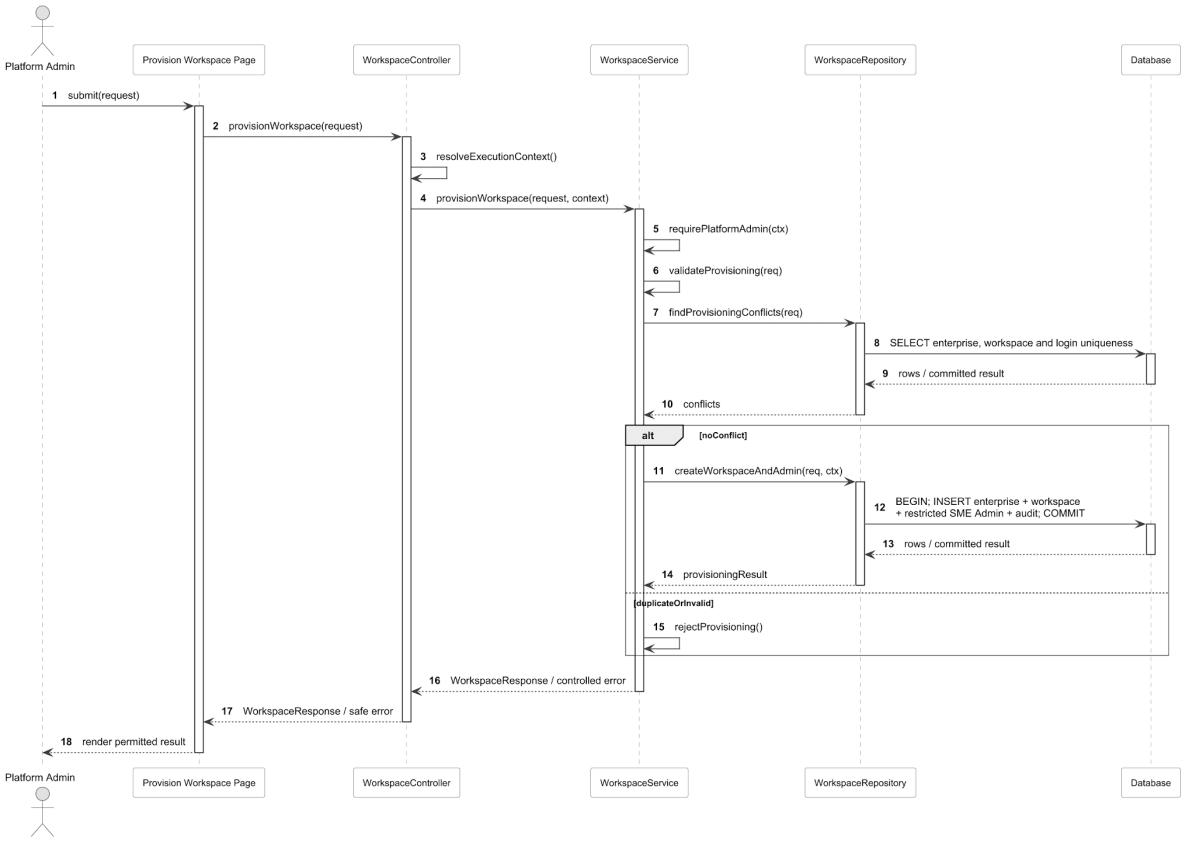


Figure 4.3.2.2.2. Provision Enterprise, Workspace and Initial SME Admin Sequence Diagram.

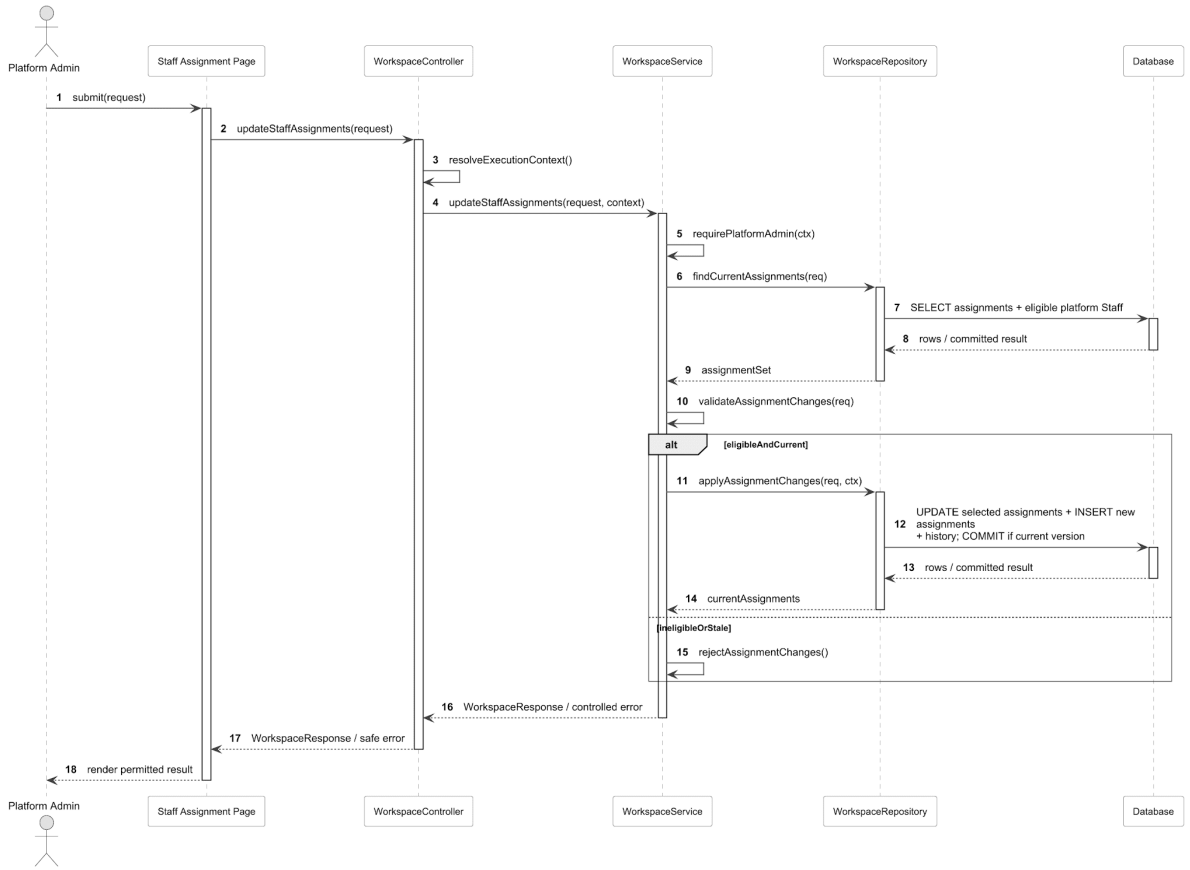


Figure 4.3.2.2.3. Add or Remove Concurrent Implementation Staff Assignments Sequence Diagram.

3.3 Plan and Subscription Management

SRS reference: 3.4.1-3.4.7. Free and Subscription have fixed entitlements. Purchases retain quotes and only verified server-side payment applies rights once. Calendar validity uses Asia/Ho_Chi_Minh, UTC storage and [start, expiry).

3.3.1 Class Diagram

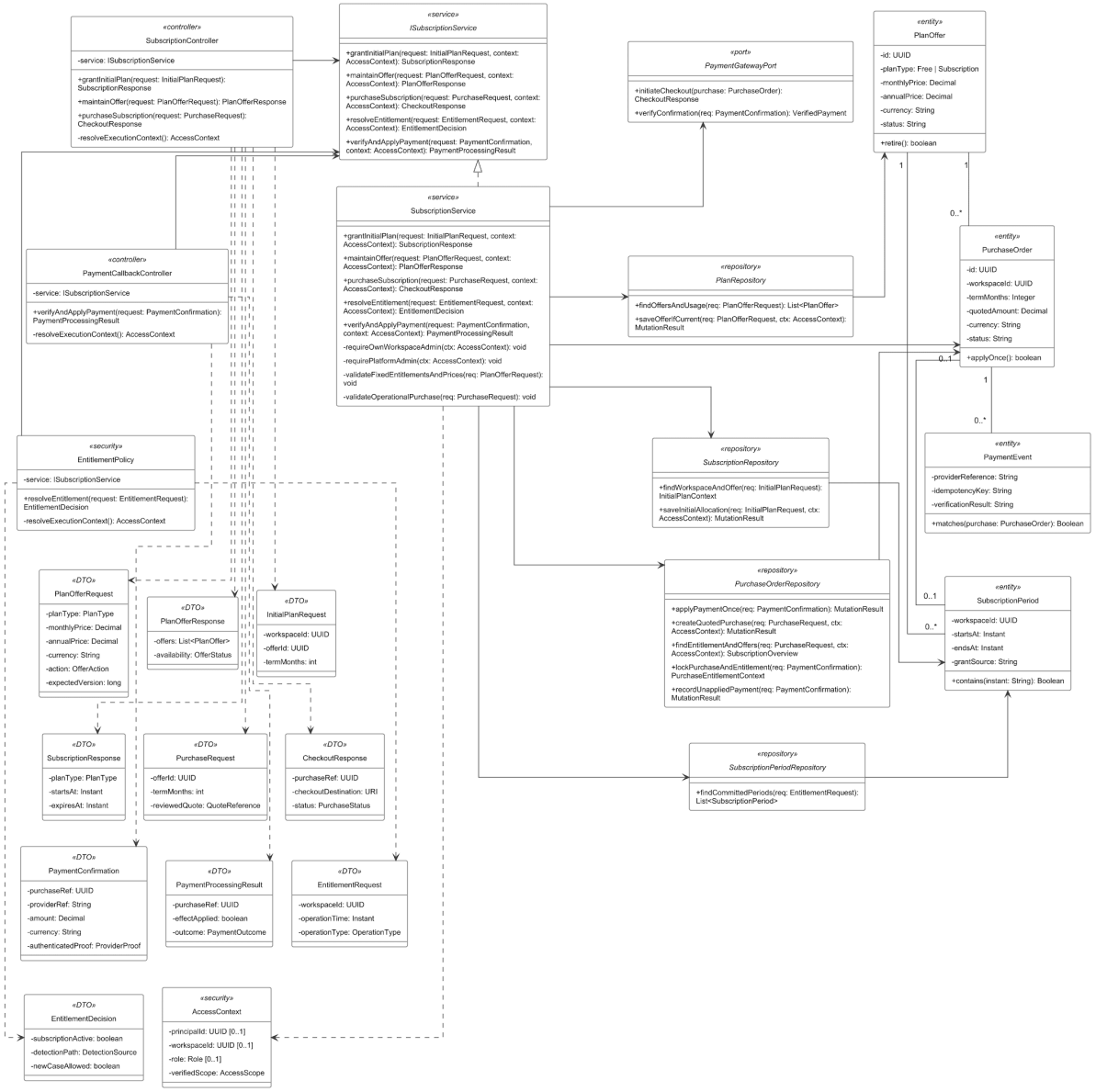


Figure 4.3.3.1.1. Plan and Subscription Management Class Diagram.

3.3.2 Sequence Diagrams

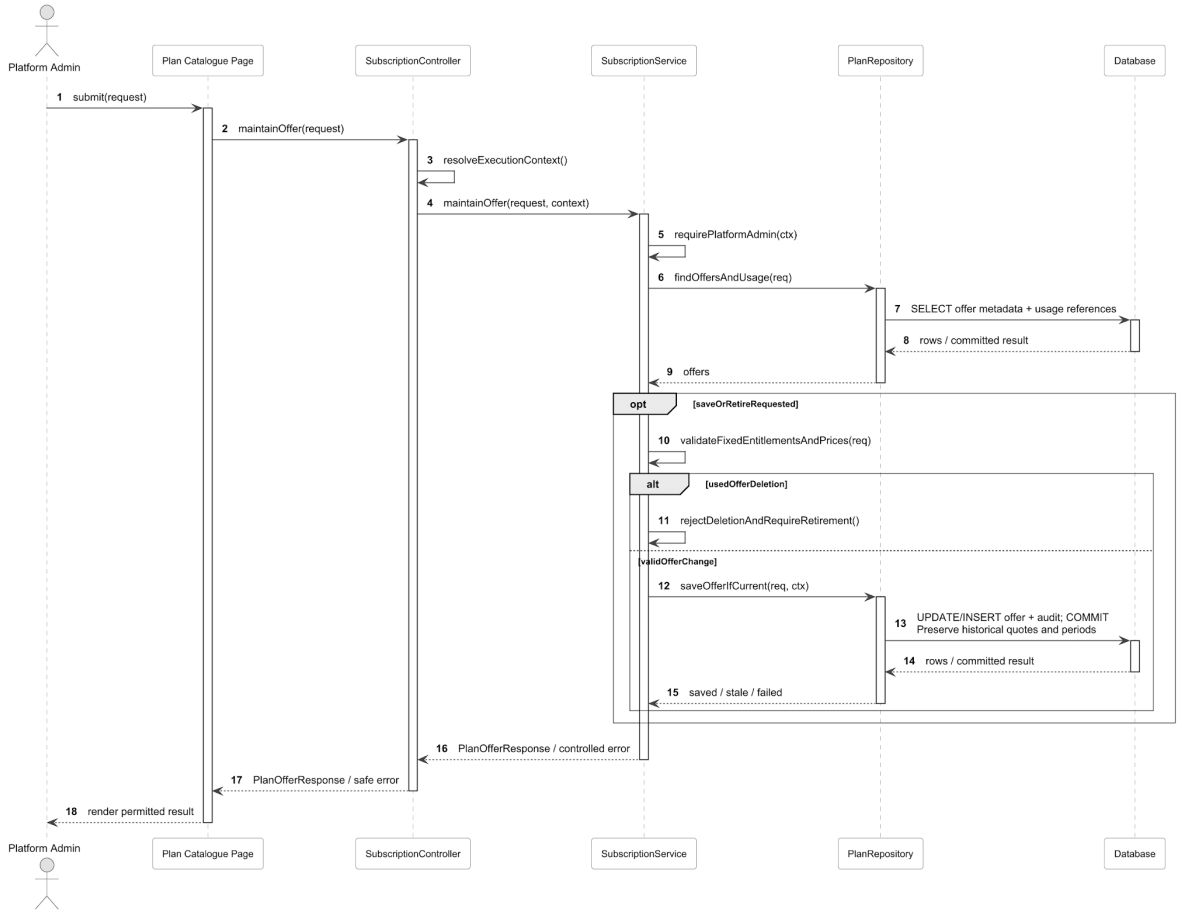


Figure 4.3.3.2.1. Review and Maintain Plan Offers Sequence Diagram.

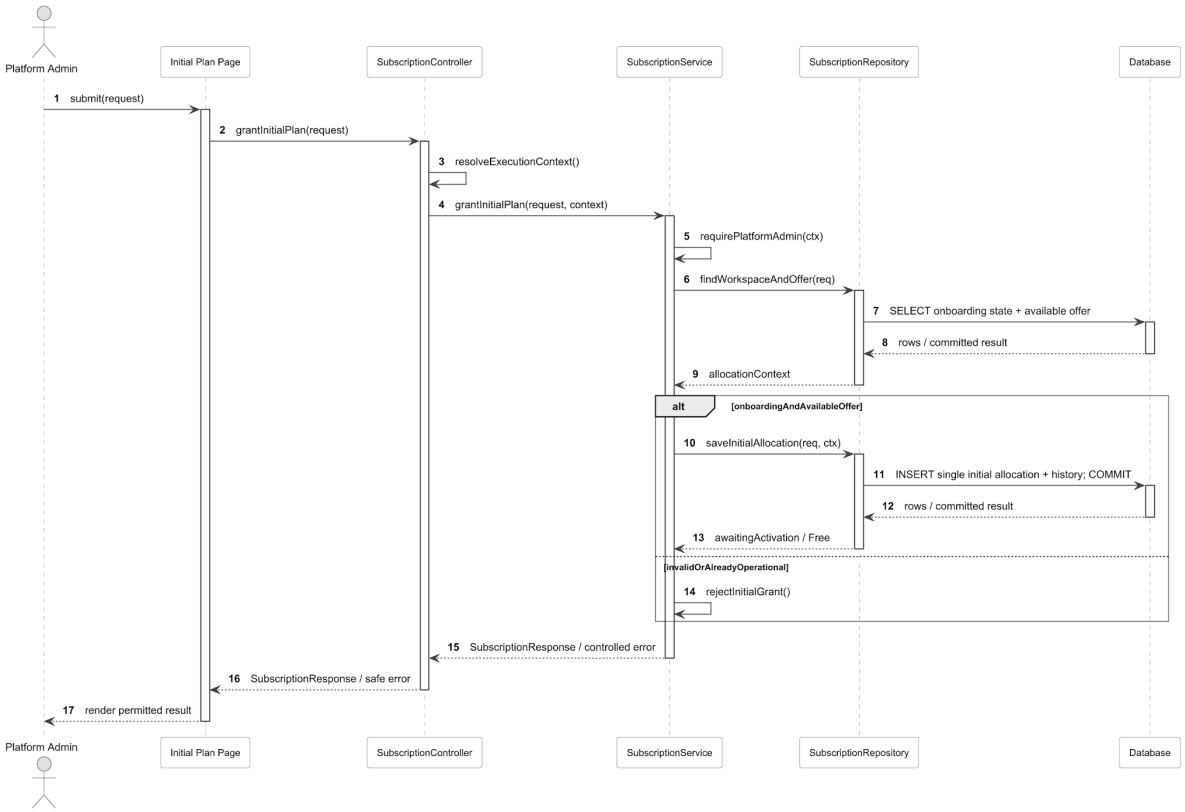


Figure 4.3.3.2.2. Grant the Initial Workspace Plan Sequence Diagram.

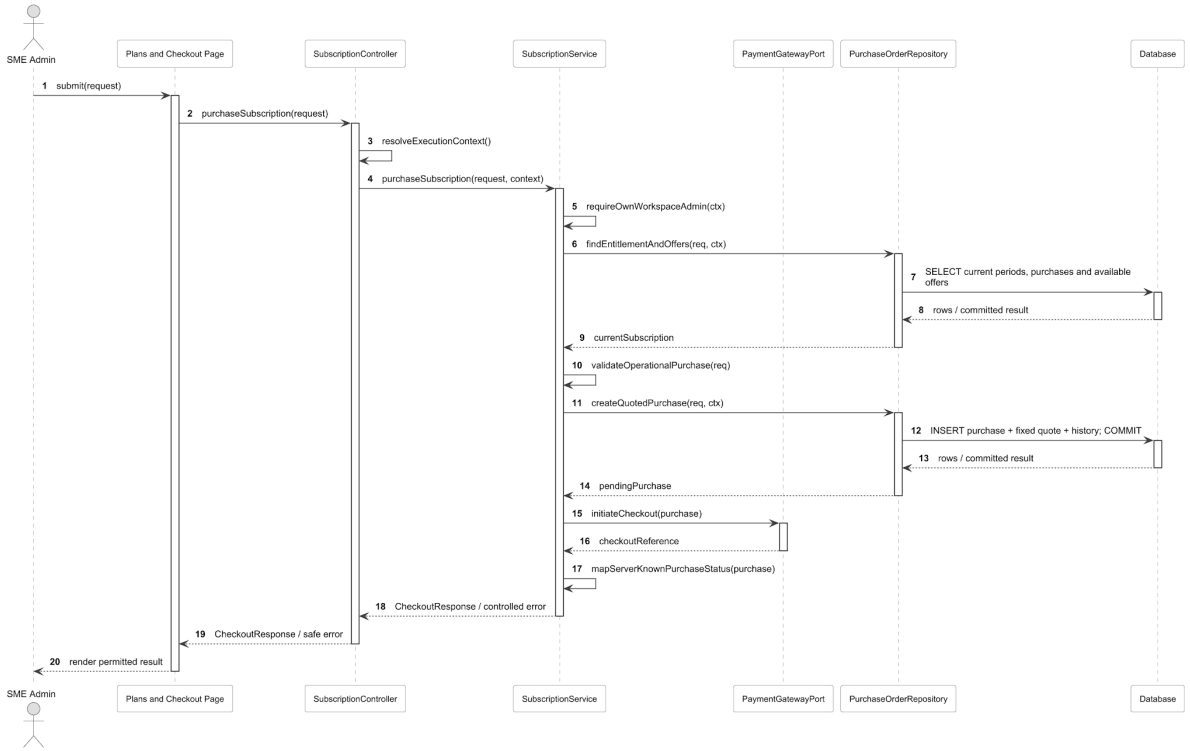


Figure 4.3.3.2.3. View Subscription and Initiate Purchase or Renewal Sequence Diagram.

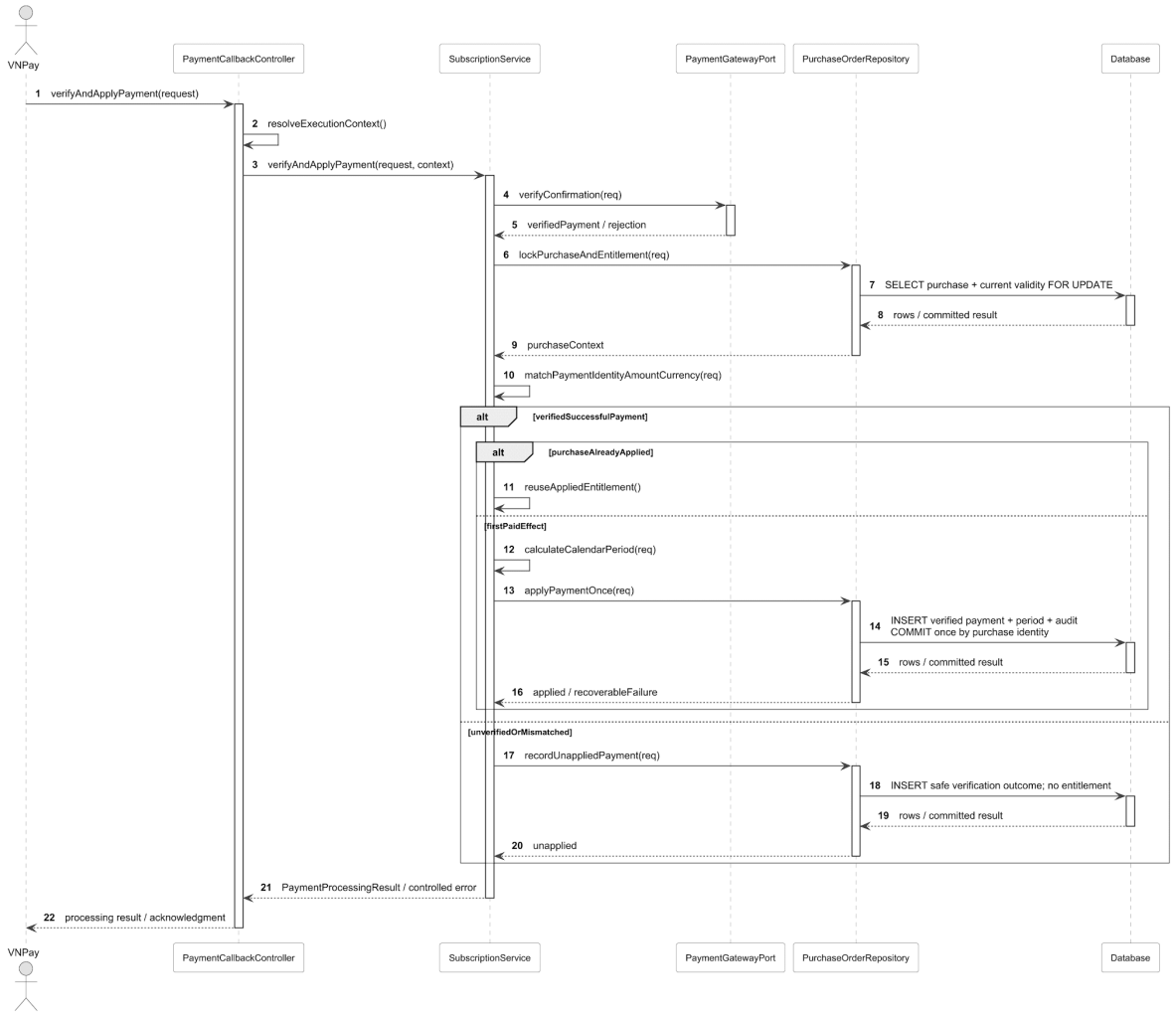


Figure 4.3.3.2.4. Verify Payment and Apply One Calendar Entitlement Effect Sequence Diagram.

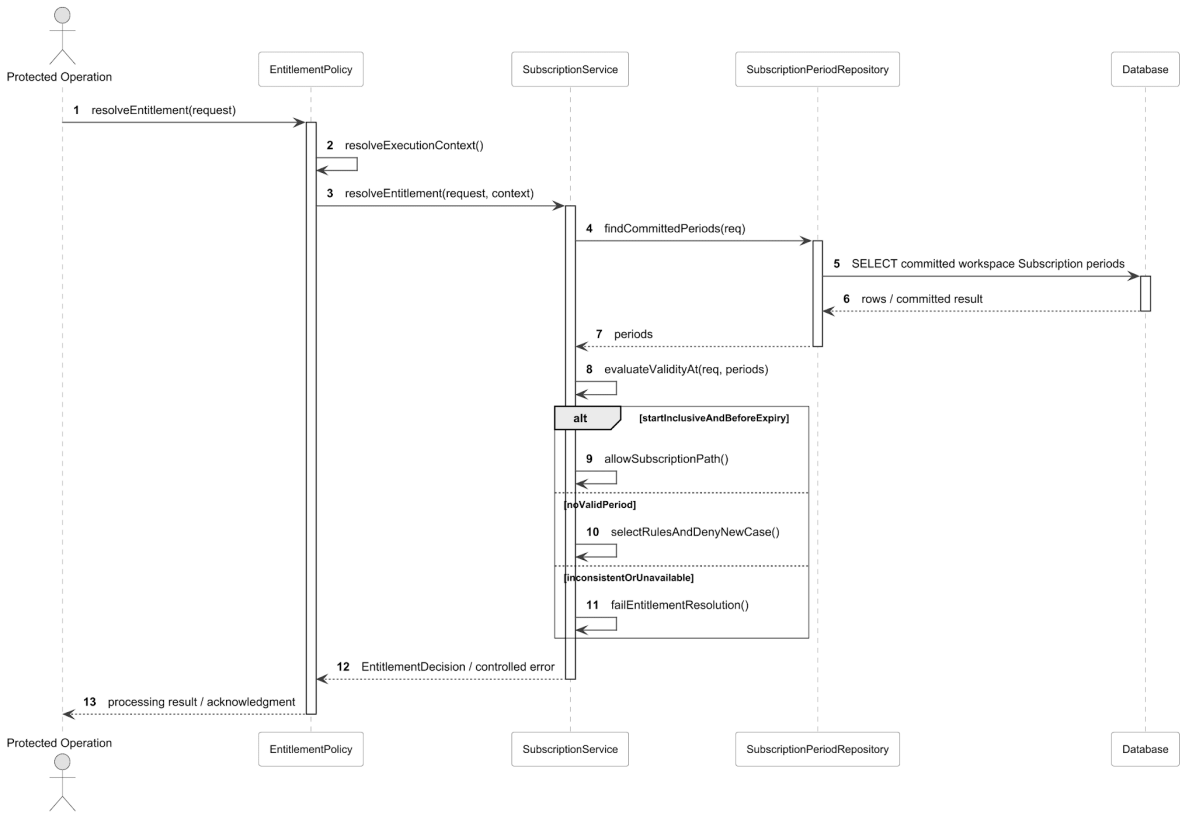


Figure 4.3.3.2.5. Enforce Expiry at New Operations and Preserve Existing Work Sequence Diagram.

3.3.3 State Diagrams

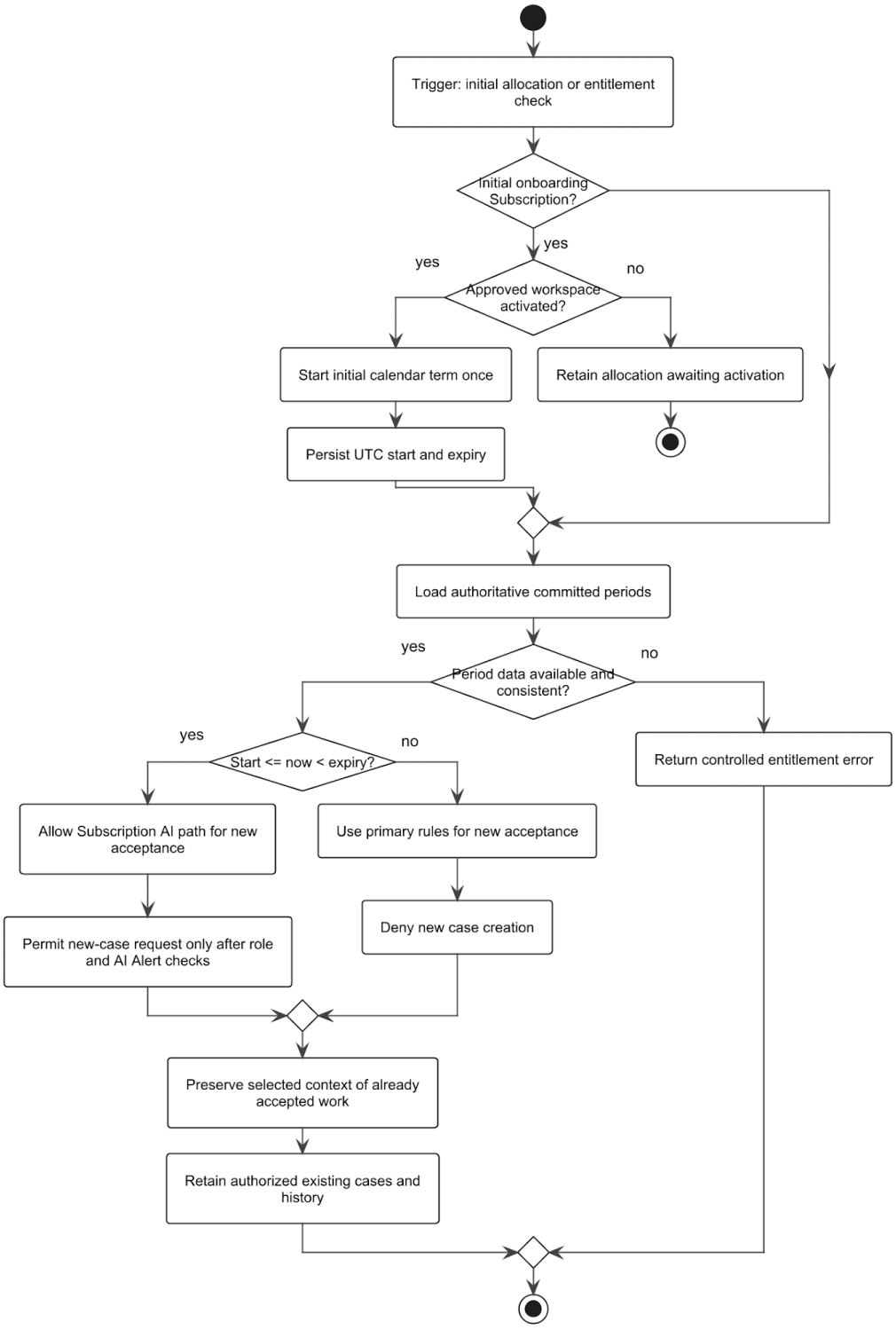


Figure 4.3.3.3.1. Subscription Entitlement Lifecycle State Diagram.

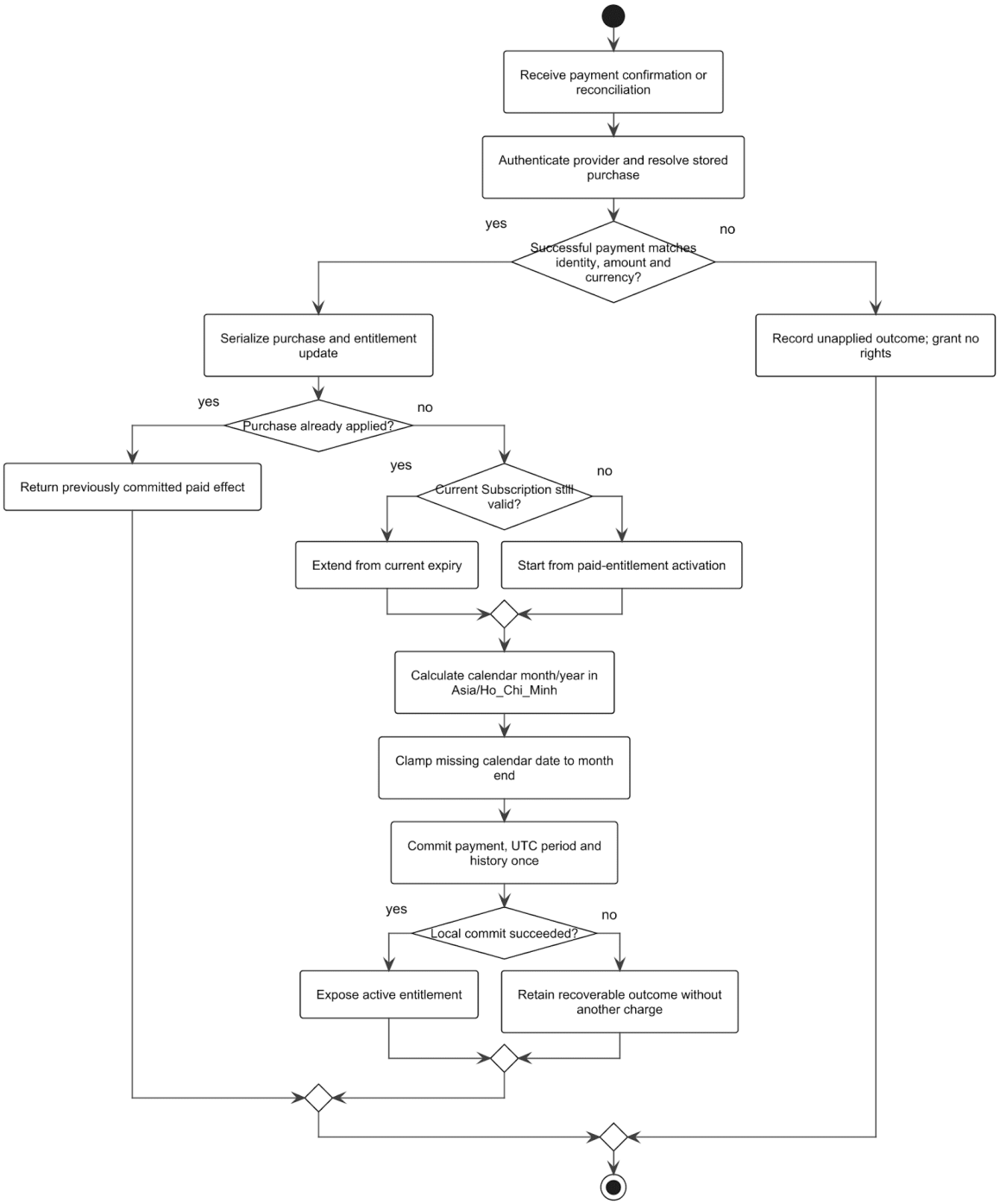


Figure 4.3.3.3.2. Purchase Verification and Local Entitlement Application State Diagram.

3.4 Workspace Onboarding and Activation

SRS reference: 3.5.1-3.5.5. Specifications are retained for human configuration work. Staff readiness, Platform Admin handover approval and SME Admin activation are distinct recorded gates; initial Subscription starts only at activation.

3.4.1 Class Diagram

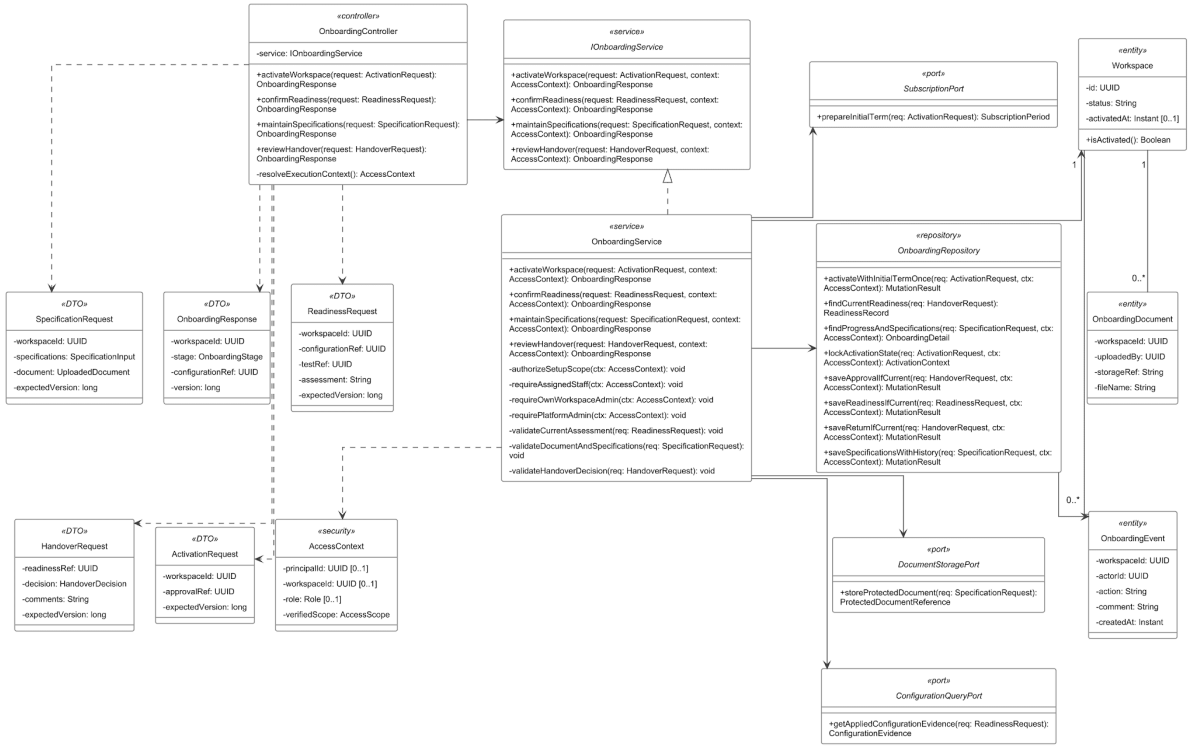


Figure 4.3.4.1.1. Workspace Onboarding and Activation Class Diagram.

3.4.2 Sequence Diagrams

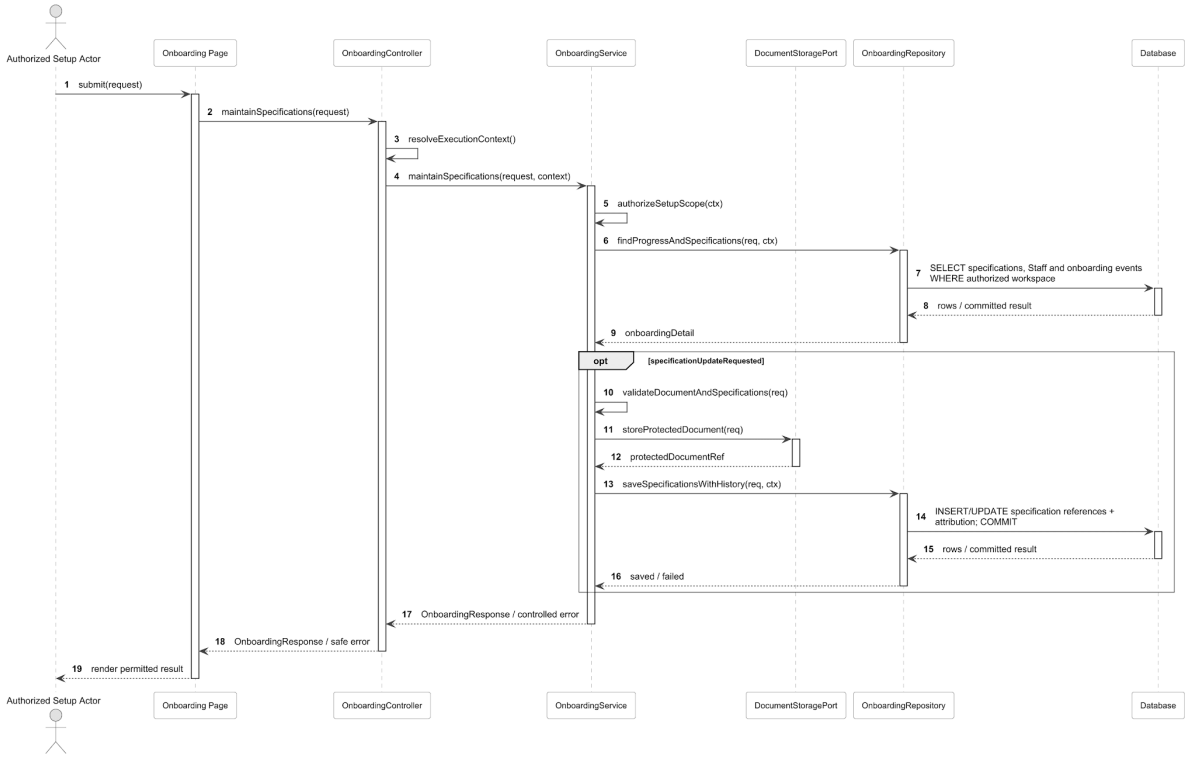


Figure 4.3.4.2.1. Maintain Specifications and Track Onboarding Progress Sequence Diagram.

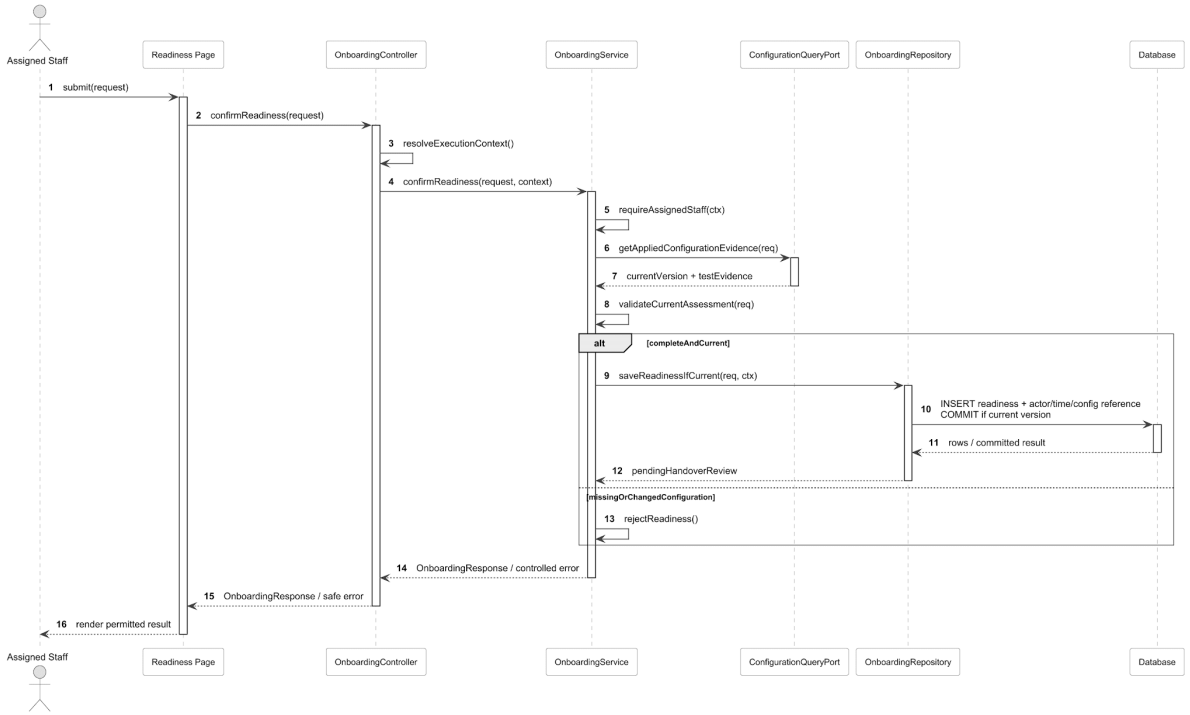


Figure 4.3.4.2.2. Confirm Implementation Readiness Sequence Diagram.

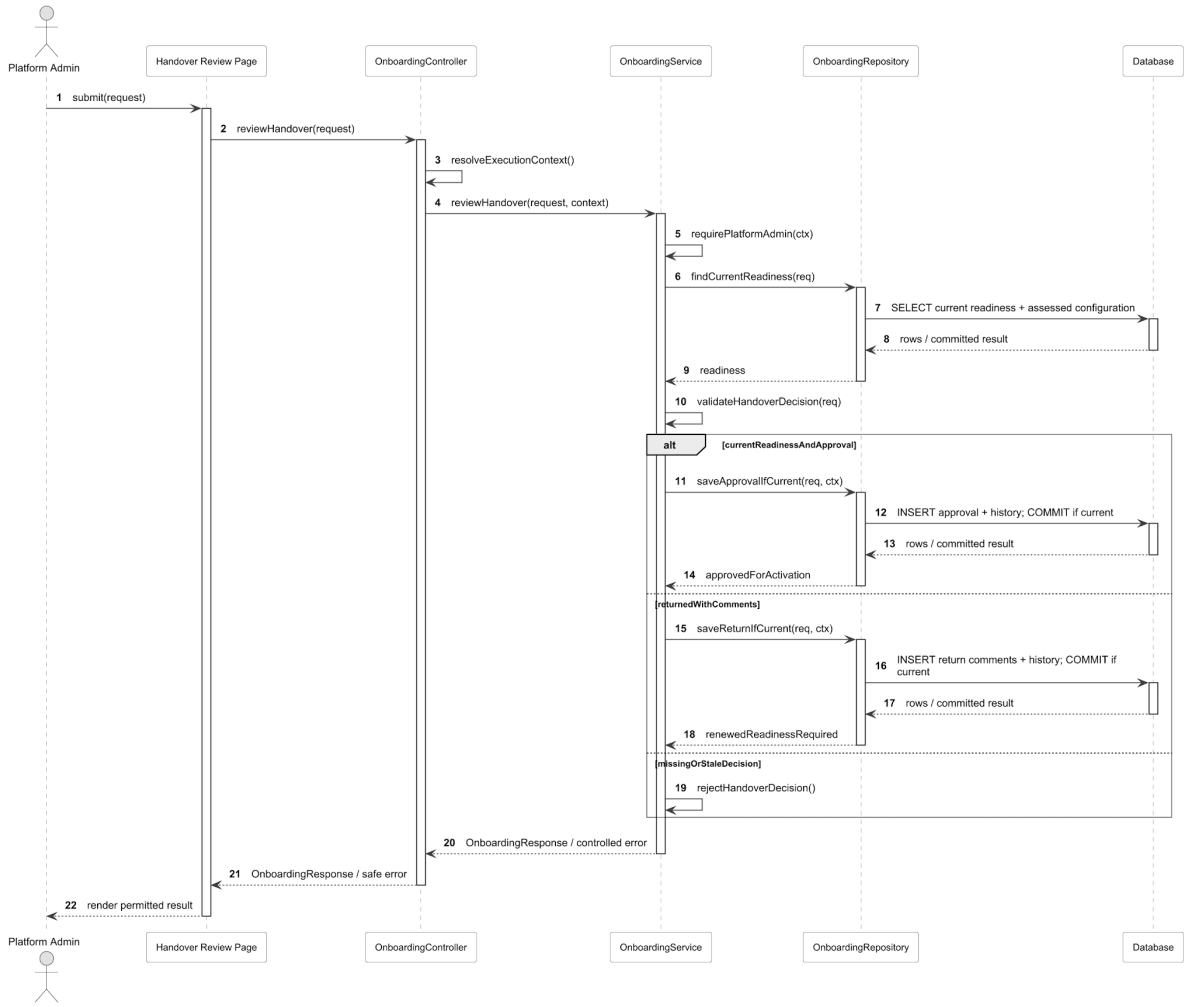


Figure 4.3.4.2.3. Approve or Return Workspace Handover Sequence Diagram.

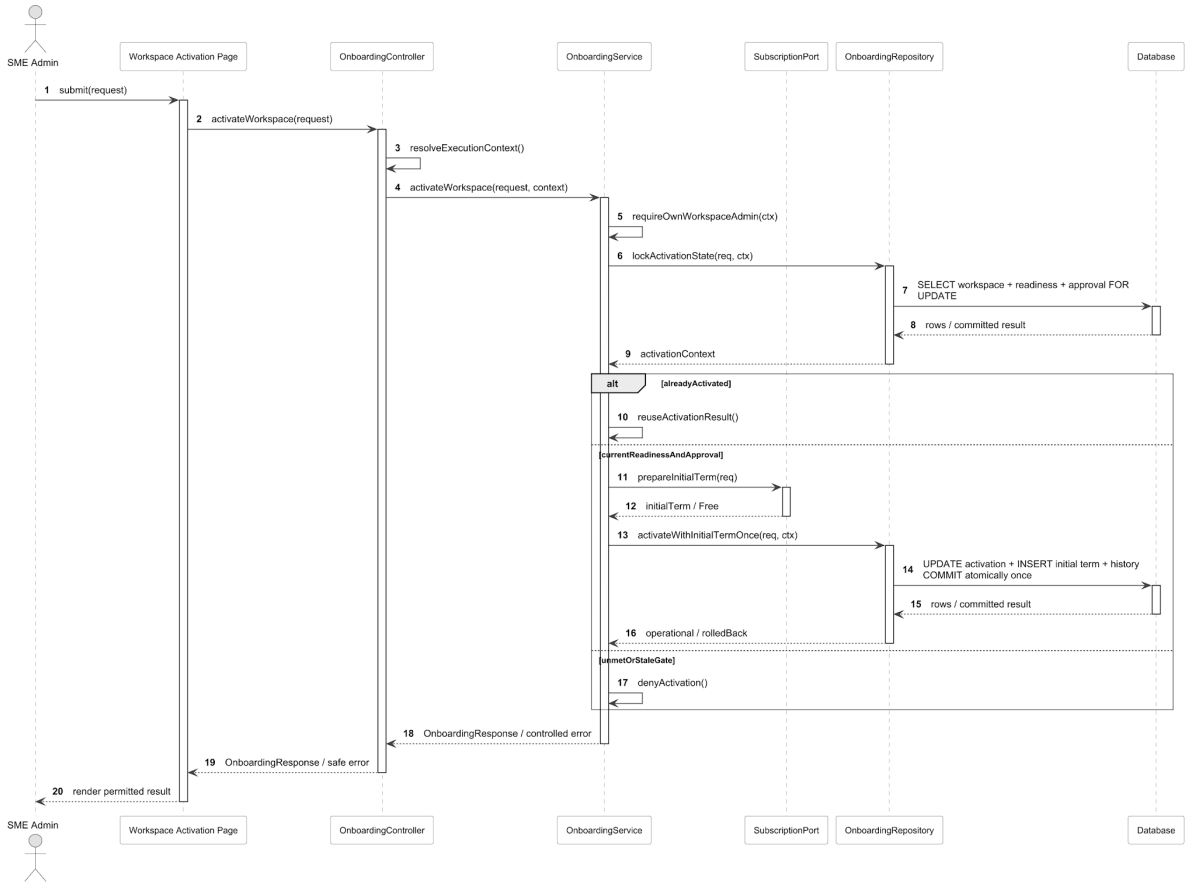


Figure 4.3.4.2.4. Activate Approved Workspace and Start the Initial Term Once Sequence Diagram.

3.4.3 State Diagrams

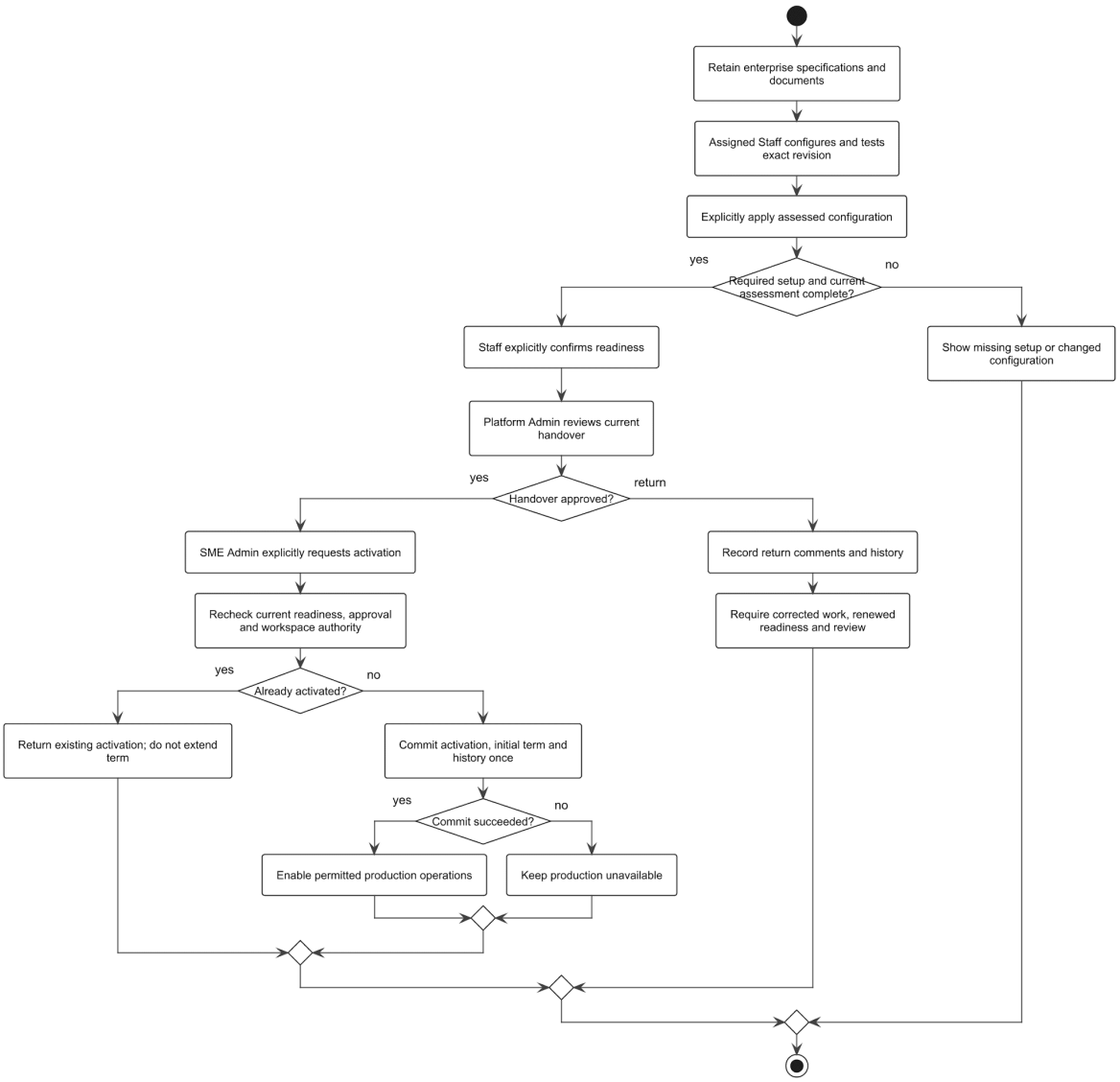


Figure 4.3.4.3.1. Workspace Readiness, Handover and Activation Lifecycle State Diagram.

3.5 Enterprise Member and Permission Management

SRS reference: 3.6.1-3.6.4. SME Admin manages only own-workspace enterprise accounts and explicit field-view grants. Permission administration does not grant SME Admin plaintext access; history survives role/access changes.

3.5.1 Class Diagram

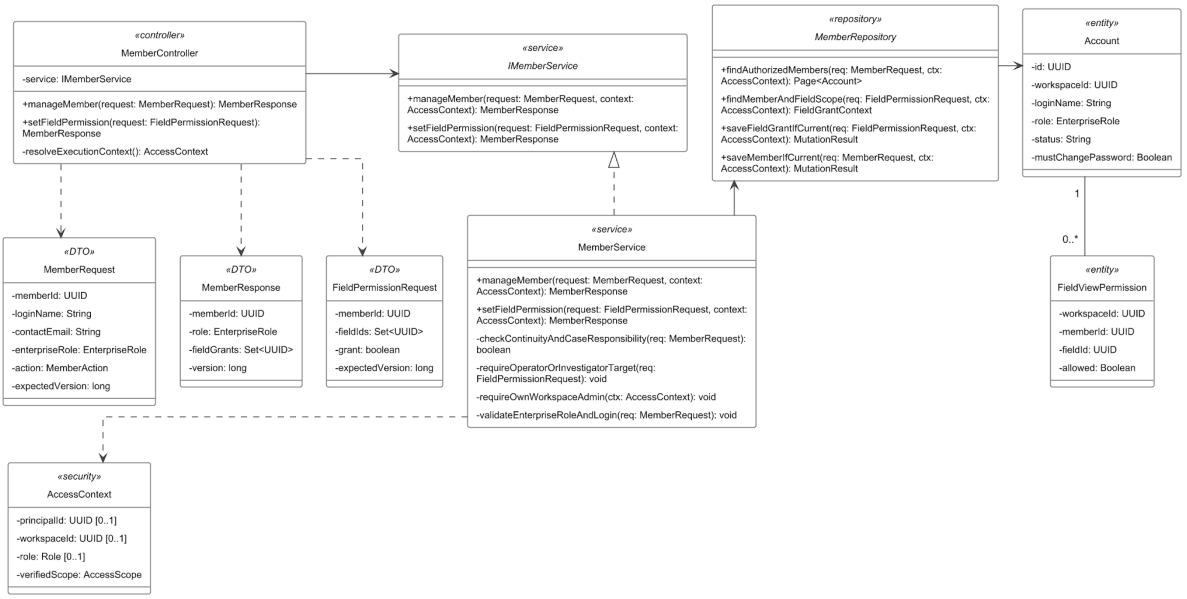


Figure 4.3.5.1.1. Enterprise Member and Permission Management Class Diagram.

3.5.2 Sequence Diagrams

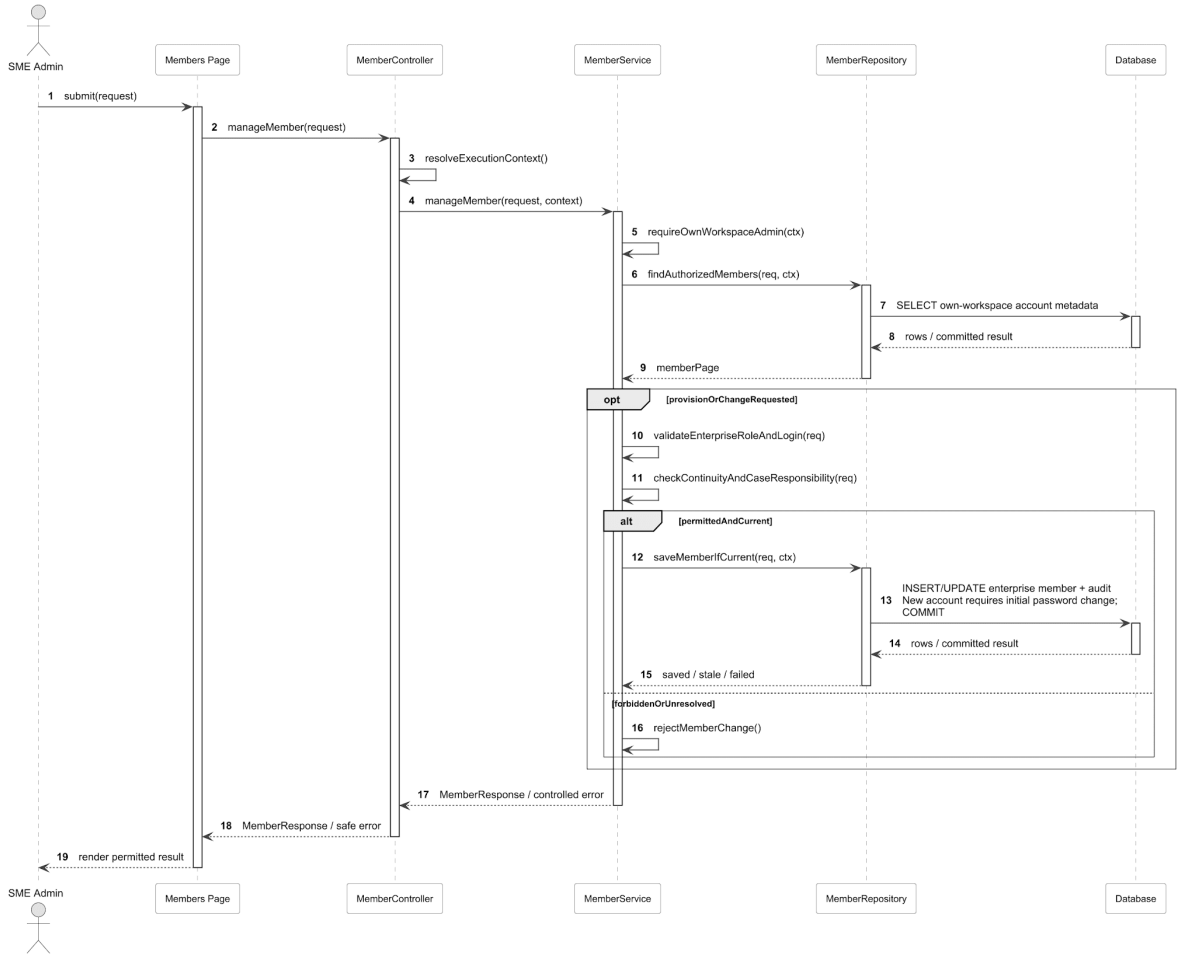


Figure 4.3.5.2.1. List, Provision and Update Enterprise Members Sequence Diagram.

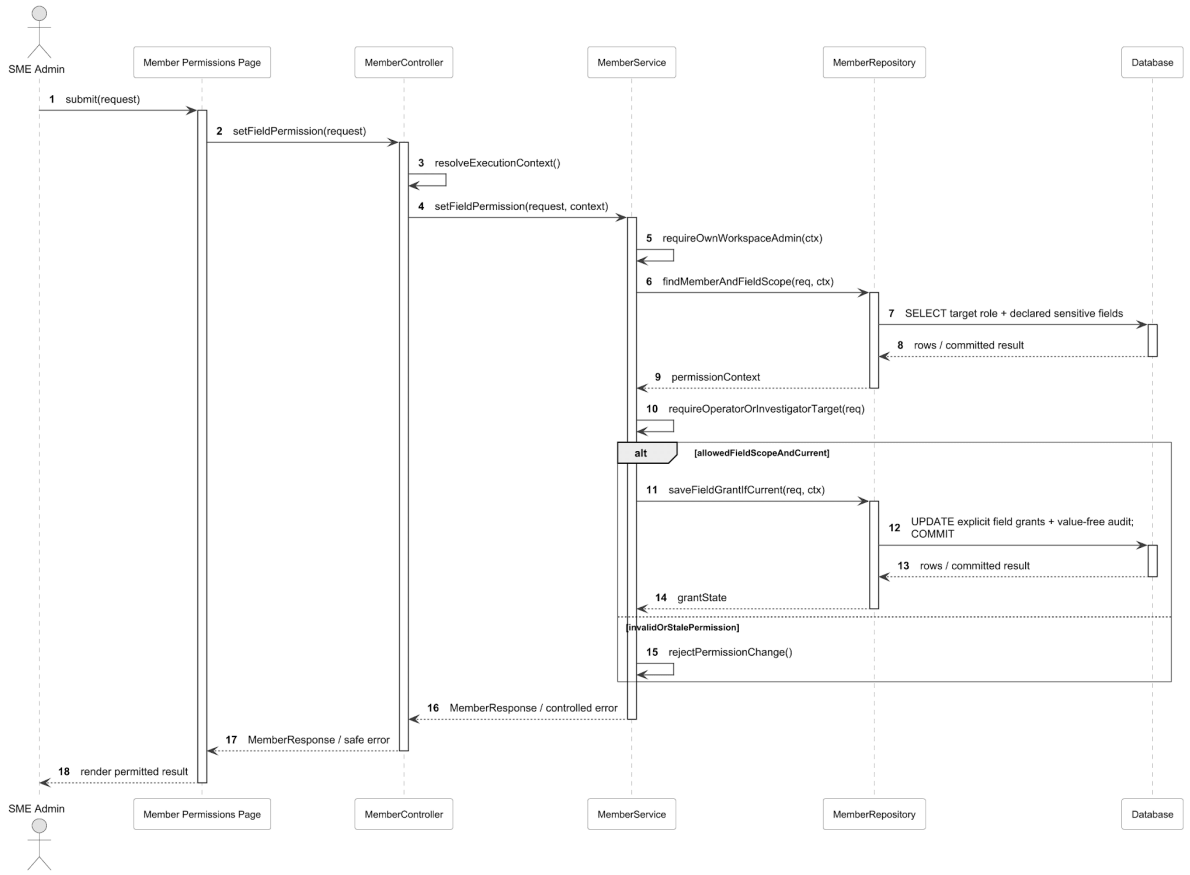


Figure 4.3.5.2.2. Grant or Revoke Explicit Sensitive-Field Viewing Permission Sequence Diagram.

3.6 Detection Configuration Management

SRS reference: 3.7.1-3.7.8. Assigned Staff edits a versioned candidate containing fields, channel mappings and rules. Testing uses the exact revision and temporary samples; explicit application atomically changes one active version for newly accepted work.

3.6.1 Class Diagram

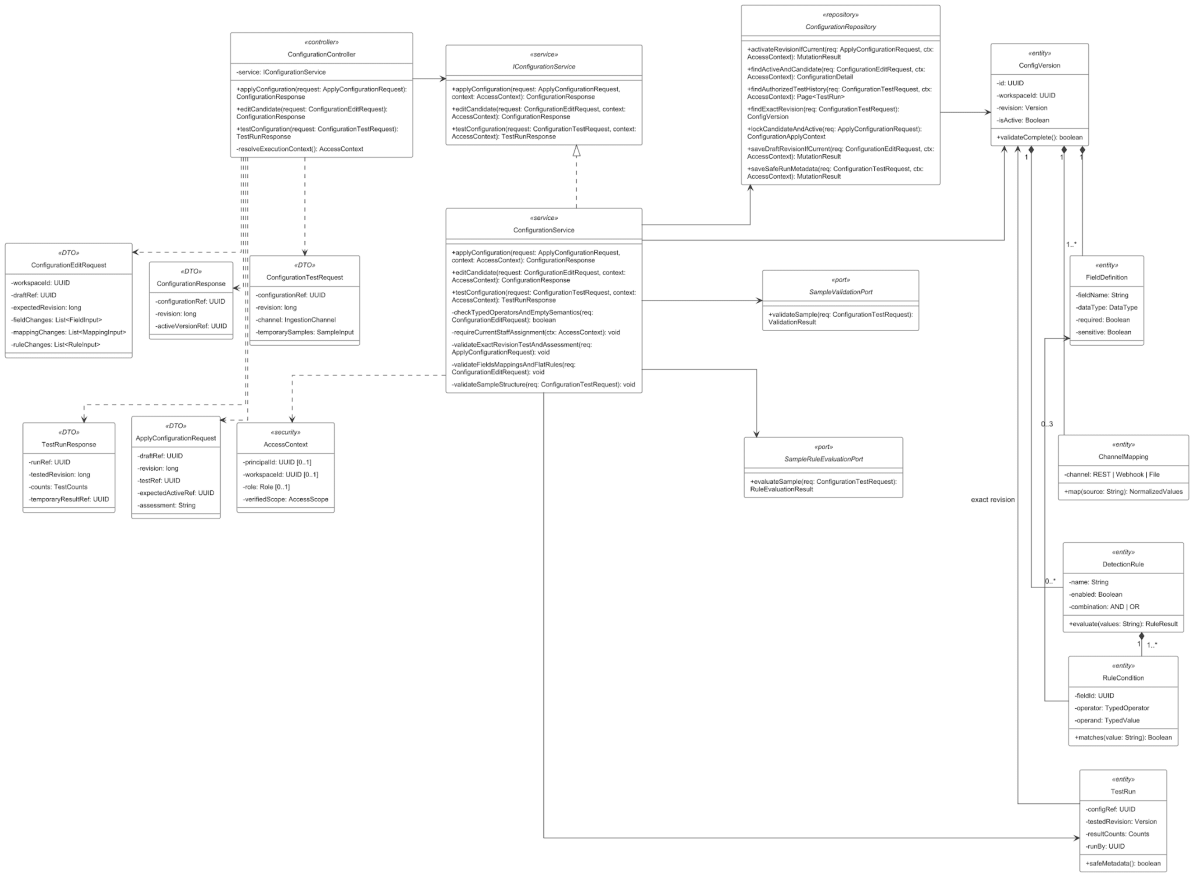


Figure 4.3.6.1.1. Detection Configuration Management Class Diagram.

3.6.2 Sequence Diagrams

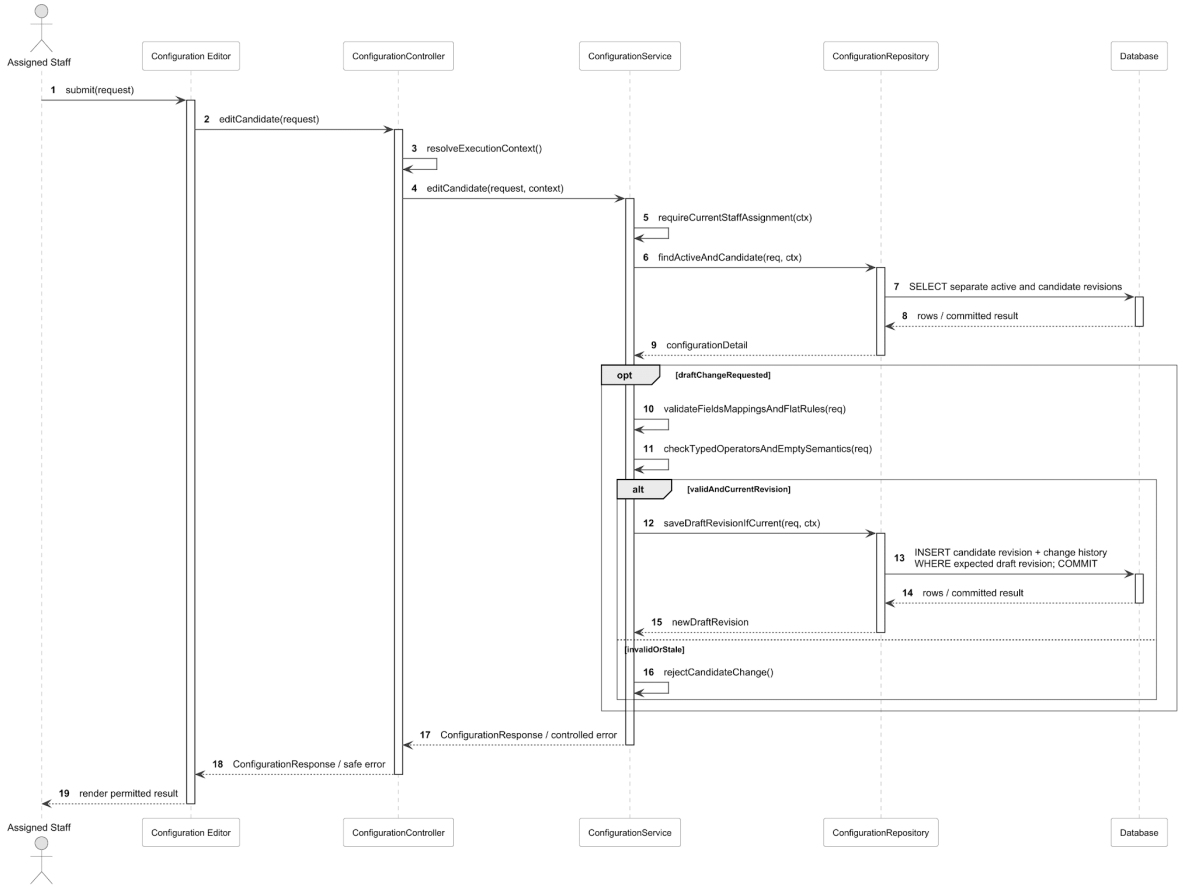


Figure 4.3.6.2.1. View and Edit Candidate Fields, Mappings and Rules Sequence Diagram.

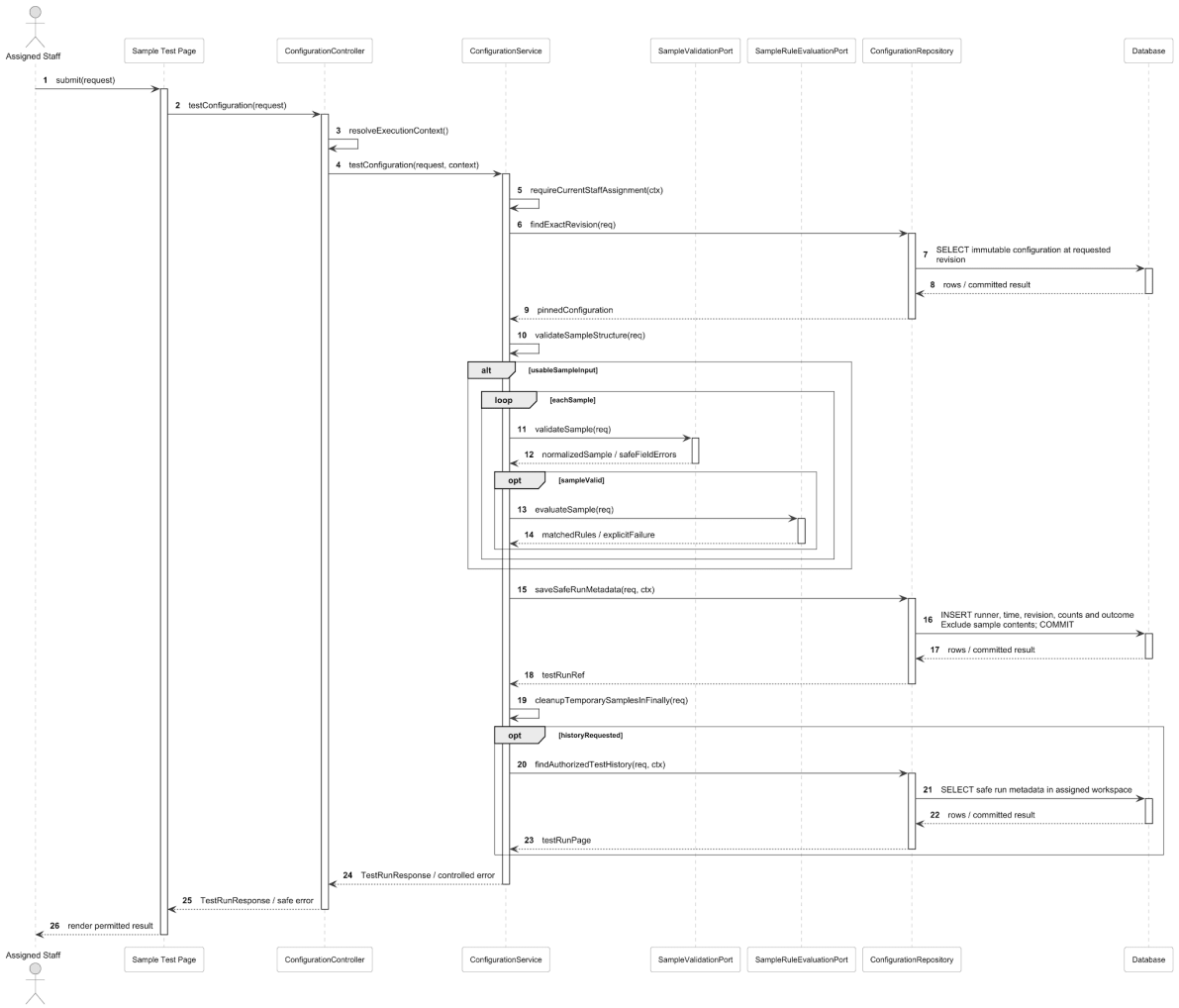


Figure 4.3.6.2.2. Test an Exact Revision and Review Safe Run History Sequence Diagram.

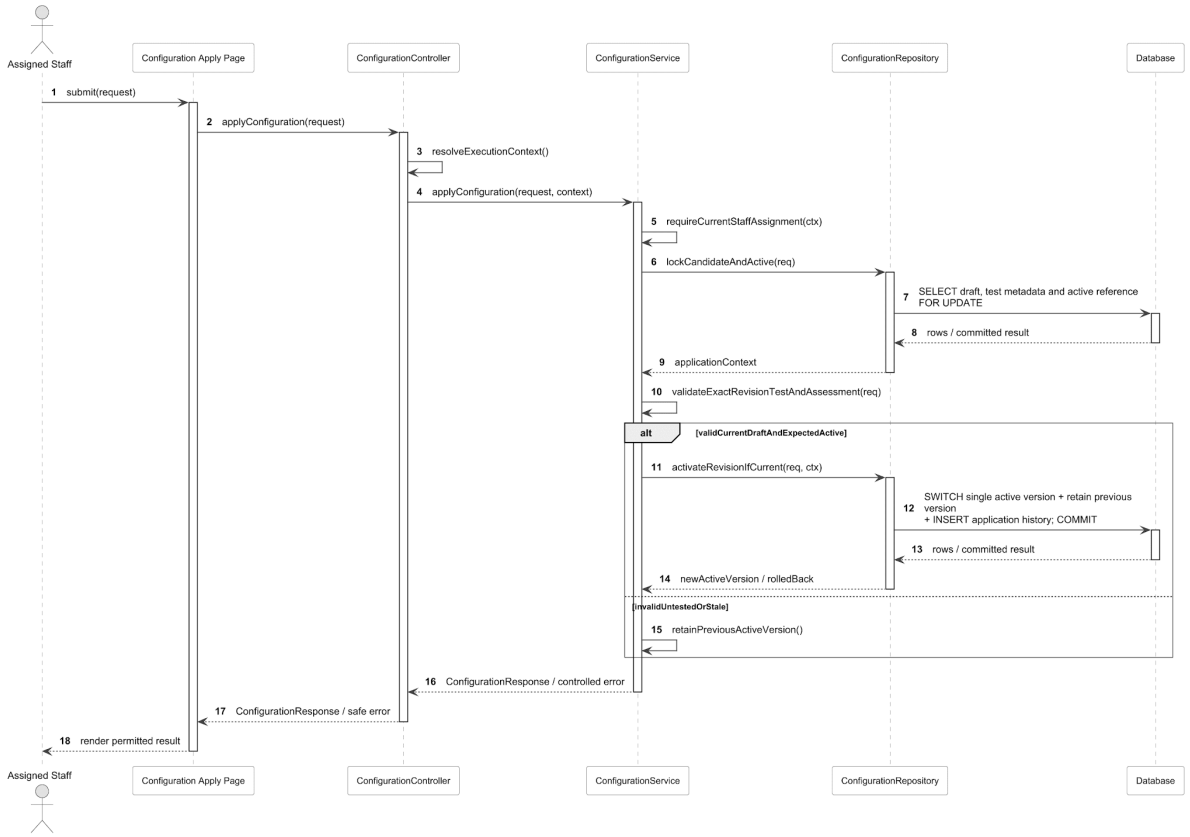


Figure 4.3.6.2.3. Apply the Tested Candidate Atomically Sequence Diagram.

3.6.3 State Diagrams

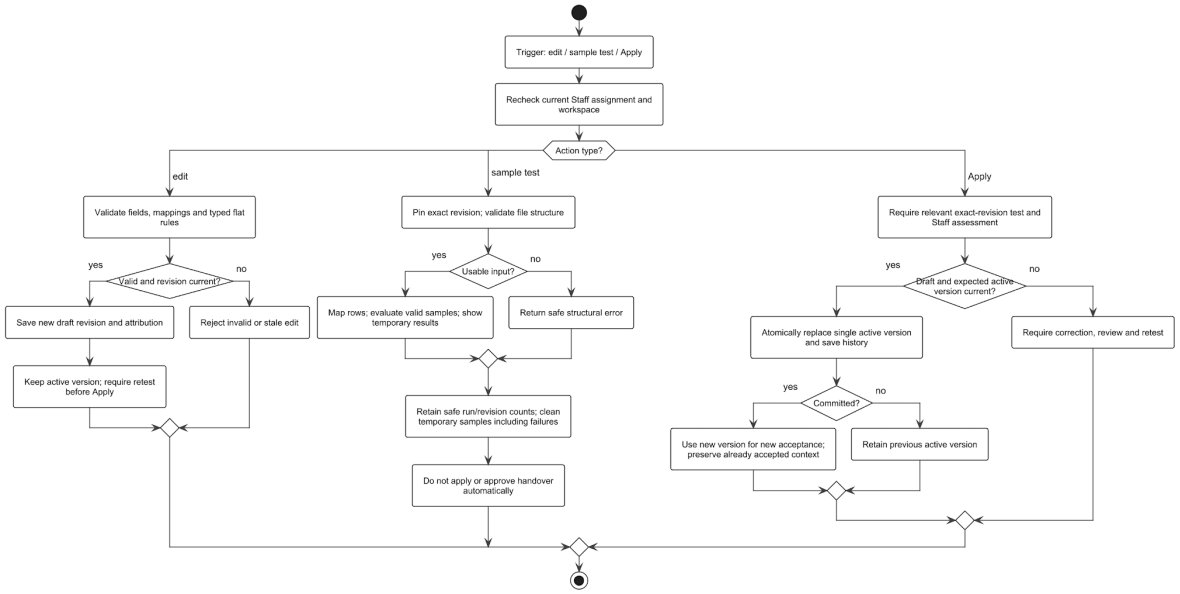


Figure 4.3.6.3.1. Candidate Configuration Testing and Application Lifecycle State Diagram.

3.7 Integration and Transaction Ingestion

SRS reference: 3.8.1-3.8.7. Credentials establish integration workspace scope before mapping. REST, Webhook and production file import share normalization and atomic cross-channel business identity. Acceptance is durable and distinct from detection completion.

3.7.1 Class Diagram

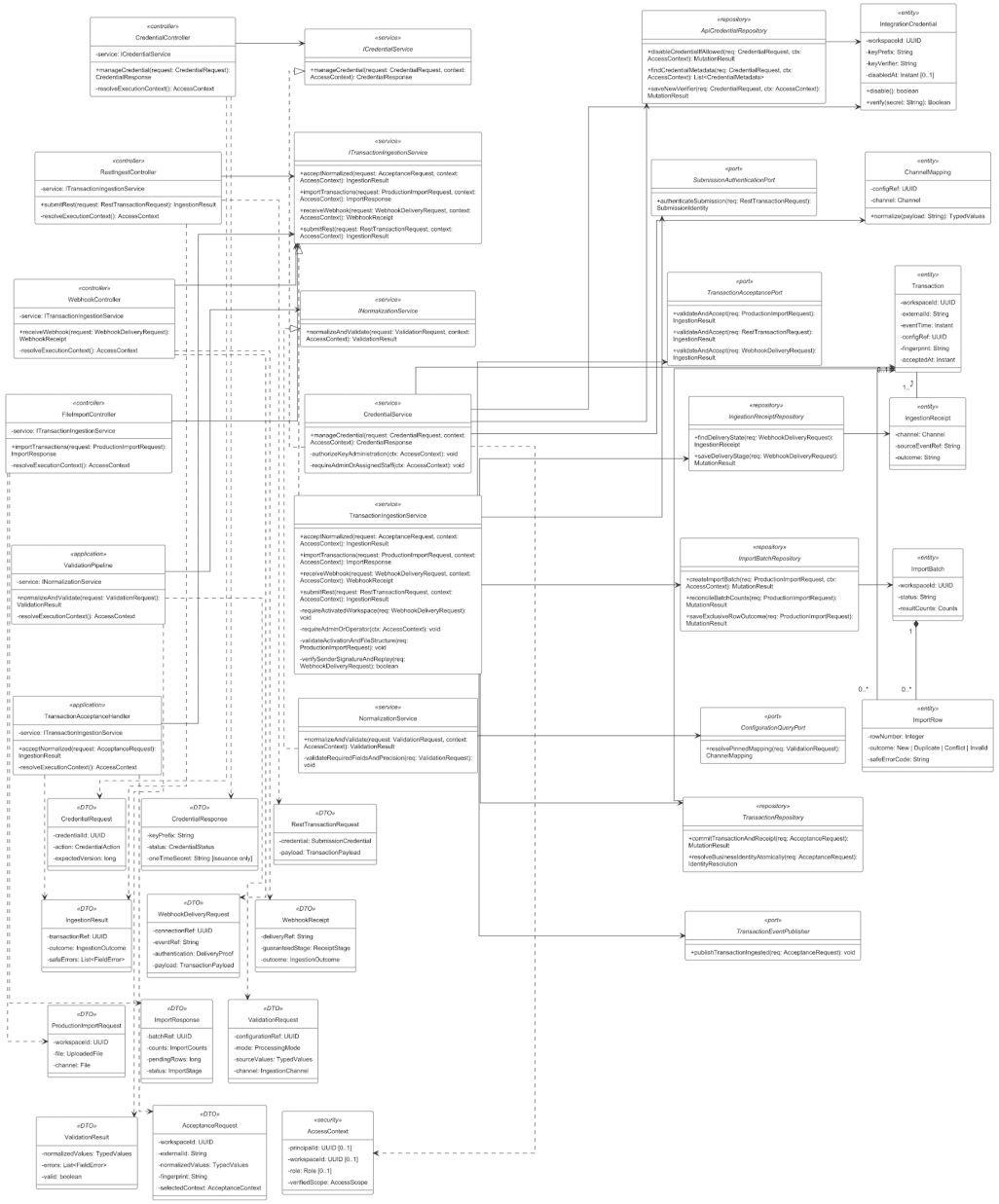


Figure 4.3.7.1.1. Integration and Transaction Ingestion Class Diagram.

3.7.2 Sequence Diagrams

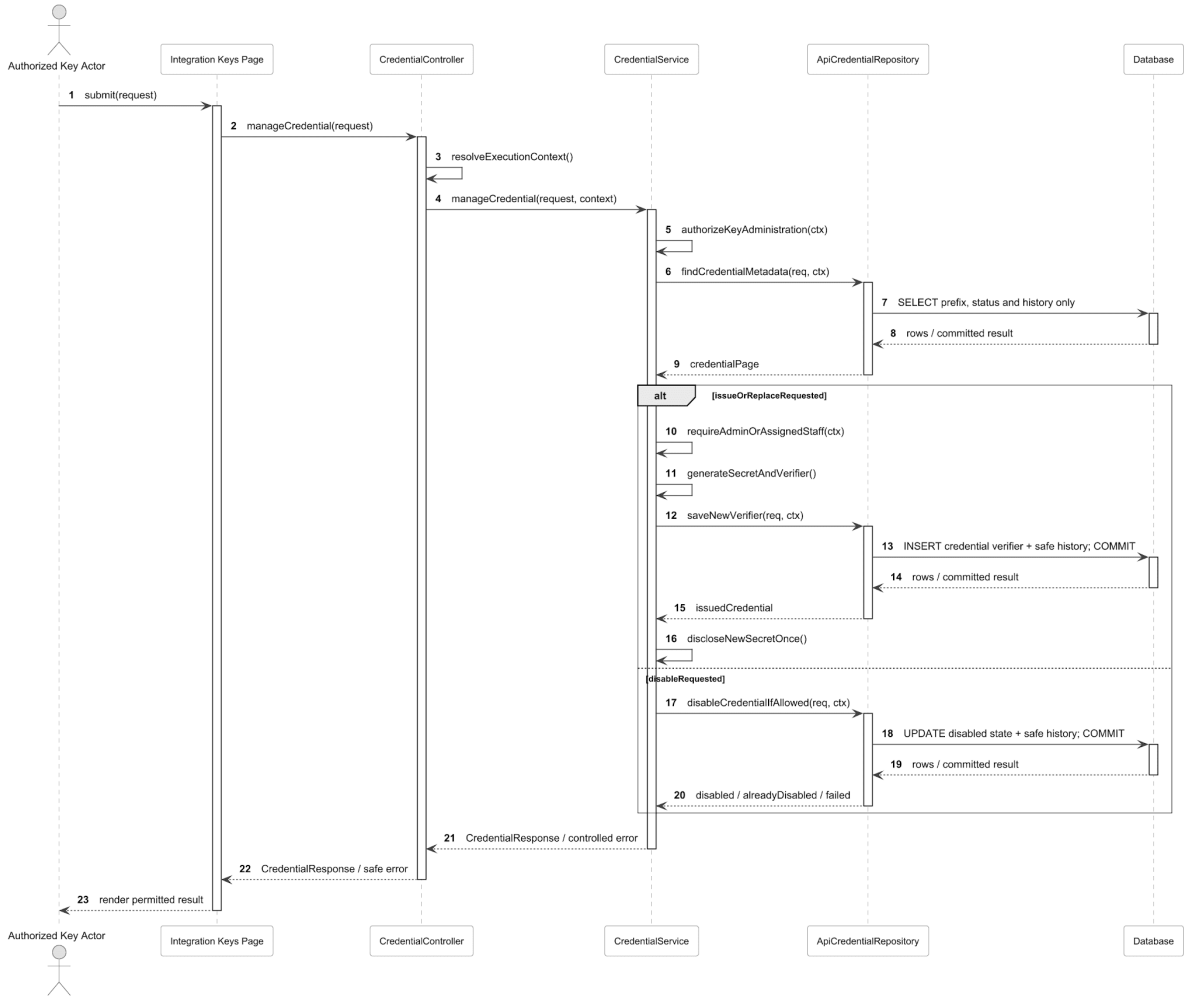


Figure 4.3.7.2.1. Inspect, Issue, Replace and Disable Workspace API Keys Sequence Diagram.

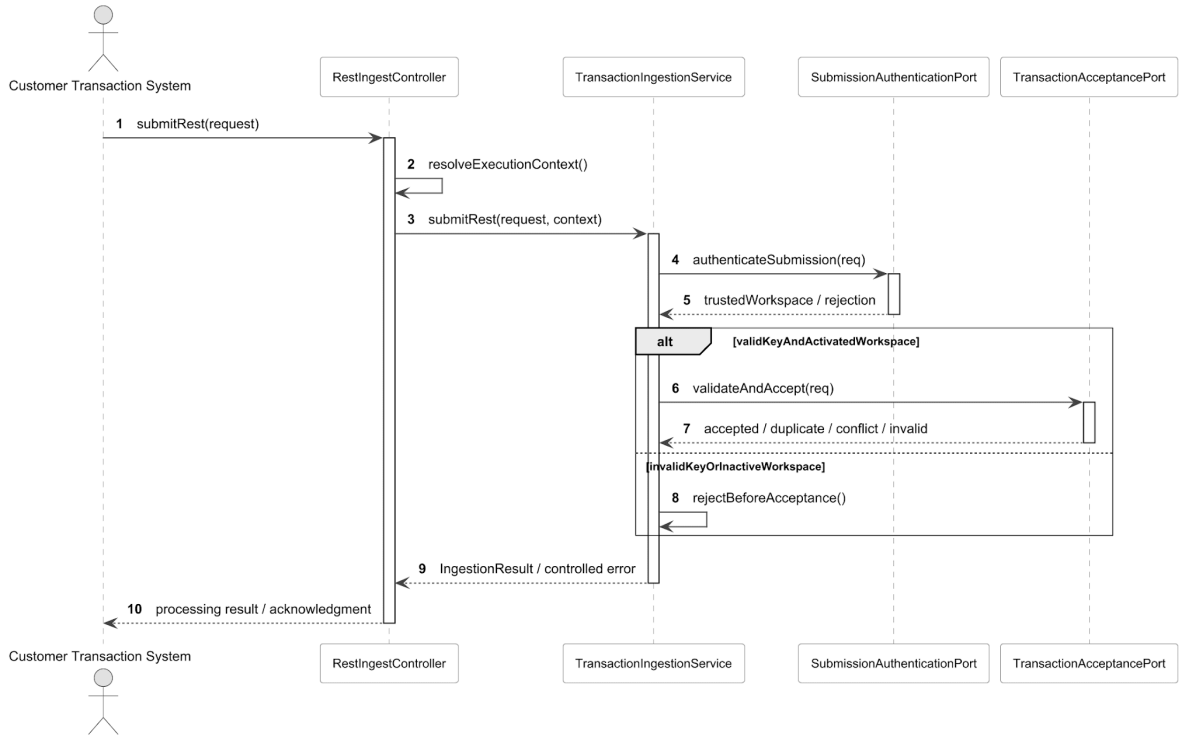


Figure 4.3.7.2.2. Submit Production Transactions through REST API Sequence Diagram.

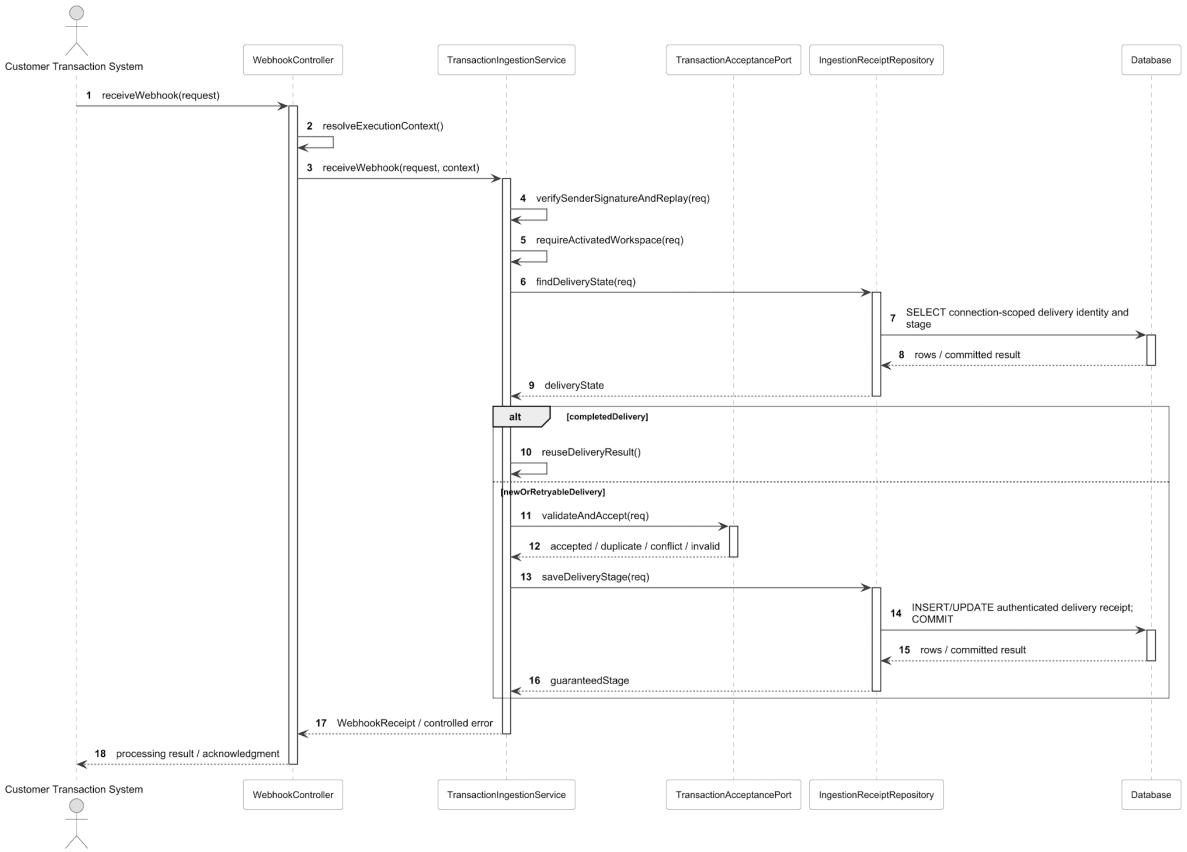


Figure 4.3.7.2.3. Authenticate Incoming Webhook and Handle Delivery Retries Sequence Diagram.

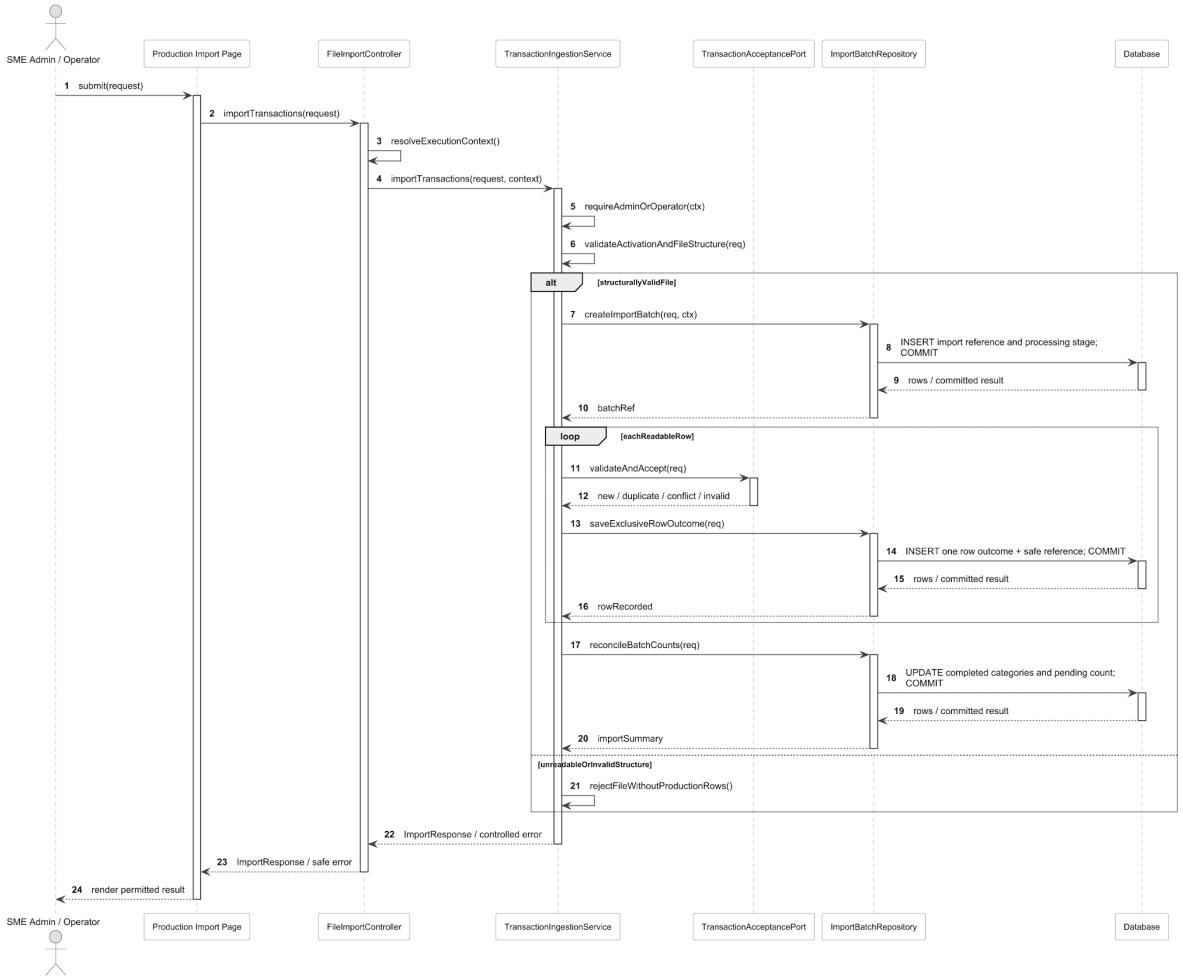


Figure 4.3.7.2.4. Import Production CSV/Excel with Reconciled Row Outcomes Sequence Diagram.

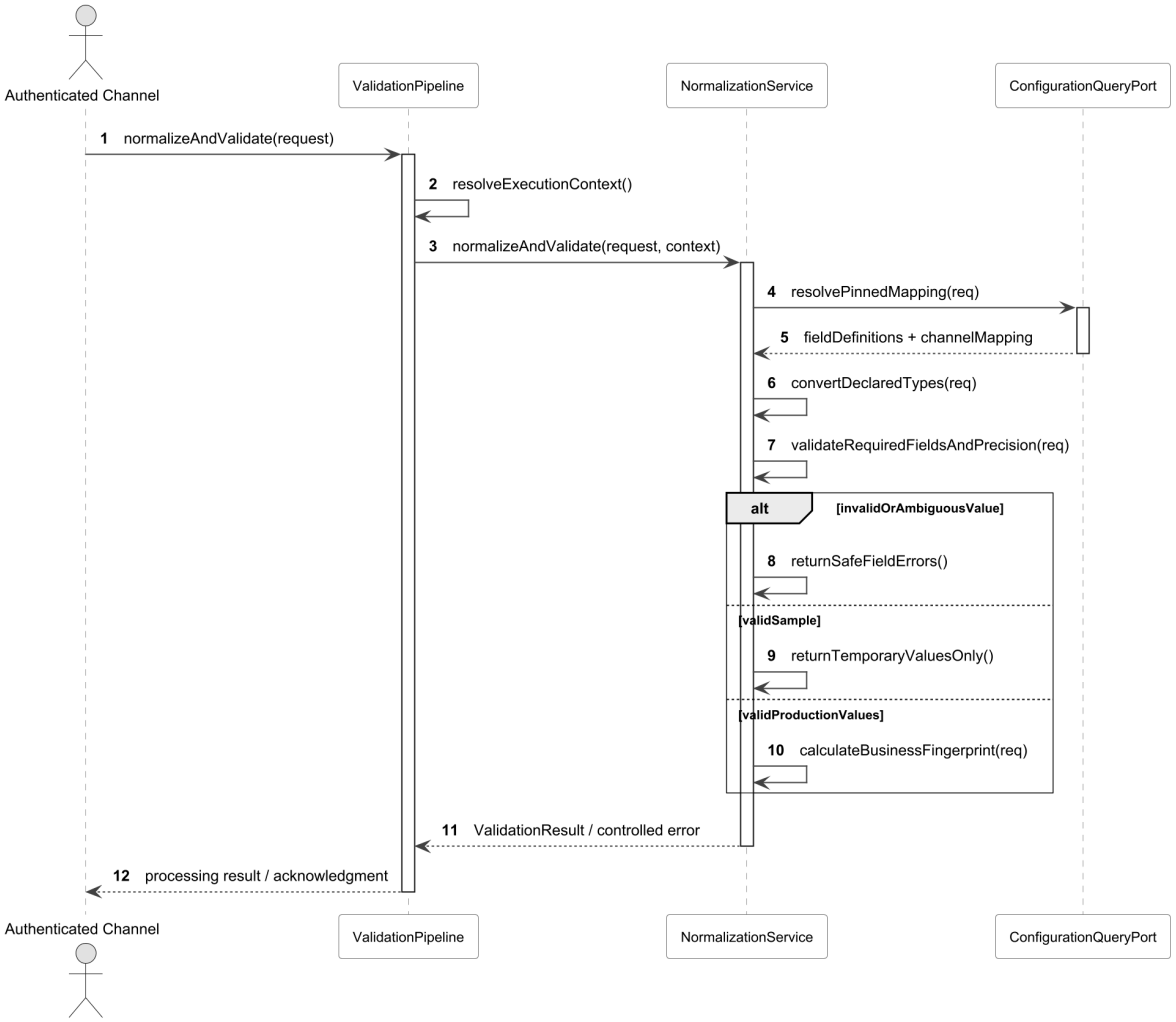


Figure 4.3.7.2.5. Normalize and Validate Declared Channel Values Sequence Diagram.

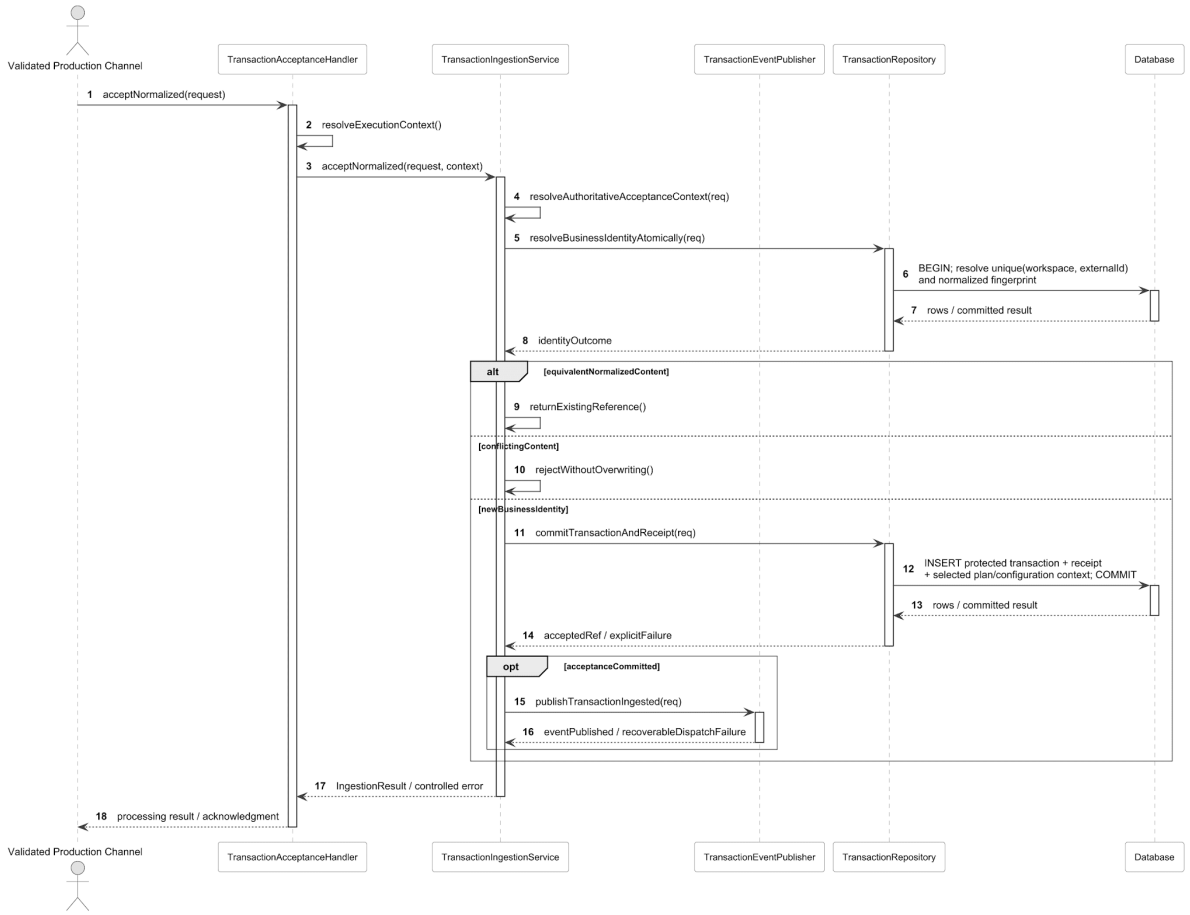


Figure 4.3.7.2.6. Atomically Resolve Cross-Channel Business Identity and Accept Once Sequence Diagram.

3.7.3 State Diagrams

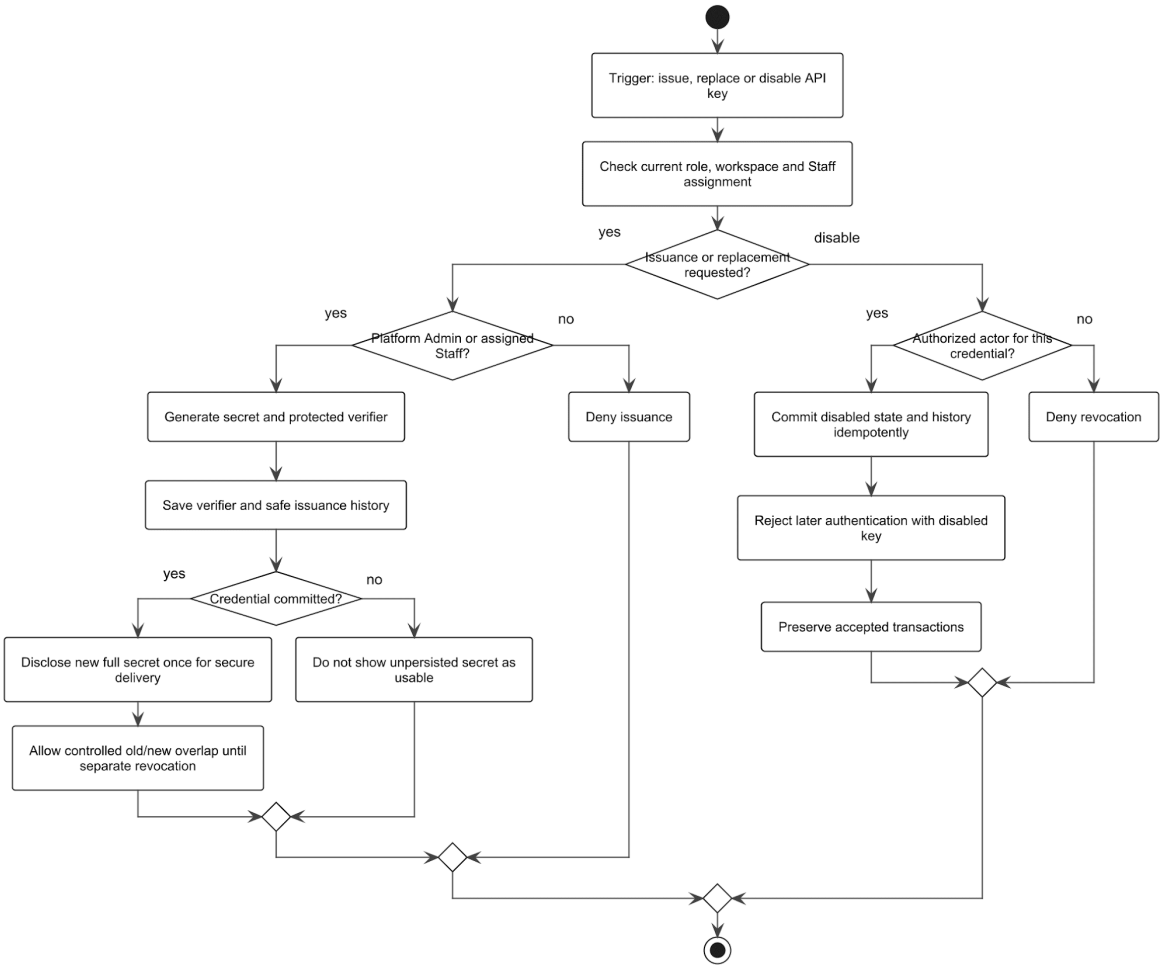


Figure 4.3.7.3.1. Workspace API Credential Lifecycle State Diagram.

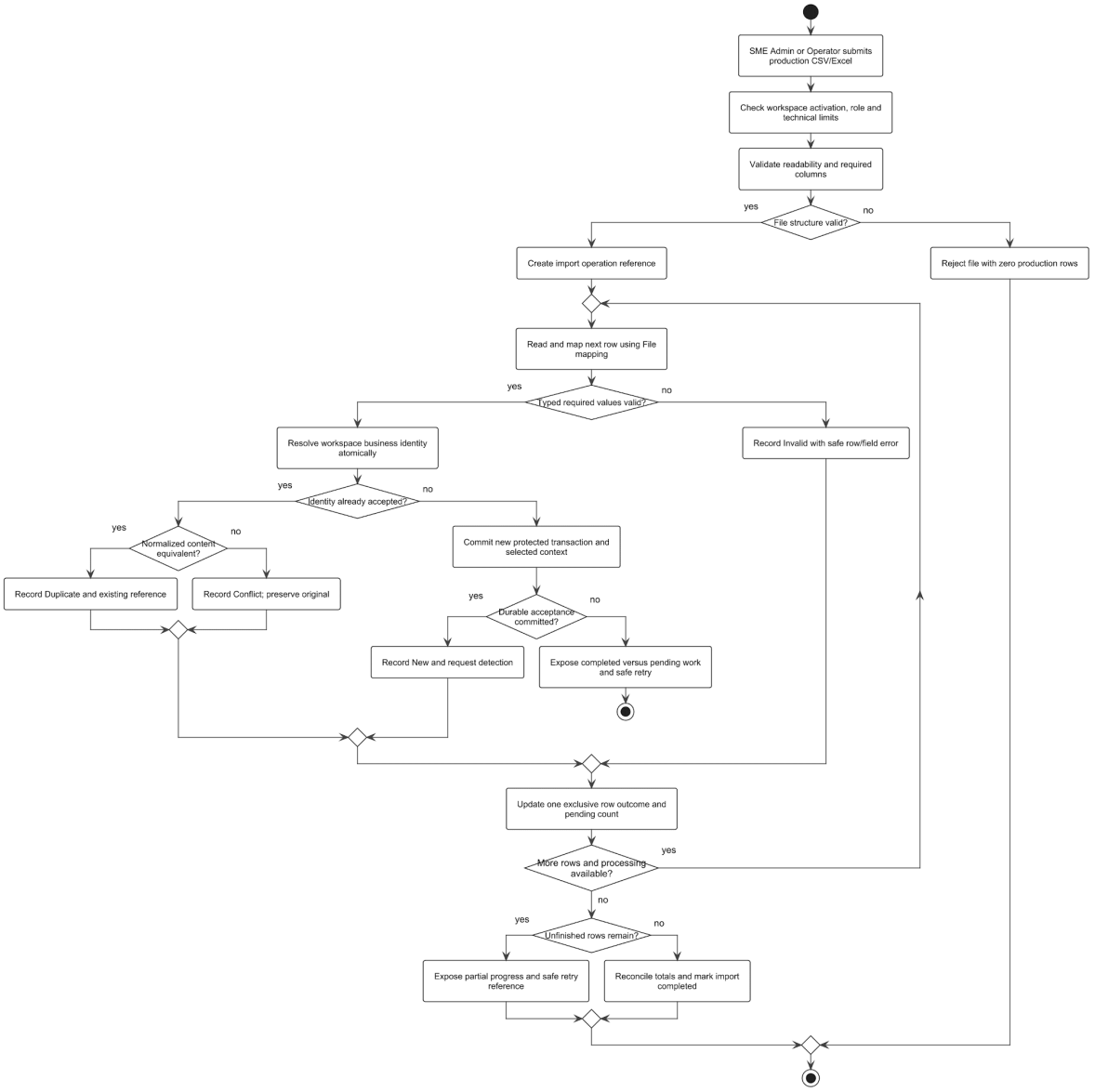


Figure 4.3.7.3.2. Production Import Processing Lifecycle State Diagram.

3.8 Rule-Based Detection

SRS reference: 3.9.1-3.9.2. Free primary detection and Subscription fallback evaluate the selected configuration with the same typed semantics. All matching rules form one Rule Alert for the completed attempt; no match is distinct from failure.

3.8.1 Class Diagram

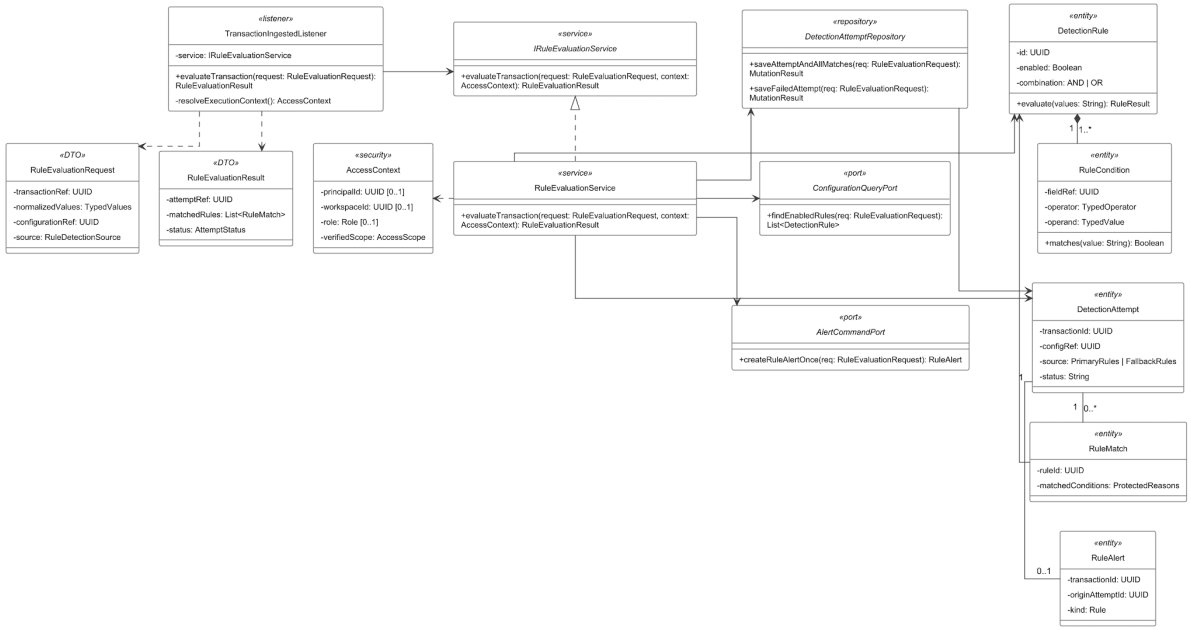


Figure 4.3.8.1.1. Rule-Based Detection Class Diagram.

3.8.2 Sequence Diagrams

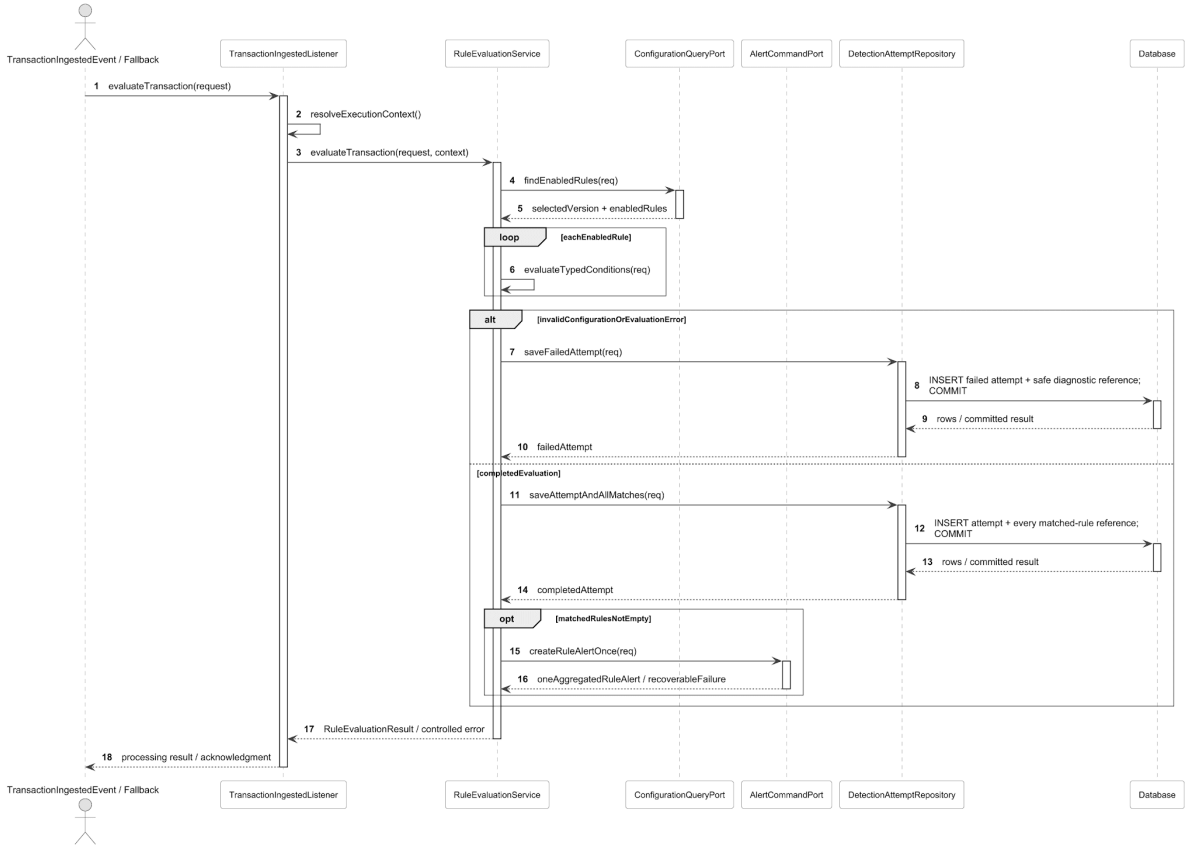


Figure 4.3.8.2.1. Evaluate All Enabled Rules and Create One Aggregated Alert Sequence Diagram.

3.9 AI Detection and Rule-Based Fallback

SRS reference: 3.10.1-3.10.3. Subscription primary processing calls AI only. Unusable AI responses trigger rules and retain both attempt sources. Recovery reuses the transaction, checks historical event-time entitlement and preserves earlier alerts and cases.

3.9.1 Class Diagram

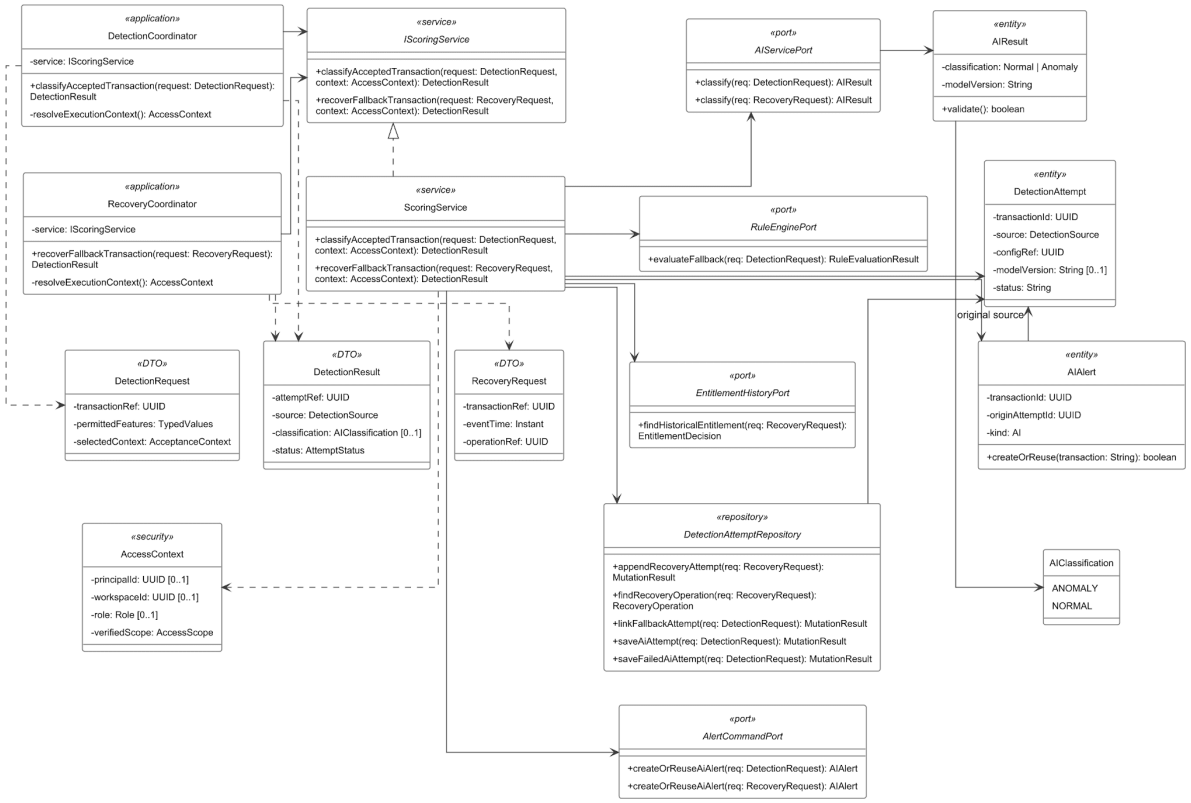


Figure 4.3.9.1.1. AI Detection and Rule-Based Fallback Class Diagram.

3.9.2 Sequence Diagrams

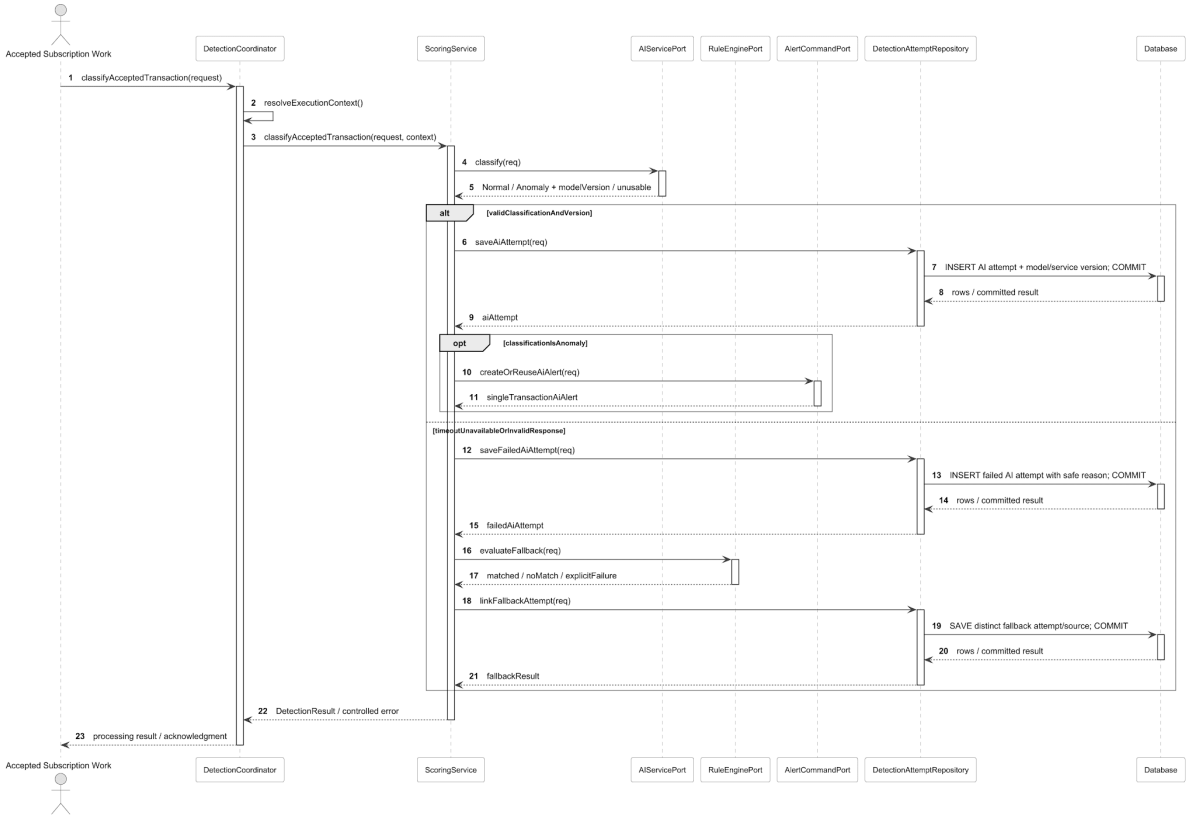


Figure 4.3.9.2.1. Classify Subscription Transaction and Fall Back on Unusable AI Sequence Diagram.

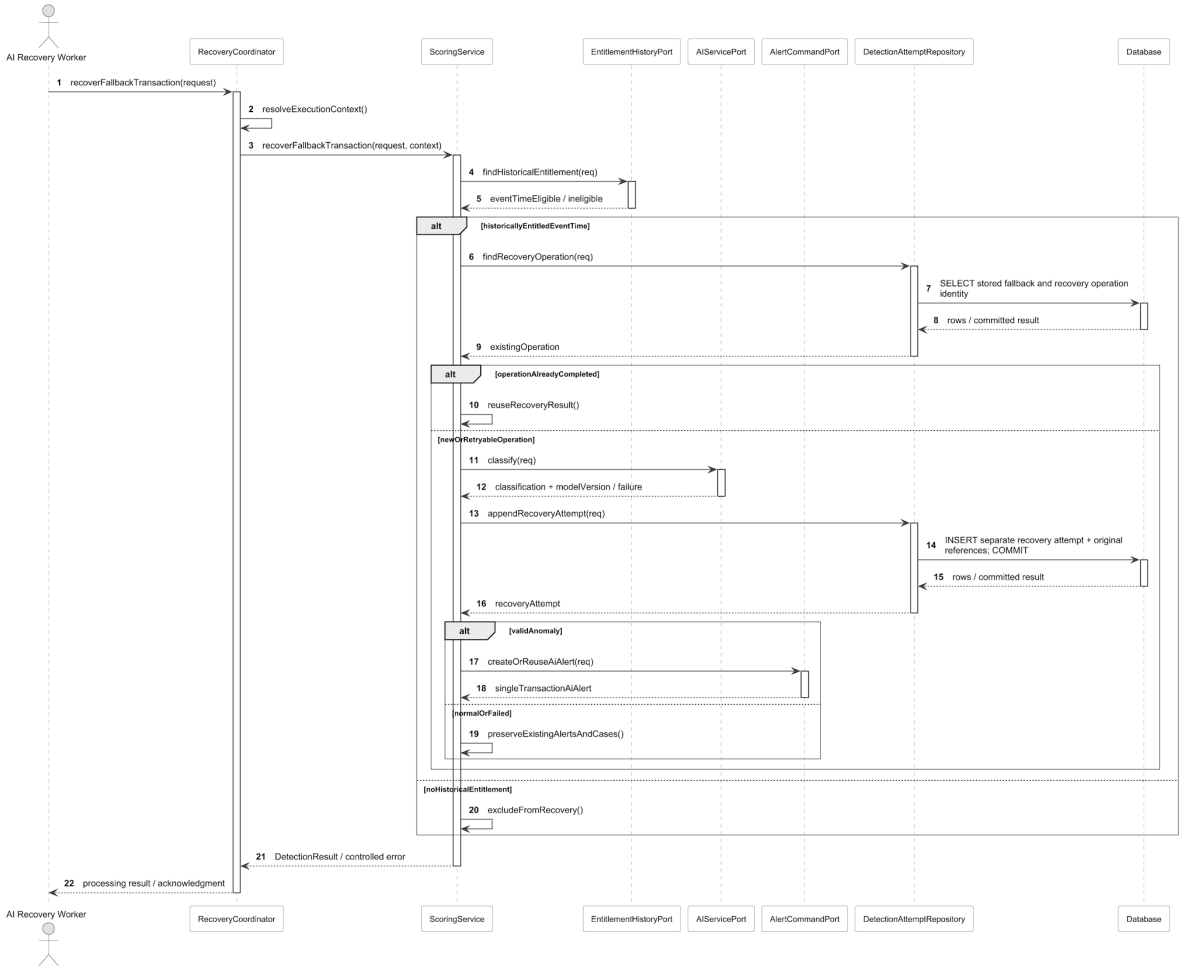


Figure 4.3.9.2.2. Recover Historically Eligible Fallback Work and Reconcile AI Alerts Sequence Diagram.

3.9.3 State Diagrams

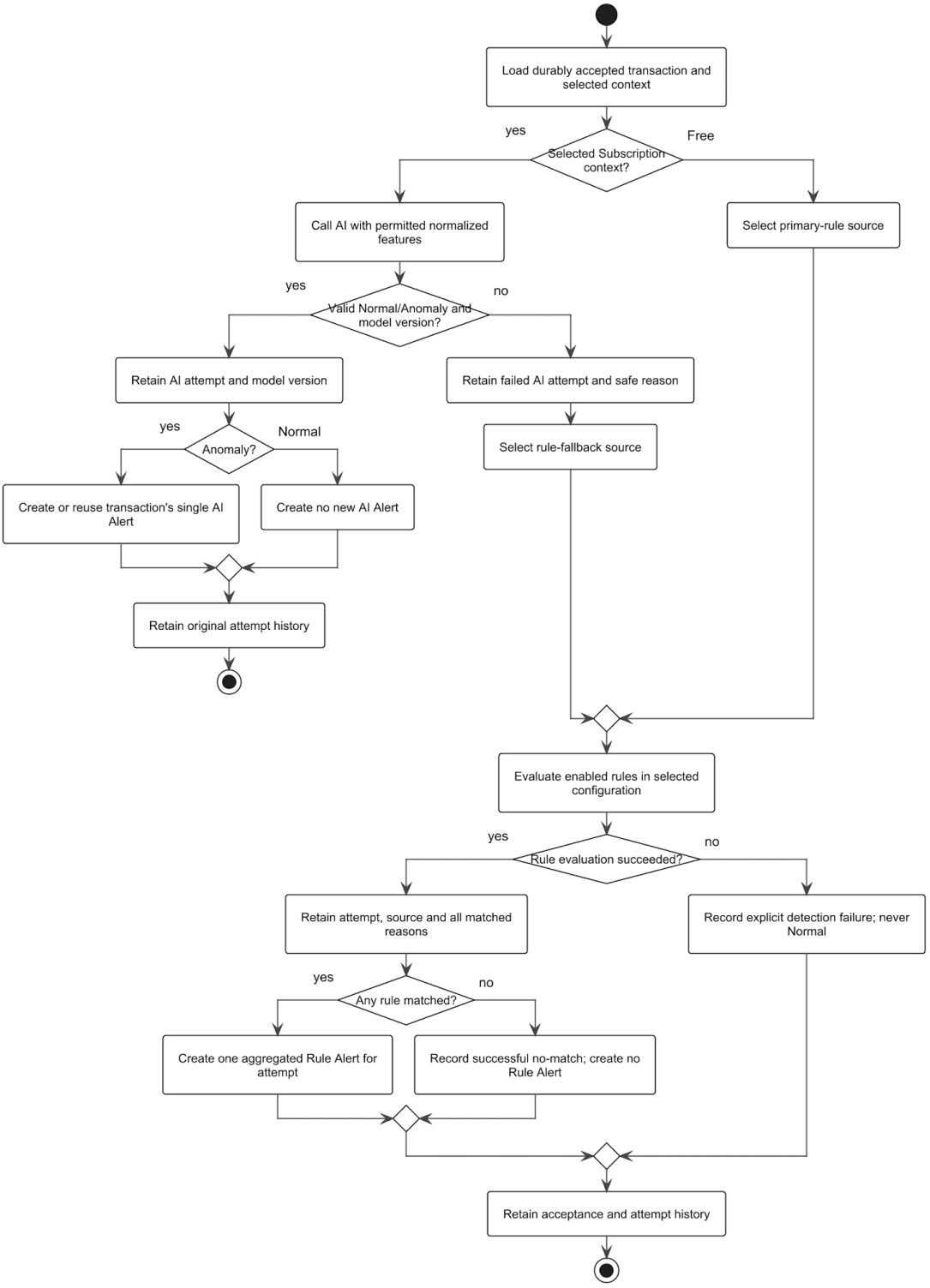


Figure 4.3.9.3.1. Transaction Detection and Rule-Based Fallback Processing State Diagram.

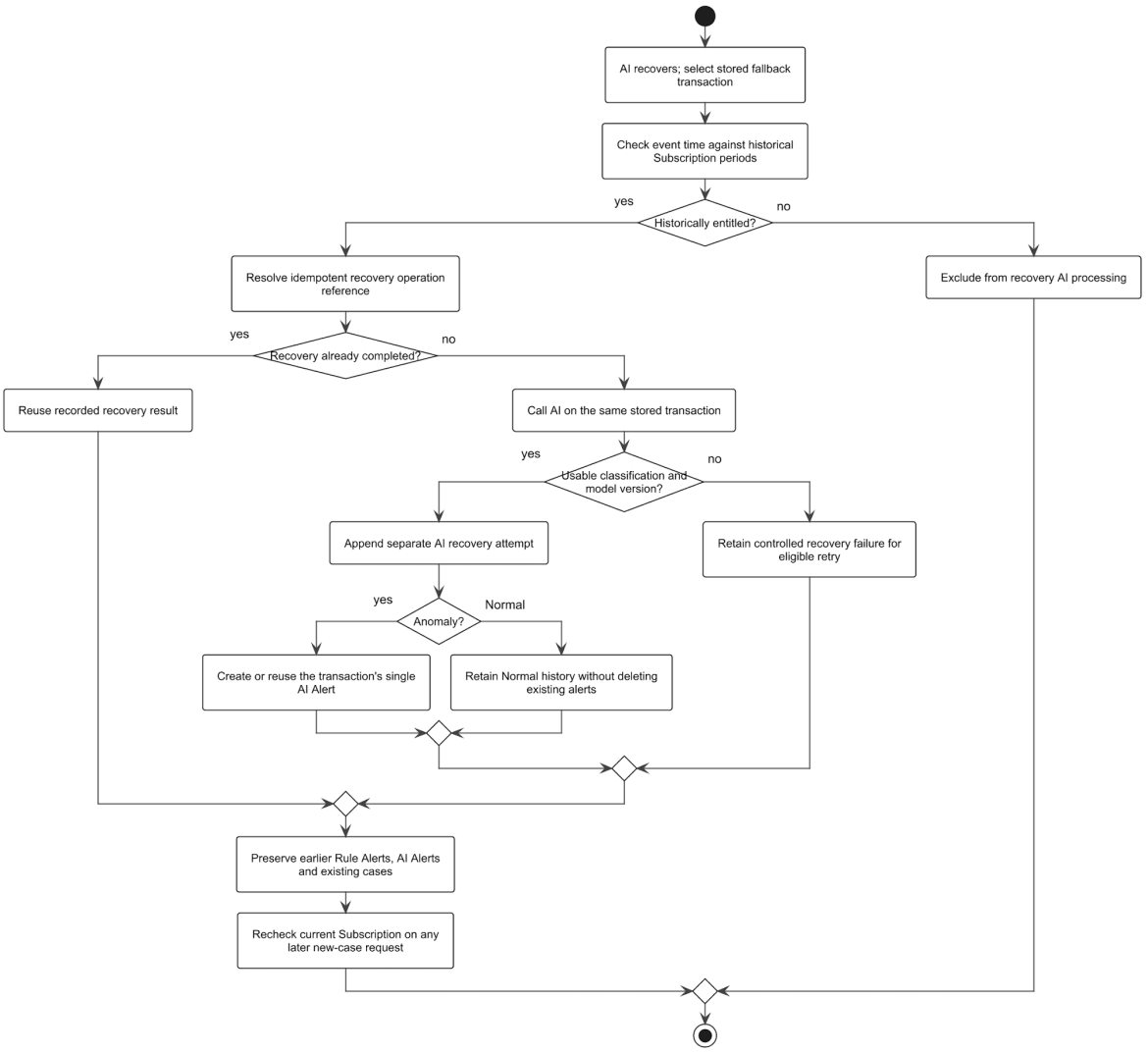


Figure 4.3.9.3.2. Historically Entitled AI Recovery Processing State Diagram.

3.10 Transaction and Alert Review

SRS reference: 3.11.1-3.11.6. Operator reviews workspace production transactions, alerts and attempt history. Import results have separate SME Admin/Operator scope. Explicit reveal requires current field grants and, for Investigator, the relevant case assignment.

3.10.1 Class Diagram

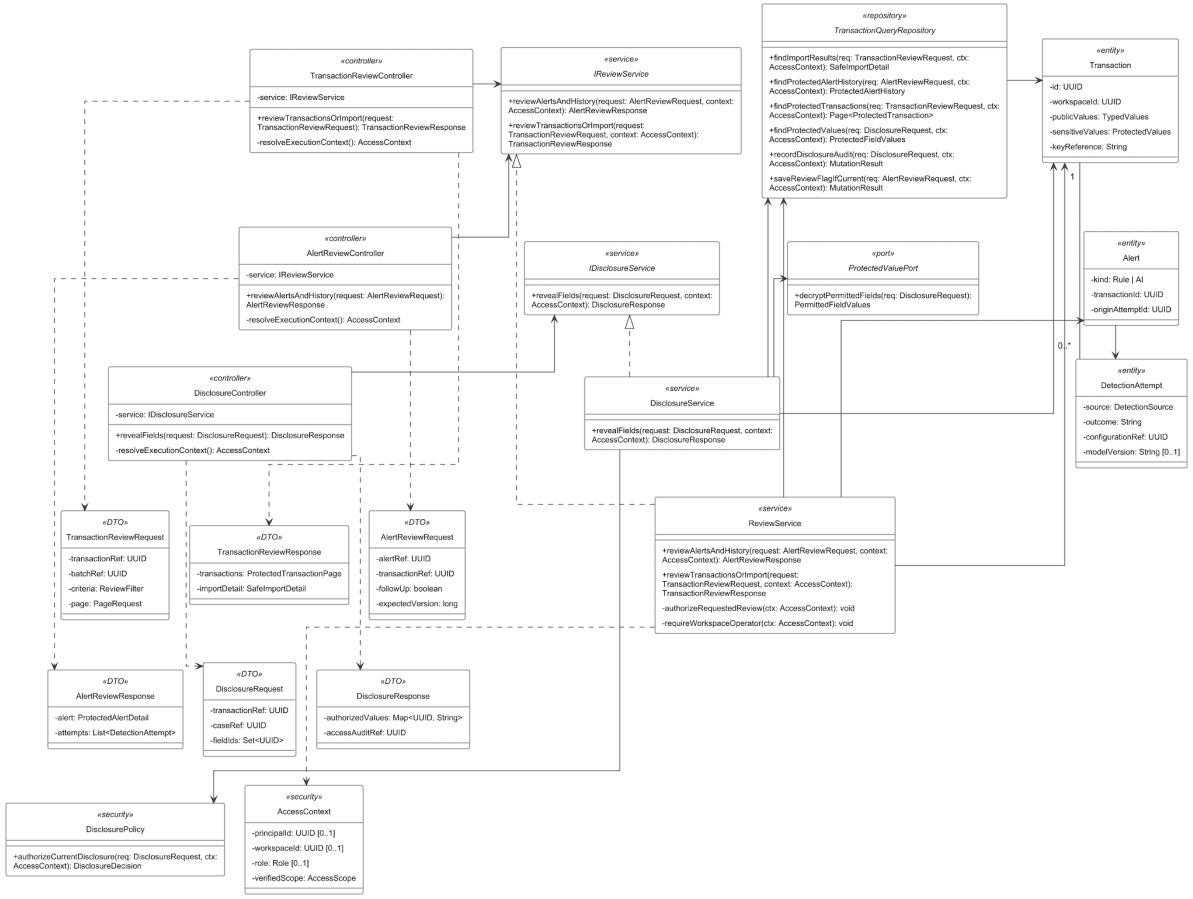


Figure 4.3.10.1.1. Transaction and Alert Review Class Diagram.

3.10.2 Sequence Diagrams

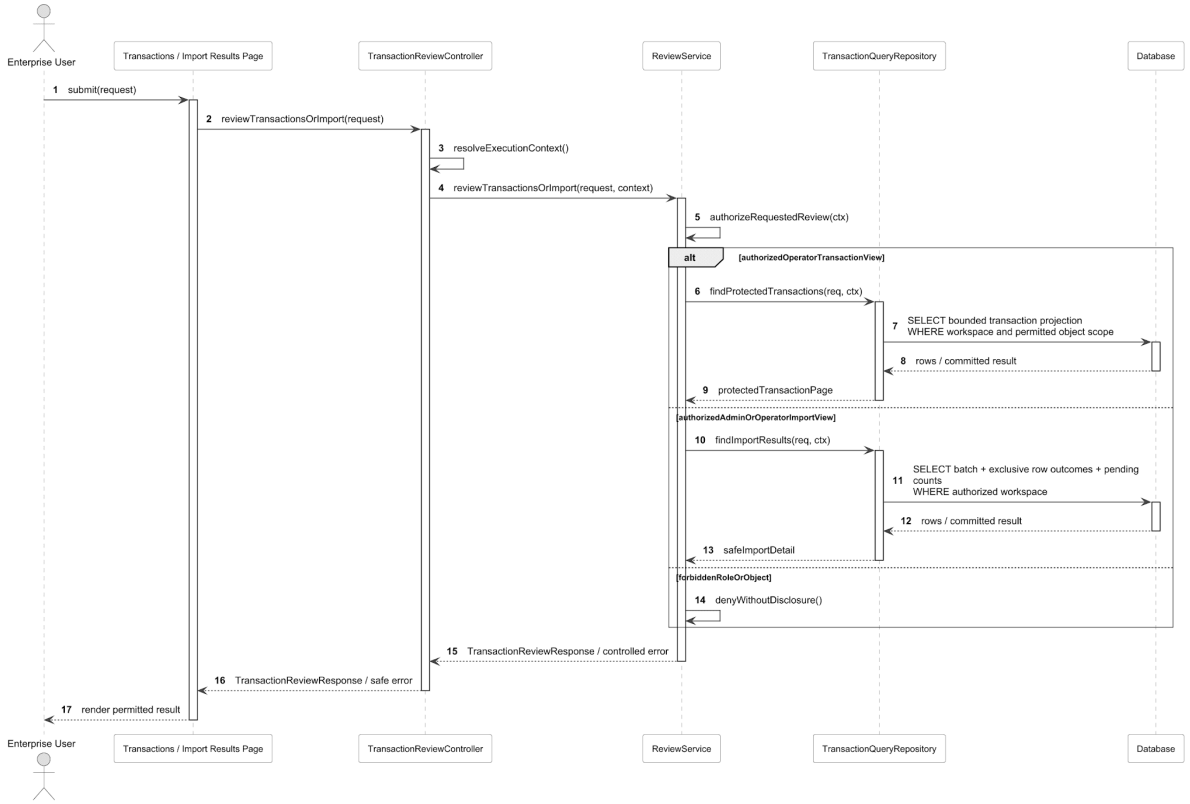


Figure 4.3.10.2.1. Review Transactions and Role-Scoped Import Results Sequence Diagram.

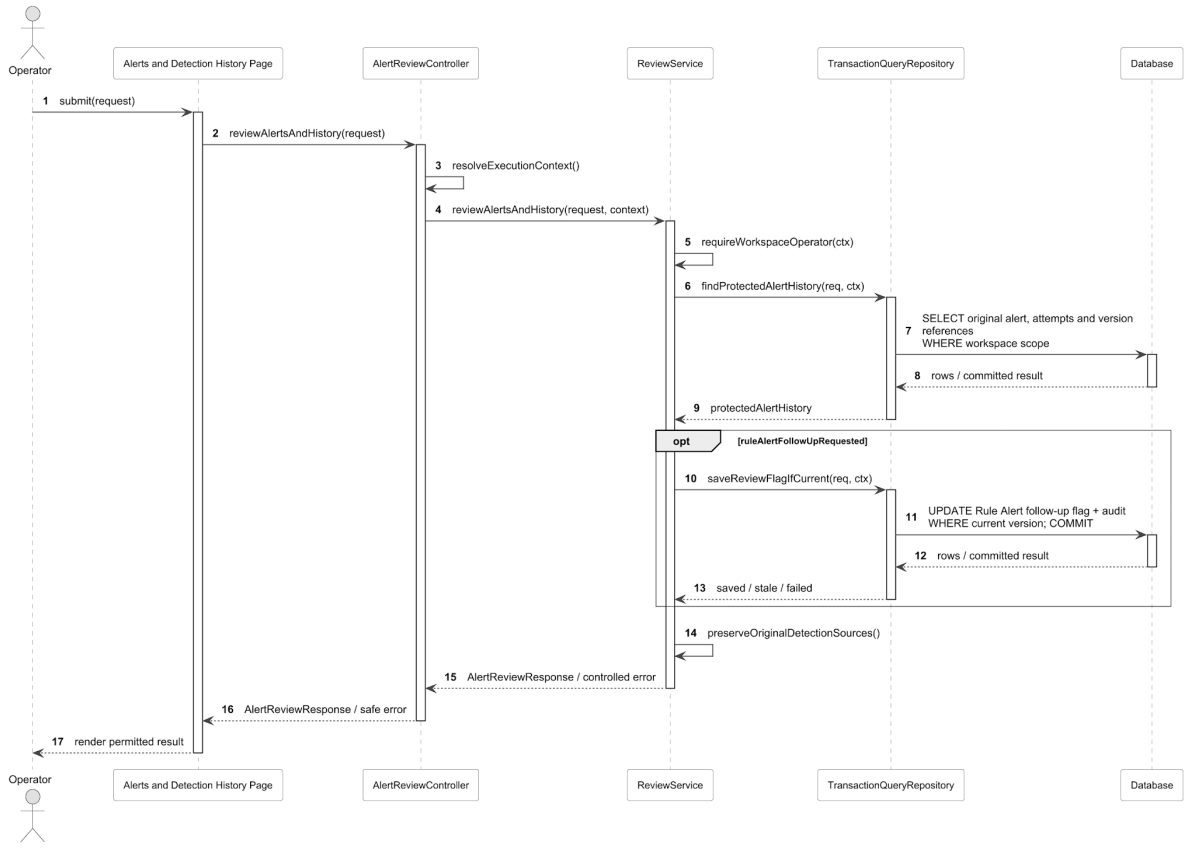


Figure 4.3.10.2.2. Review Rule/AI Alerts, Follow-Up Flags and Attempt History Sequence Diagram.

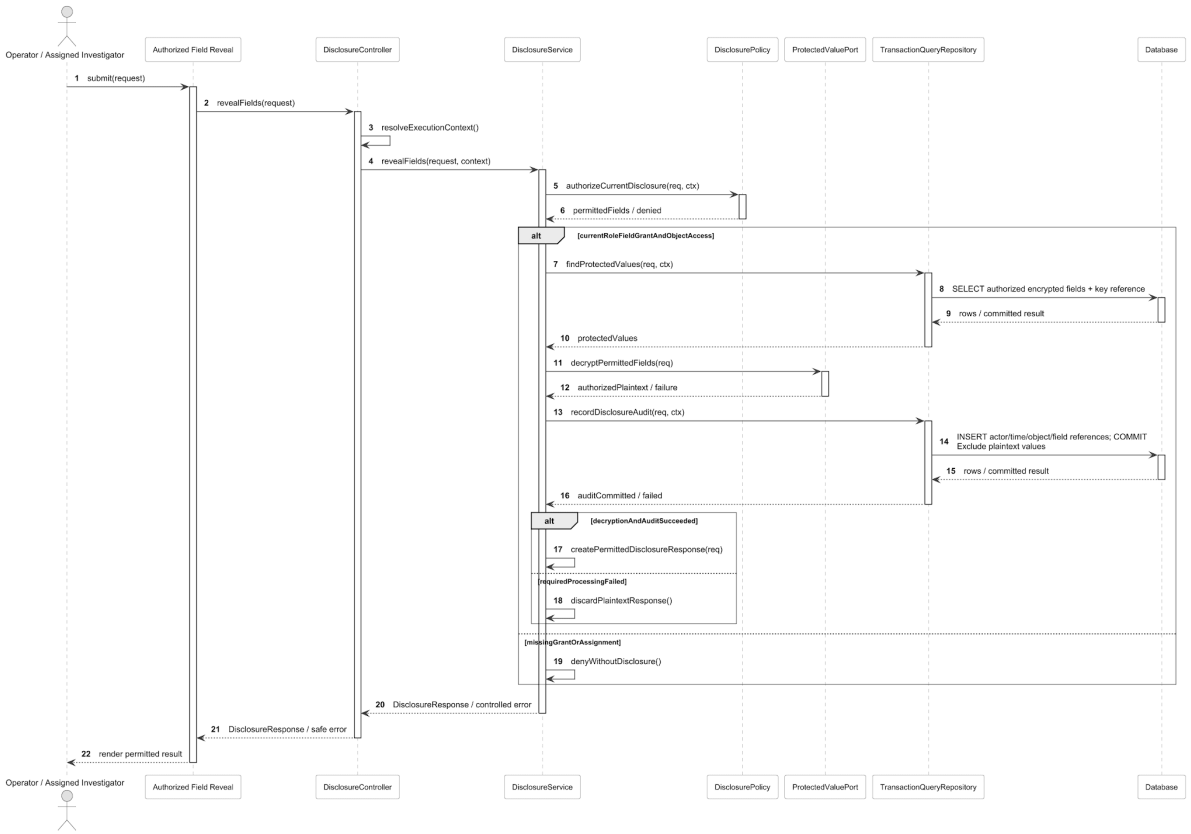


Figure 4.3.10.2.3. Explicitly Reveal Only Currently Authorized Sensitive Fields Sequence Diagram.

3.11 Case and Investigation Management

SRS reference: 3.12.1-3.12.7. Only Operator opens cases from same-workspace AI Alerts under current Subscription. Assignment is separate from lifecycle. Existing case work continues after expiry; Operator may investigate, submit and confirm without a second person.

3.11.1 Class Diagram

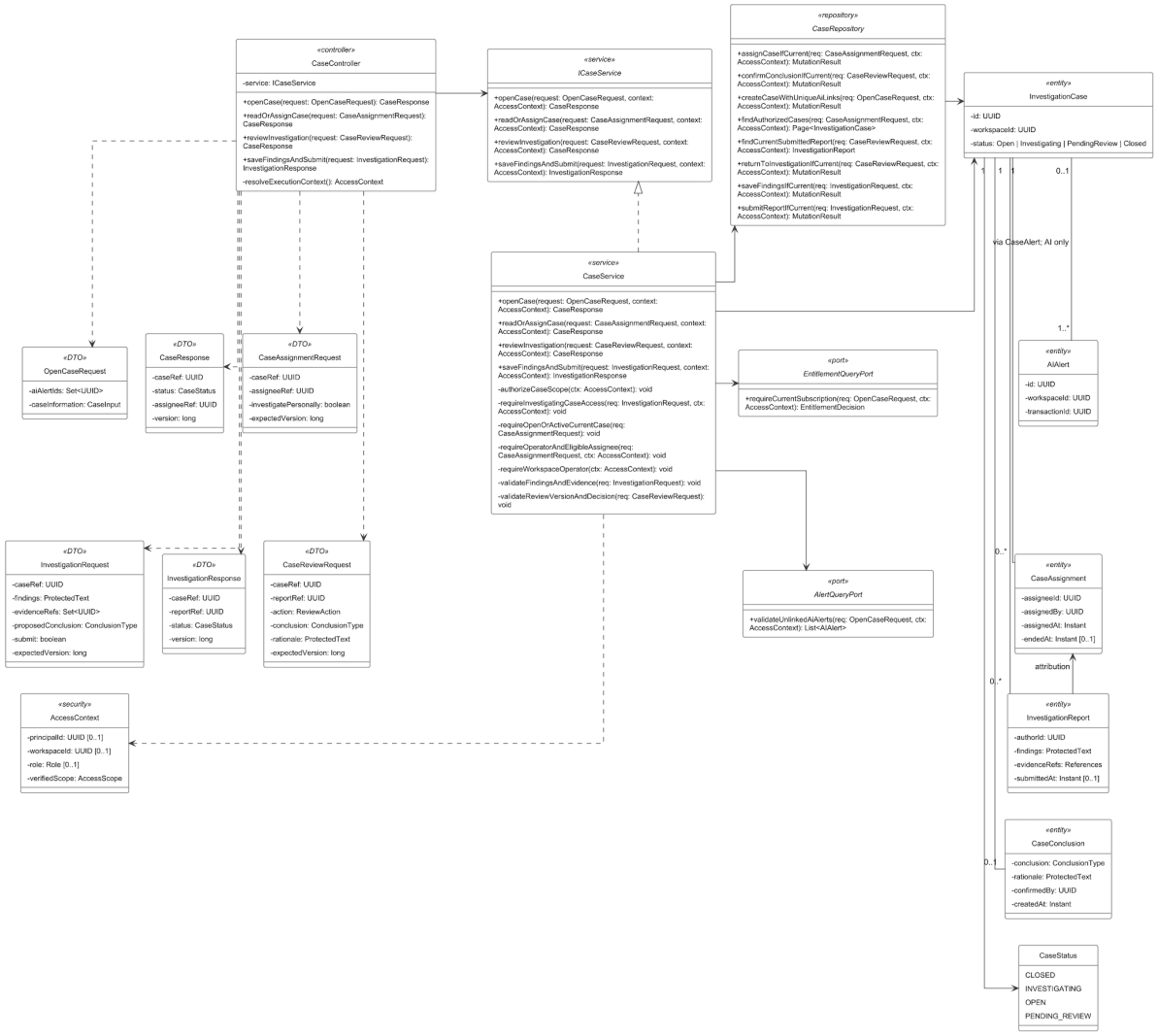


Figure 4.3.11.1.1. Case and Investigation Management Class Diagram.

3.11.2 Sequence Diagrams

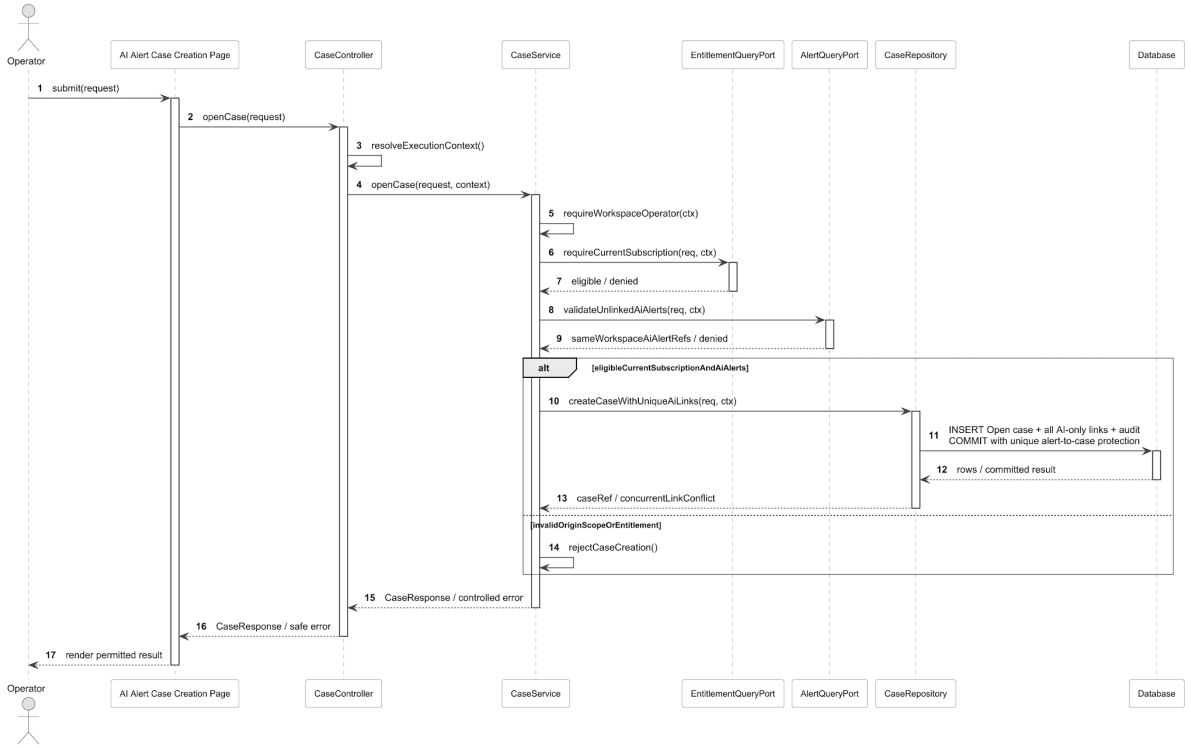


Figure 4.3.11.2.1. Open One Case from Eligible AI Alert References Sequence Diagram.

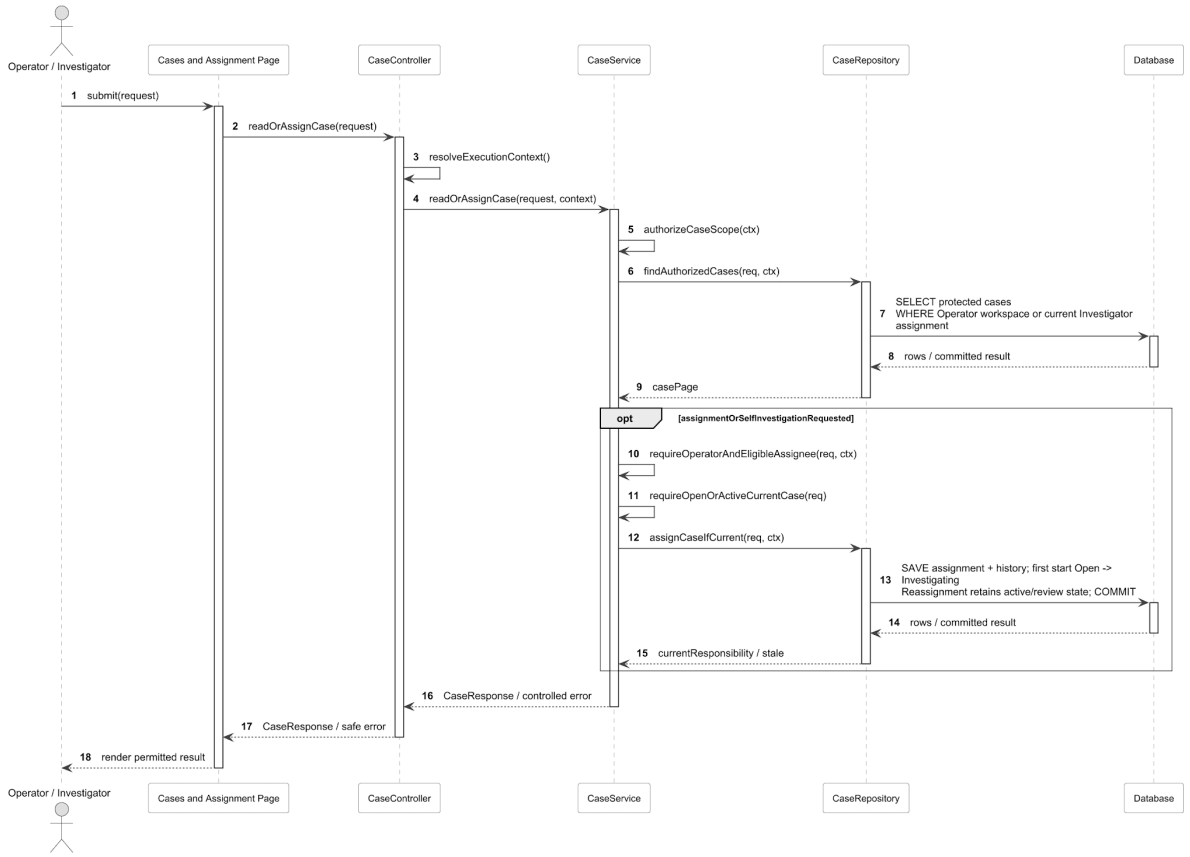


Figure 4.3.11.2.2. View Cases and Assign or Start Self-Investigation Sequence Diagram.

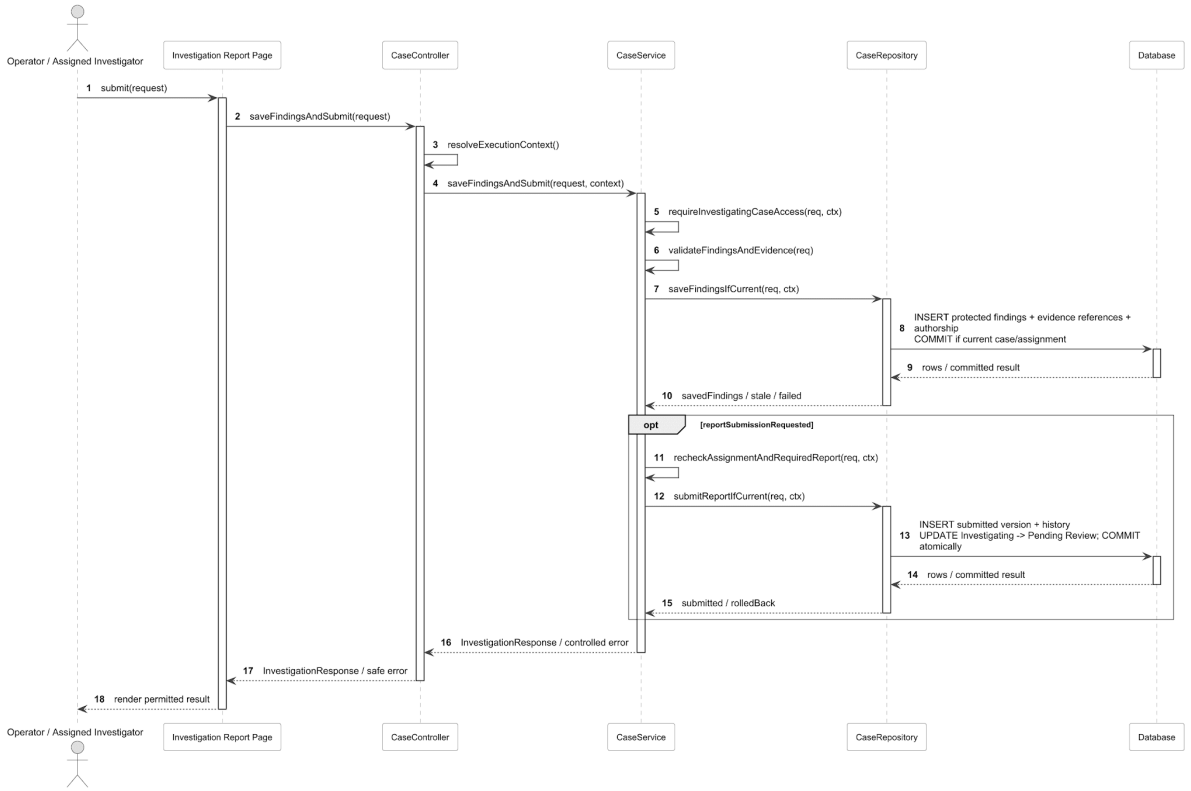


Figure 4.3.11.2.3. Record Findings and Submit an Investigation Report Sequence Diagram.

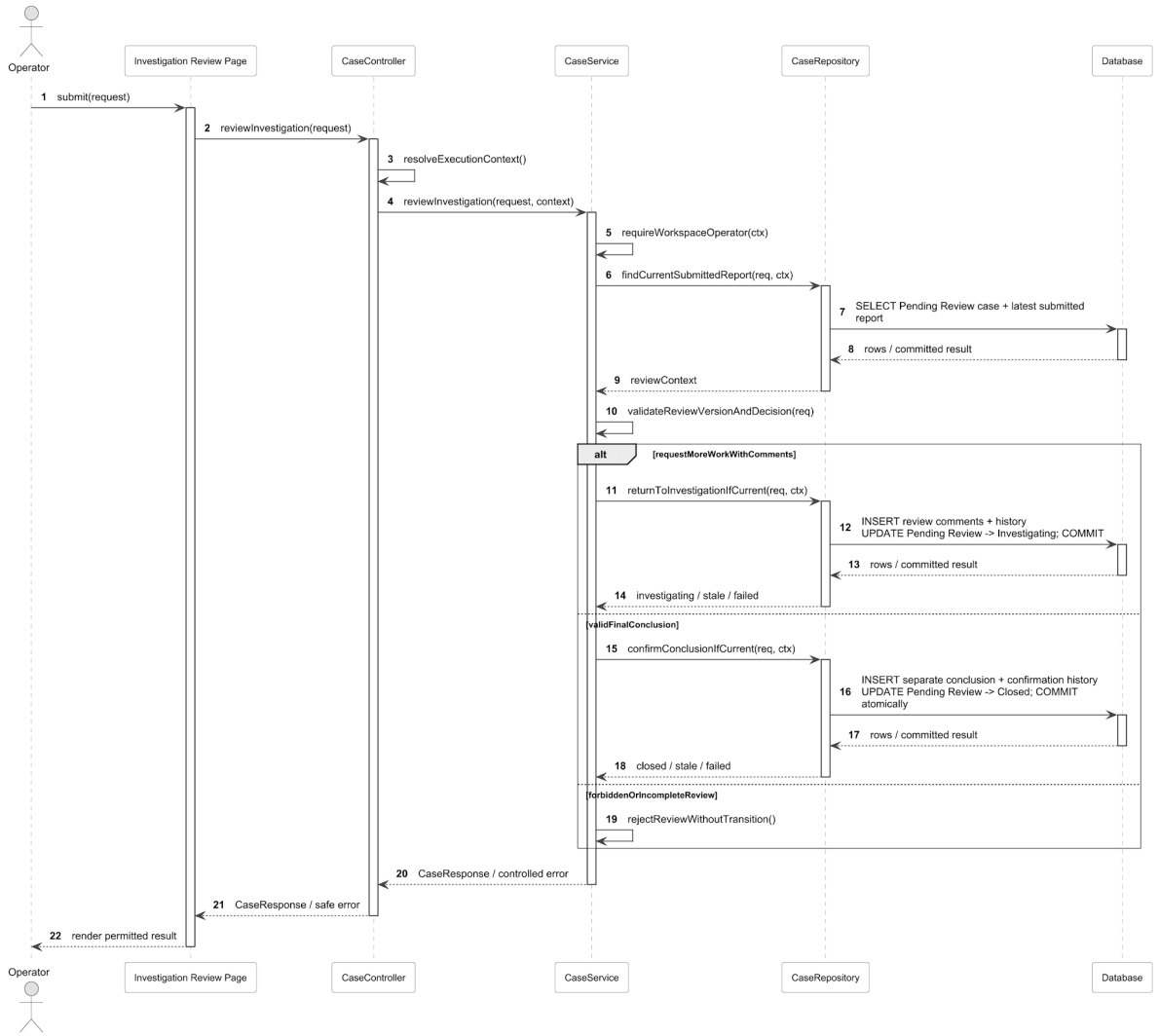


Figure 4.3.11.2.4. Request Additional Investigation or Confirm the Final Conclusion Sequence Diagram.

3.11.3 State Diagrams

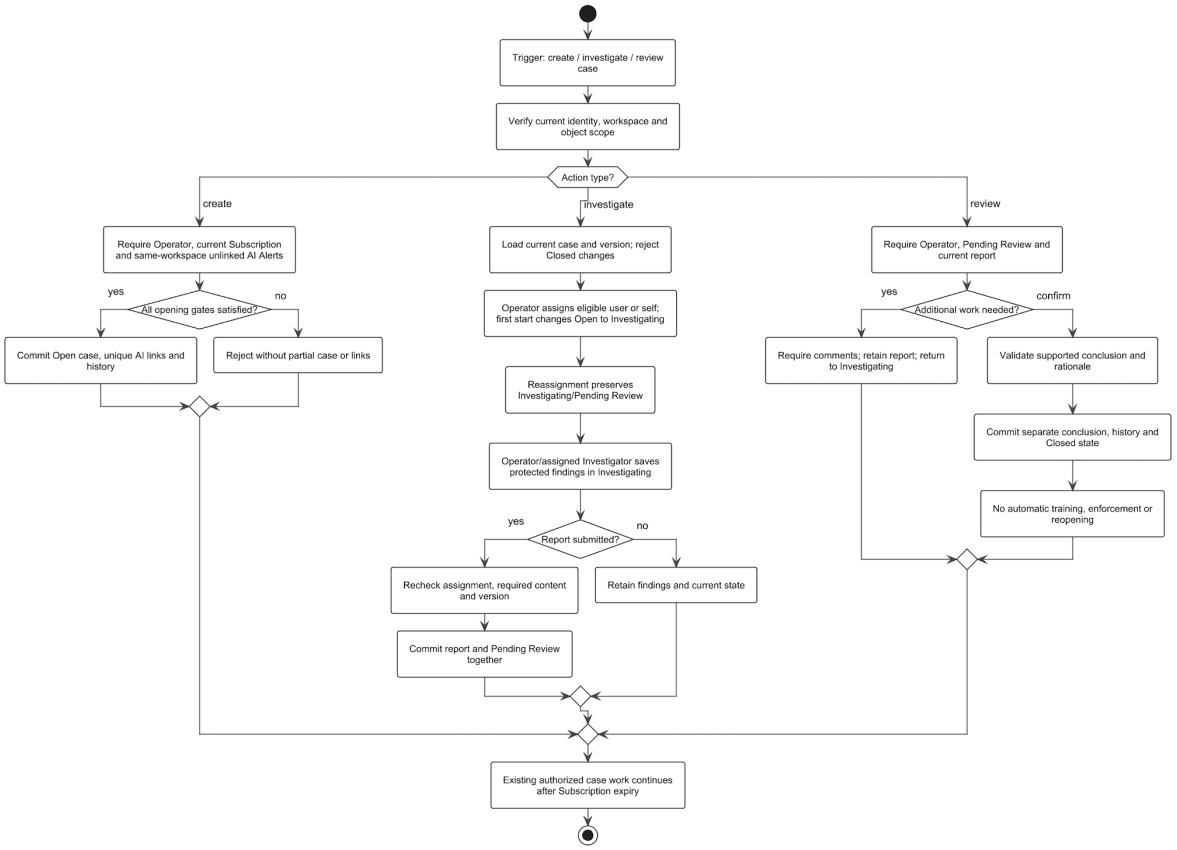


Figure 4.3.11.3.1. Investigation Case Lifecycle State Diagram.

3.12 Dashboard and Reporting

SRS reference: 3.13.1-3.13.4. Aggregates, reports and exports preserve workspace, role, assignment and field restrictions. Generation and download each recheck access; plaintext reveal does not authorize plaintext export.

3.12.1 Class Diagram

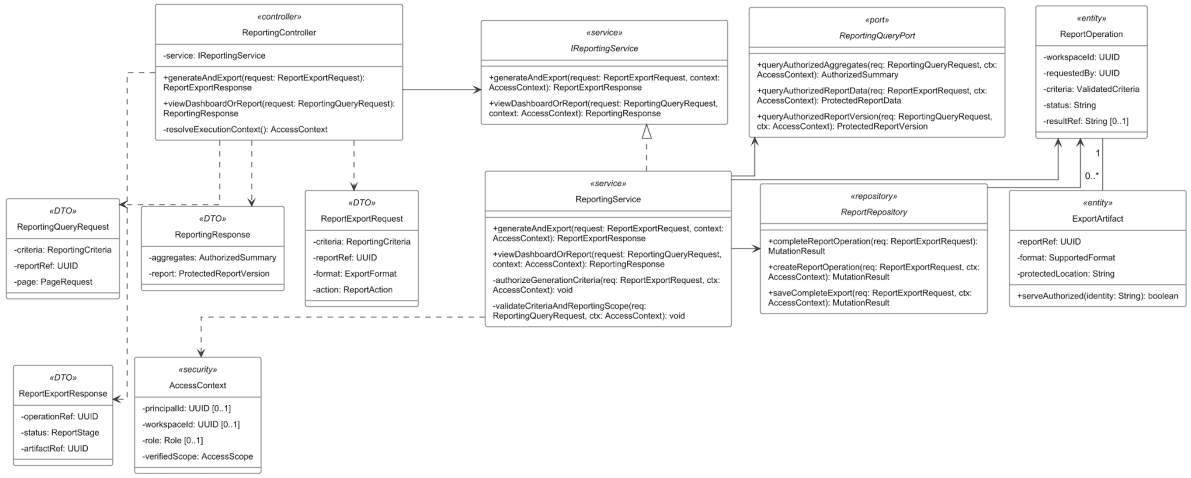


Figure 4.3.12.1.1. Dashboard and Reporting Class Diagram.

3.12.2 Sequence Diagrams

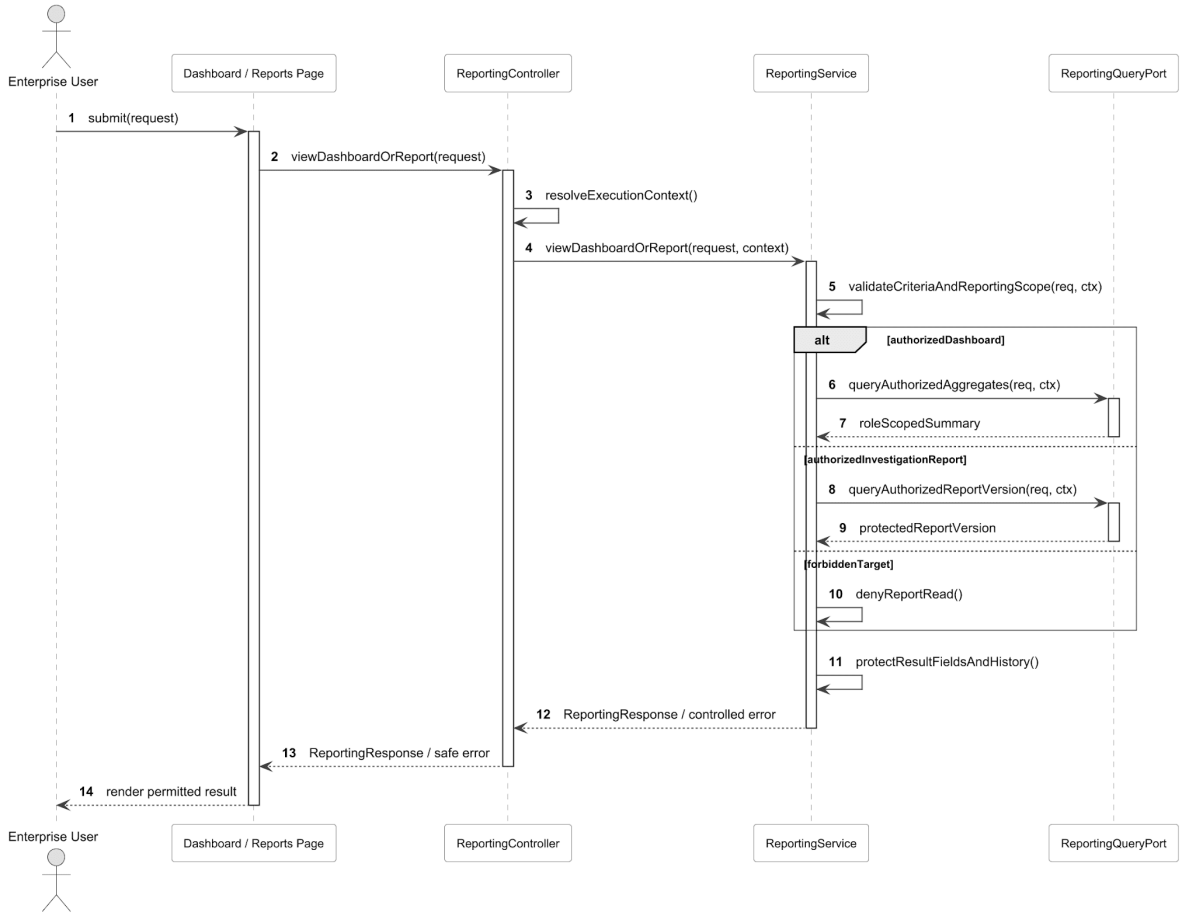


Figure 4.3.12.2.1. View Authorized Dashboard and Investigation Reports Sequence Diagram.

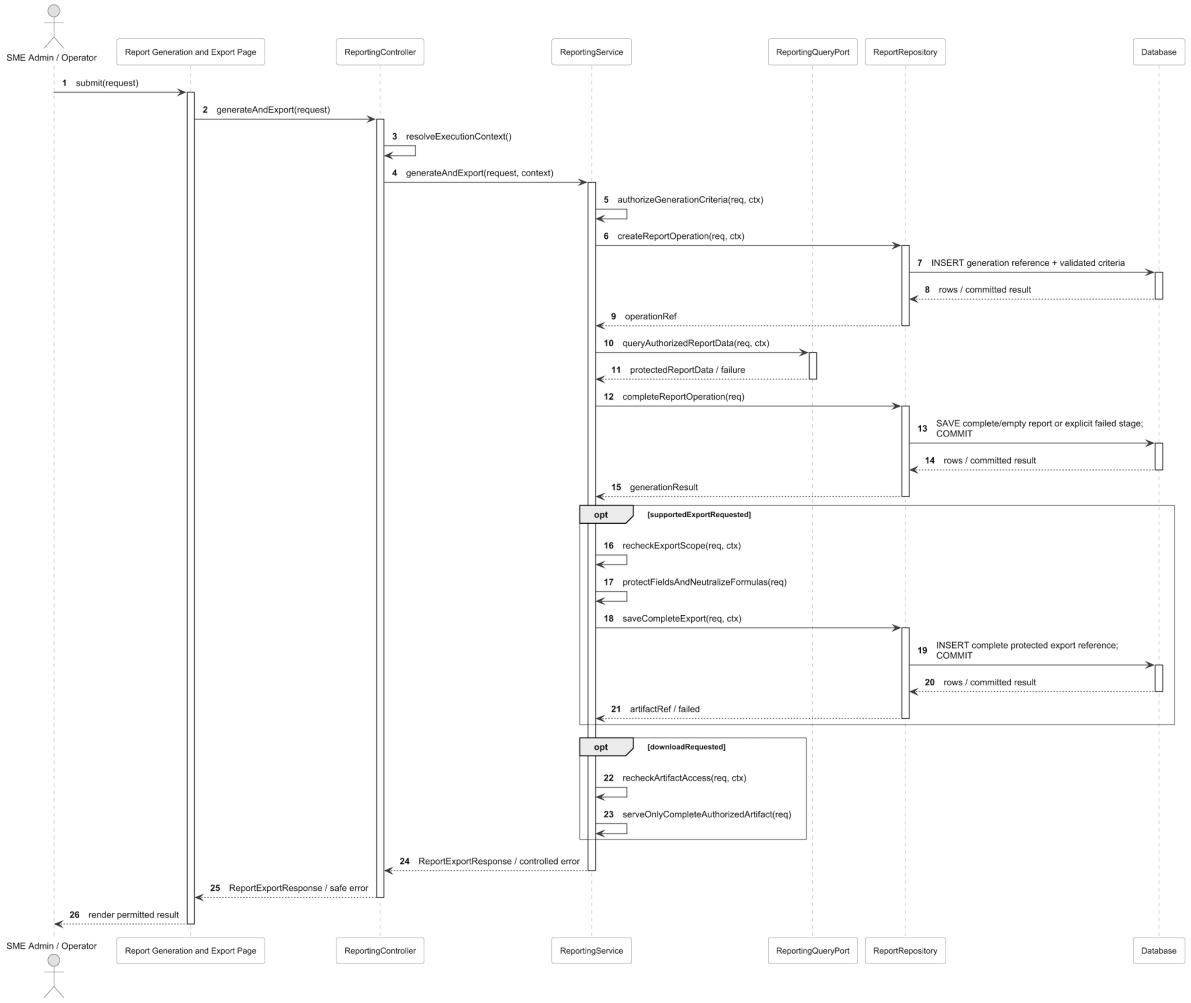


Figure 4.3.12.2.2. Generate and Export a Complete Authorized Enterprise Report Sequence Diagram.

3.12.3 State Diagrams

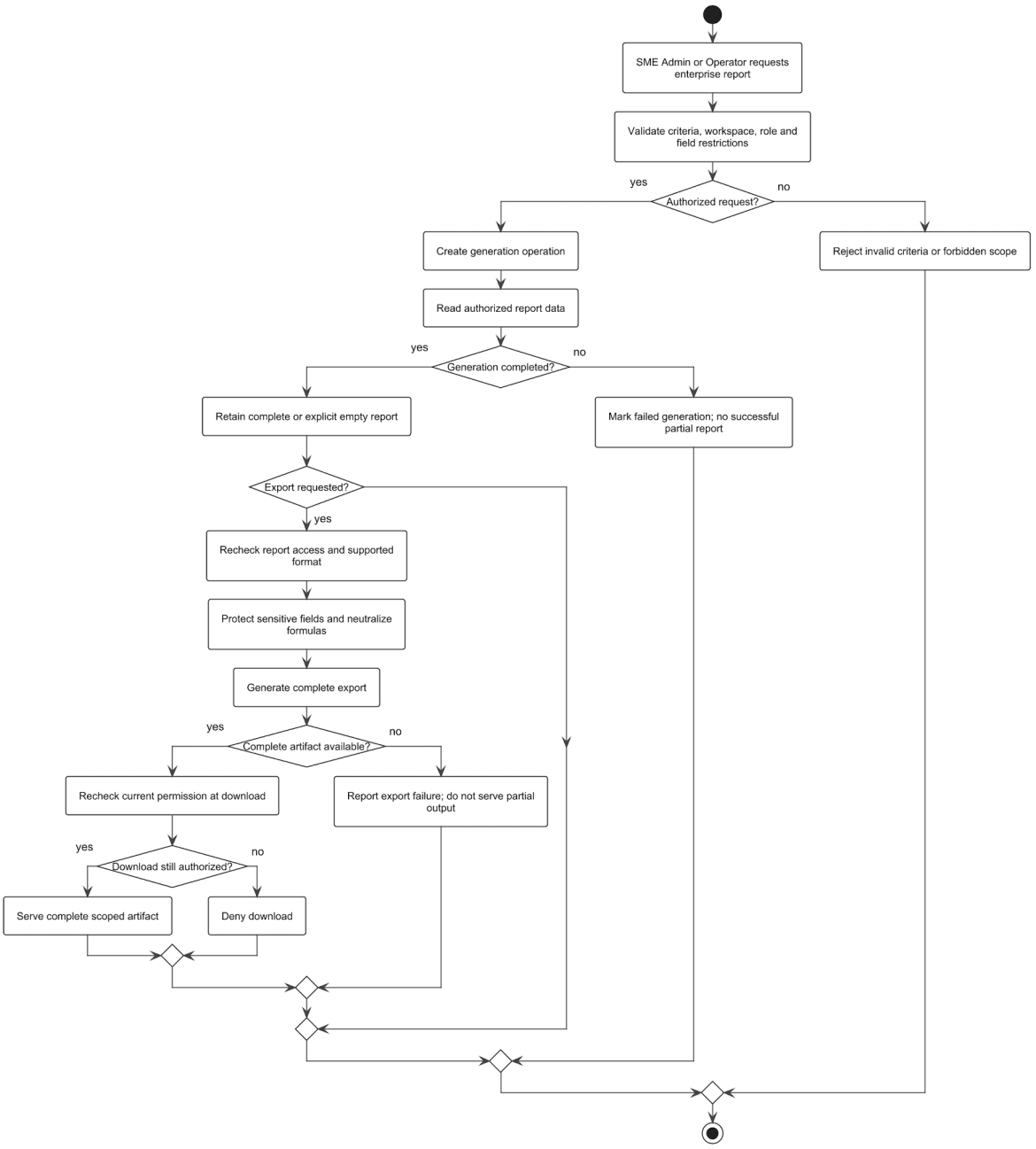


Figure 4.3.12.3.1. Enterprise Report Generation Lifecycle State Diagram.

3.13 Notifications

SRS reference: 3.14.1-3.14.2. Relevant enterprise events produce minimal in-app/email content. Recipients and linked records remain scoped; external delivery is retried independently and never reverses or repeats the committed business effect.

3.13.1 Class Diagram

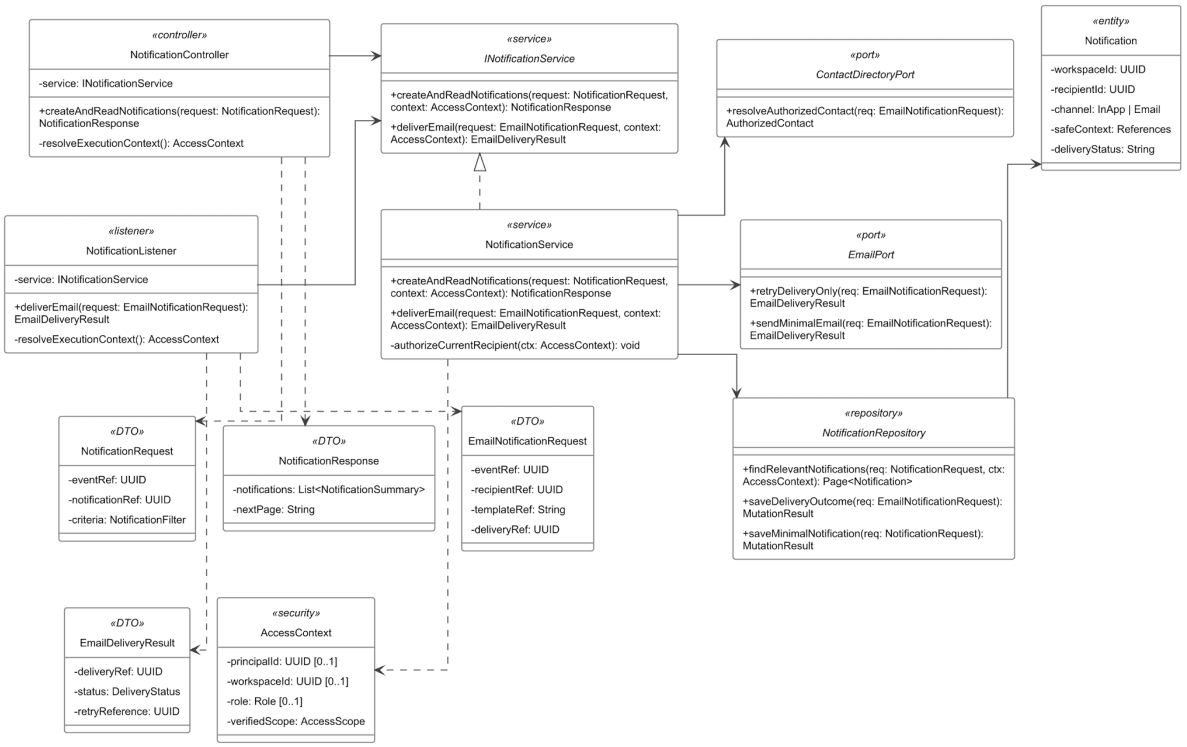


Figure 4.3.13.1.1. Notifications Class Diagram.

3.13.2 Sequence Diagrams

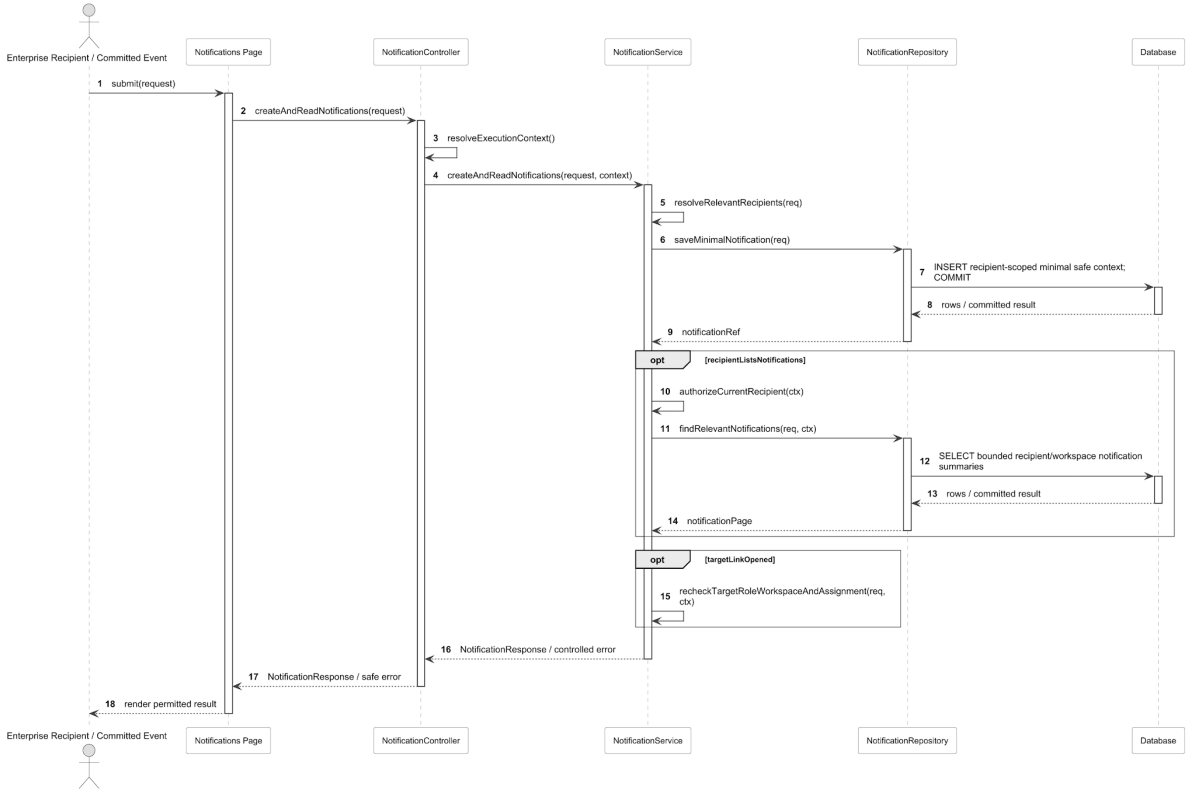


Figure 4.3.13.2.1. Create and View Relevant In-App Notifications Sequence Diagram.

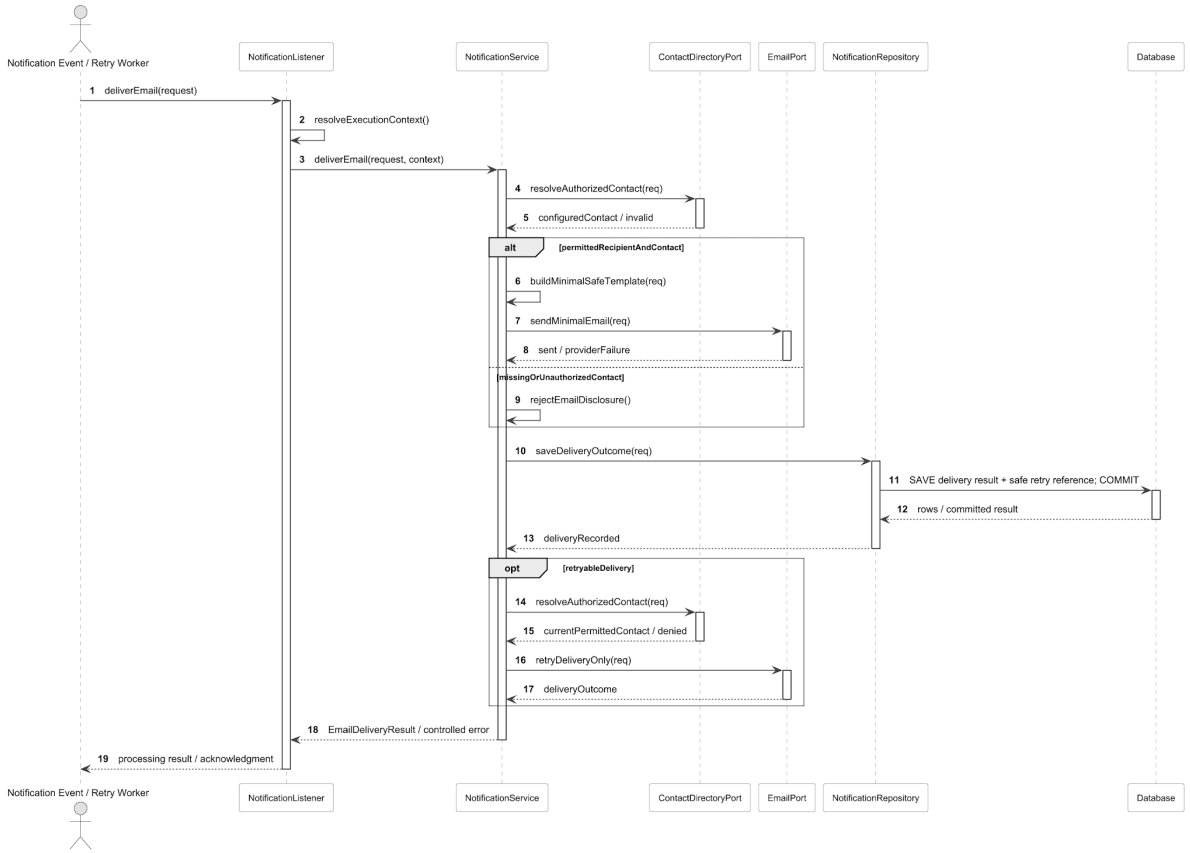


Figure 4.3.13.2.2. Send Minimal Email and Retry Delivery Independently Sequence Diagram.

3.14 Platform Monitoring and Audit

SRS reference: 3.15.1-3.15.2. Platform Admin sees operational metadata only. Object history follows the underlying administrative, Staff-assignment or enterprise/case scope. Required local change history commits with the business action and excludes values/secrets.

3.14.1 Class Diagram

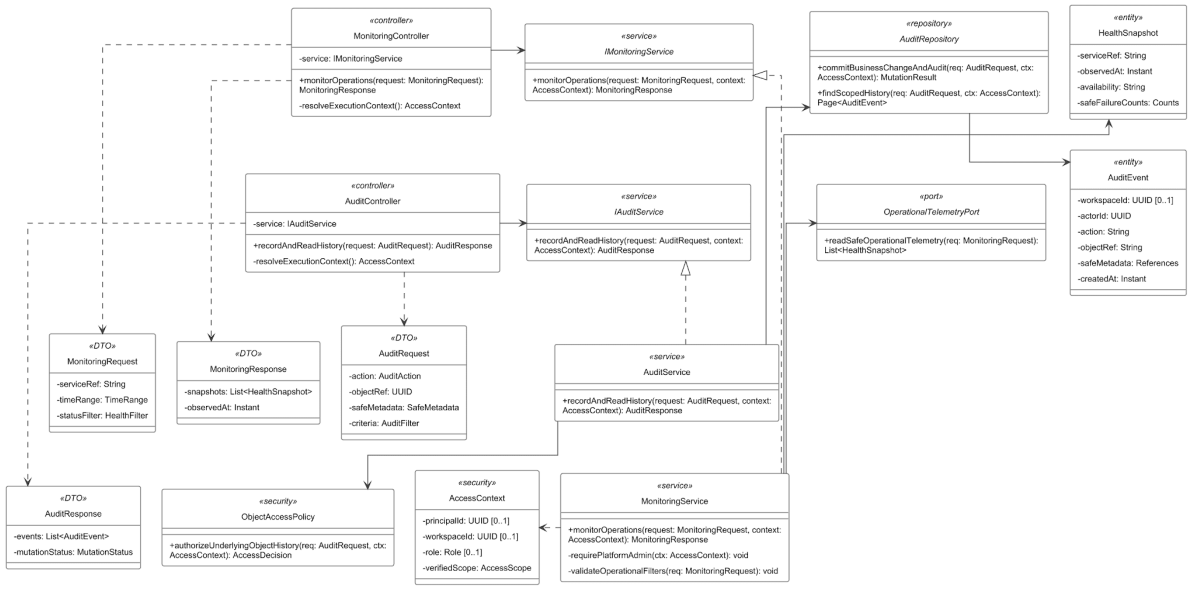


Figure 4.3.14.1.1. Platform Monitoring and Audit Class Diagram.

3.14.2 Sequence Diagrams

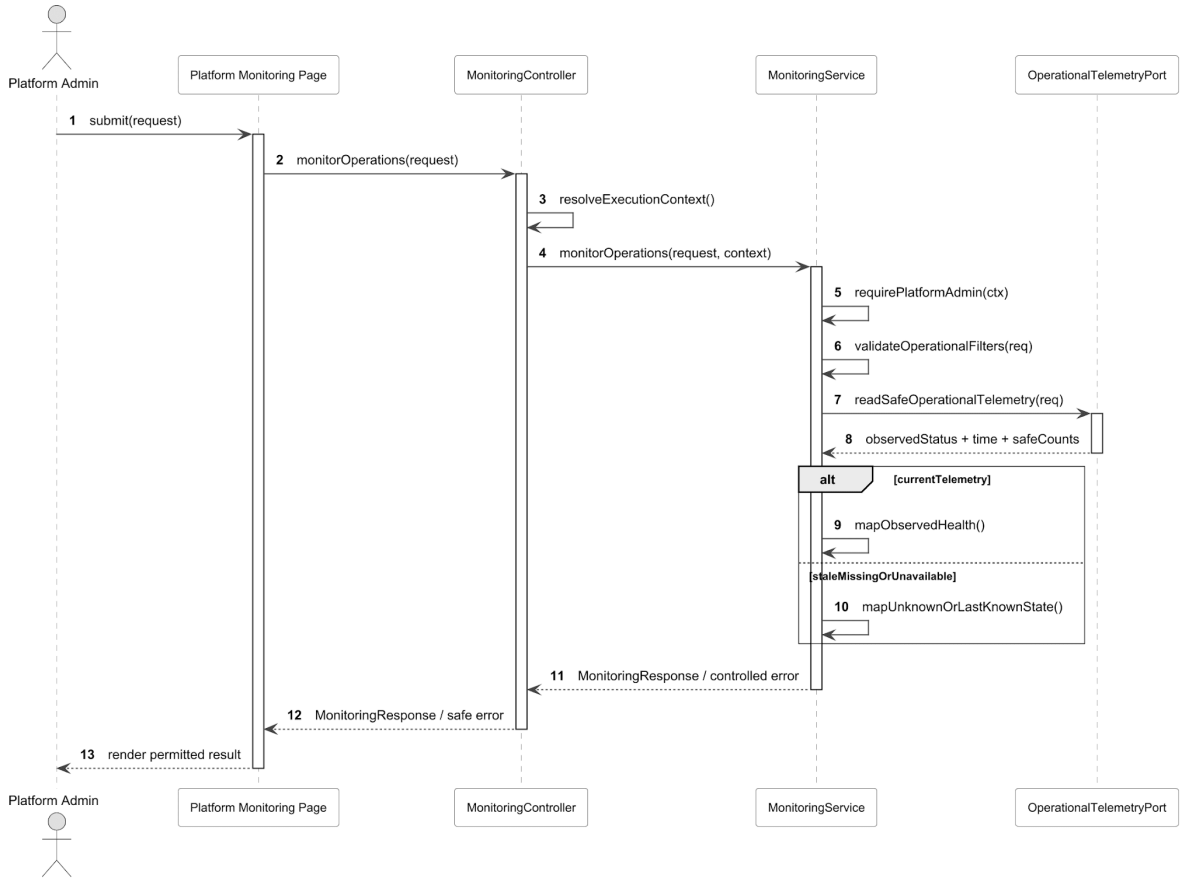


Figure 4.3.14.2.1. Monitor Platform Operations without Production Content Sequence Diagram.

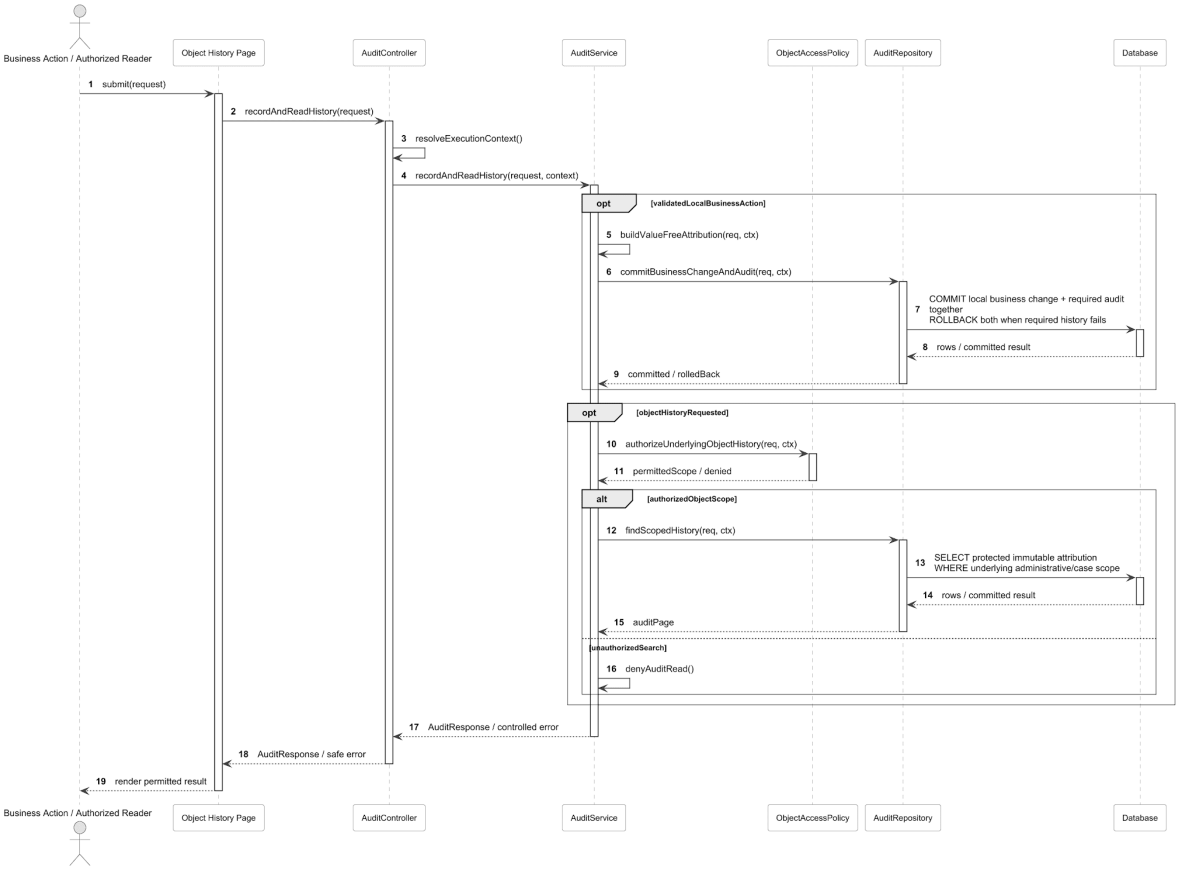


Figure 4.3.14.2.2. Commit Required Audit Attribution and Read Scoped Object History Sequence Diagram.